



Capstone Report

Formula SAE Electric

Mechanical Engineering Capstone Project

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Abstract

This report documents the design and optimization of an electric race car following the Formula SAE EV rulebook. Major design decisions focused around complying to strict technical requirements, while prioritizing safety, performance and manufacturability.

The chassis pairs a carbon-fiber front monocoque with a welded steel rear sub-frame that keeps fabrication manageable. A double-wishbone push-rod suspension optimizes the car's dynamics, and independent four-wheel drive is delivered through custom uprights and compound planetary gear reducers. The 600V accumulator and the in-hub motors are cooled with two independent cooling circuits. The driver interface features an adjustable pedal box and custom steering system for driver comfort, and downforce is generated by multi-element wings and a venturi undertray to keep the car stable.

The design process was finalized with a custom MATLAB optimization pipeline application that unified all the subsystems under a single performance framework, which enabled faster iteration. The app has a GUI for which users can input desired performance targets and driver specifications and get an instant design summary and updated CAD assembly. The report concludes with a case study on an AI-generated impact attenuator, finding that safety-critical components require empirical testing and additional oversight as a part of the responsibility of a professional engineer.



Table of contents

1. Project Charter	1
1.1 Mandate	1
1.2 Requirements	1
1.3 Constraints	1
1.4 Criteria	1
1.5 Design Optimization	1
2. Design Solution	2
2.1 Solution Description	2
2.1.1 Chassis Design	2
2.1.2 Suspension Design	4
2.1.3 Drivetrain Design	7
2.1.4 Steering Design	8
2.1.5 Pedal Box Design	9
2.1.6 Accumulator Design	12
2.1.7 Aerodynamics Design	13
2.1.8 Overall Vehicle Mass	16
2.2 Discussion on interconnectivity	17
3. Optimization	18
3.1 Outline	18
3.1.1 Optimized Frame Components	19
3.1.2 Optimized Suspension components	19
3.1.3 Optimized Drive Train components	20
3.1.4 Optimized Pedal Box components	21
3.1.5 Optimized Accumulator components	21
3.1.6 Optimized Aerodynamic components	21
3.2 Component optimization	22

3.3 Discussion on Results	23
3.3.1 Impact Attenuator Results.....	23
3.3.2 Suspension Results	23
3.3.3 Drive Train Results	25
3.3.4 Pedal Box Results	26
3.3.5 Accumulator Results.....	27
3.3.6 Aerodynamic Results.....	27
4. Ethics case study.....	29
4.1 Impact Attenuator Case Study	29
4.2 Cost and Budget Analysis.....	30
4.2.1 Itemized Cost Estimate for the Impact Attenuator (per unit):	30
4.2.2 Itemized Cost Estimate for the Testing Setup:	30
5. References.....	31
Appendix A. Datasheet.....	34
Appendix B. Additional Requirements.....	60
B.1 General Requirements.....	60
B.2 Frame Requirements	60
B.3 Suspension Requirements	60
B.4 Steering Requirements	61
B.5 Pedal Box Requirements.....	61
B.6 Drivetrain Requirements	61
B.7 Accumulator Requirements.....	62
B.8 Cooling Requirements.....	62
B.9 Aerodynamics and Bodywork Requirements.....	62
Appendix C. Additional Constraints.....	63
C.1 General Constraints.....	63
C.2 Frame Constraints	63
C.3 Suspension Constraints	63

C.4 Steering Constraints	63
C.5 Pedal Box Constraints	64
C.6 Drivetrain Constraints	64
C.7 Battery Box Constraints	64
C.8 Cooling Constraints.....	64
C.9 Aerodynamics and Bodywork Constraints.....	65
Appendix D. Additional Criteria	66
D.1 Mass and Center of Gravity	66
D.2 Durability	66
D.3 Manufacturability.....	66
D.4 Packaging.....	67
D.5 Safety	67
D.6 Assembly	67
D.7 Repairability, Serviceability and Maintainability	67
D.8 System Stiffness (Stability).....	67
D.9 Ergonomics	67
D.10 Reliability.....	68
D.11 Precision and Backlash	68
D.12 Tunability.....	68
D.13 Efficiency.....	68
D.14 Thermal Management Difficulty	69
D.15 Dynamic Event Criteria	69
D.15.1. Acceleration (75m sprint)	69
D.15.2. Skidpad (figure-eight circuit).....	69
D.15.3. Autocross (technical sprint)	69
D.15.4. Endurance (22km race).....	69
D.15.5. Efficiency (evaluated during endurance)	70
Appendix E. Additional Designs	71

E.1 Chassis Design – Monocoque	71
E.2 Suspension Design - Mounting	72
E.3 Drivetrain Design – Additional exploded views	73
E.4 Steering Design – Mounting & Column Assembly.....	74
E.5 Pedal Box Design - Throttle Pedal	75
E.6 Pedal Box Design - Platform.....	75
E.7 Pedal Box Design - Angular play	76
E.8 Aerodynamics Design	77
E.9 Cooling Design.....	81
E.10 Overall Vehicle Mass	84
Appendix F. Subsystem Flowchart.....	87
F.1 Kinematics Modelling Flowchart	87
F.2 Frame Flowchart.....	88
F.3 Suspension Flowchart.....	90
F.4 Drive Train Flowcharts.....	93
F.5 Pedal Box Flowchart	95
F.6 Accumulator Flowchart	97
F.7 Aerodynamics Flowchart.....	98
Appendix G. Ethics Case Study.....	101
Appendix H. Read Me	103
H.1 Location of Assembly/GUI files	104
H.2 Recommended Usage Procedure:	105
H.3 Main GUI Functions	105
H.4 Interface Inputs	105
H.5 Defaults values and Ranges	106
H.6 How does the program work.....	107
H.7 Important Implementation notes	107
H.8 Troubleshooting	107

1. Project Charter

1.1 Mandate

With electric vehicles (EV) sales increasing [1], countries promoting their adoption [2], and high-performance applications gaining popularity [3], the global shift from combustion to EVs is accelerating. Our team is tasked to develop a fully integrated, competition-ready electric vehicle architecture by creating a feasible and manufacturable FSAE-compliant electric race car by applying mechanical design, vehicle dynamics, and reliable manufacturing solutions. To achieve this, our team must develop and integrate several critical subsystems, including a chassis, suspension, powertrain, driver controls, and an aerodynamic package.

1.2 Requirements

The car must safely complete all FSAE events and pass mandatory inspections such as mechanical tech, tilt, rain, and brake tests. Additionally, all systems must satisfy the general safety and performance requirements defined in the FSAE Electric rulebook.

1.3 Constraints

The design is constrained by the FSAE rulebook which includes limits on geometry, wheelbase, cockpit space, ground clearance, and driver visibility. Other components such as critical fasteners, structural joints, and interfaces must follow engineering standards. Some components are required to remain accessible for inspection, withstand worst-case loading, and be manufacturable with typical resources.

1.4 Criteria

The prioritized criteria are safety, stiffness, reliability, ease of assembly and maintenance. Other design criteria considered a low mass and center of gravity and efficient packaging. All the additional relevant requirements, constraints, and criteria can be found in the Appendix B, Appendix C and Appendix D.

1.5 Design Optimization

The optimization strategy leverages key FSAE performance metrics to drive both structural and functional design across all vehicle subsystems. By processing event-specific inputs such as longitudinal acceleration, lateral G-loads, and minimum turn radii, the model calculates the dynamic requirements for components like the frame, suspension members, and drivetrain gears. This data-driven approach focuses on maximizing performance and structural integrity while simultaneously minimizing vehicle mass. The resulting workflow ensures that every geometric output, from wing angles of attack to master cylinder bore diameters, is mathematically tuned to the specific physical and regulatory constraints of the competition. See section 3.2 Component optimization for a table of GUI inputs and Results.

2. Design Solution

Below is an isometric exploded view of all the relevant sub system discussed in this report. The design of the cooling system can be found in appendix E.9.

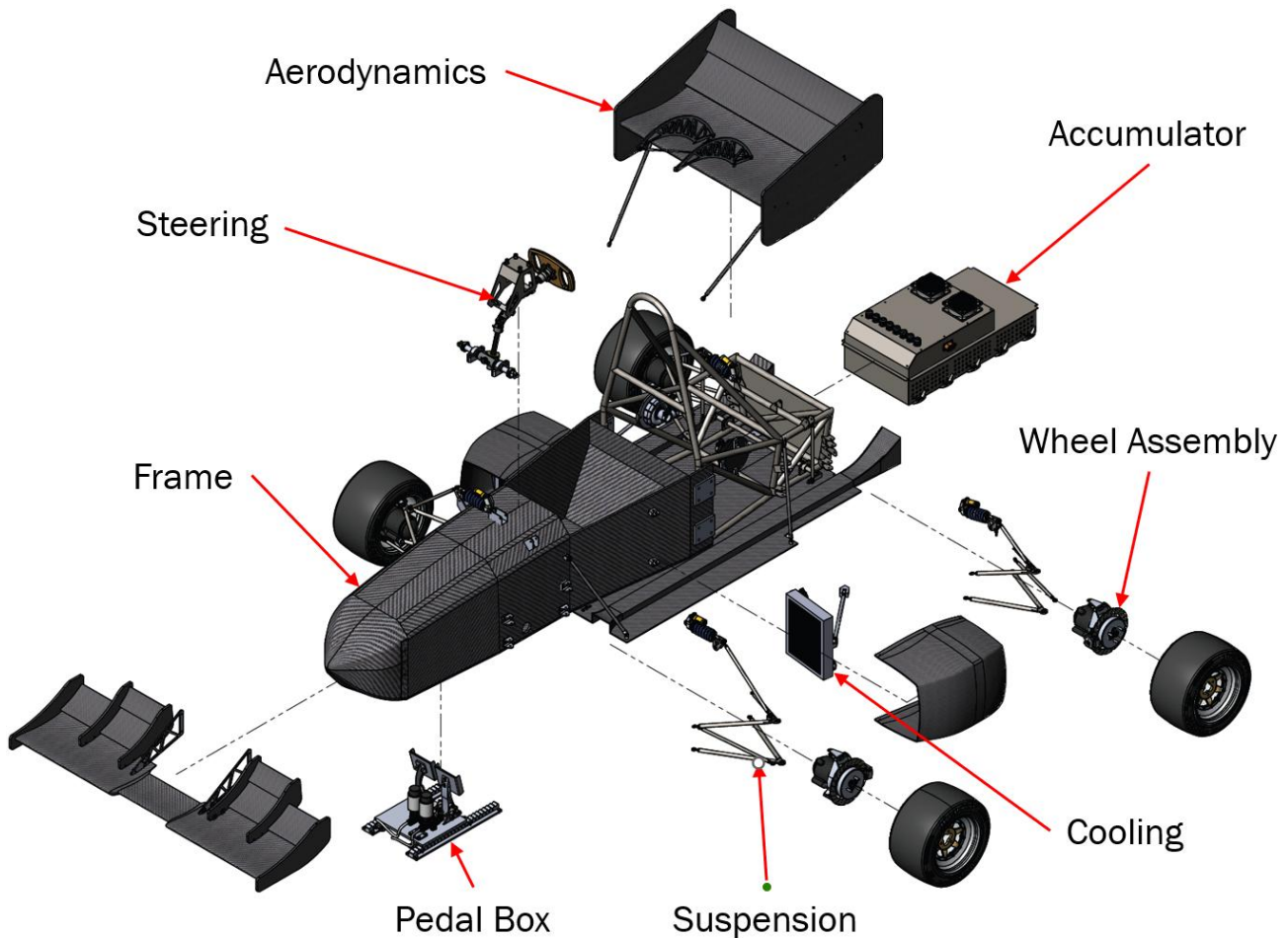


Figure 1: Semi-Exploded View of the Full Design Solution

2.1 Solution Description

2.1.1 Chassis Design

This solution proposes the integration of a front section made of composite materials and a rear section made of welded tubes forming a rear sub-frame connecting to the composite section at the main roll hoop. This solution offers excellent performance and good bending stiffness due to the composite section of the chassis. The use of a rear sub-frame allows the simplification of the manufacturing process and the decrease in the costs associated with the construction of the composite panels since only the front section needs to be made of composite materials. The tubes will be welded or bolted to the inserts that are going to be bonded into the composite panels during the curing process. The rear sub-frame will be welded to the main roll hoop as in a space frame configuration. In addition, the front composite section is designed with an opening to provide

access to the pedal box for maintenance and adjustment purposes without removing the entire front structure. This opening requires custom bolts and nuts that follow the curvature of the monocoque. Alternatively, custom potted inserts can be incorporated in the opening of the monocoque and the removable panel. However, this requires more resources during the manufacturing of the monocoque. The front of the composite panel is also reinforced to provide a mounting point for the impact attenuator, ensuring a secure and rigid attachment capable of absorbing and dissipating frontal crash energy effectively. The impact attenuator is secured using a bolted connection to the front bulkhead composite panel. The rear anti-intrusion plate is welded to the rear bulkhead. The front nose cone is bonded to the front bulkhead using an adhesive compound to safely secure it on the monocoque.

The frame represents the structural mounting platform for the different components of the car. Therefore, dimensions should be chosen in a way that ensures enough room is available for all systems while guaranteeing safety and performance. These dimensions must also comply with the specified geometric rules stated in the rulebook (i.e. side impact structure geometry). The following drawing showcases a detailed view of the chassis with the different preliminary dimensions.

For more details regarding the mounting solution on the front monocoque section, refer to Appendix E.1.

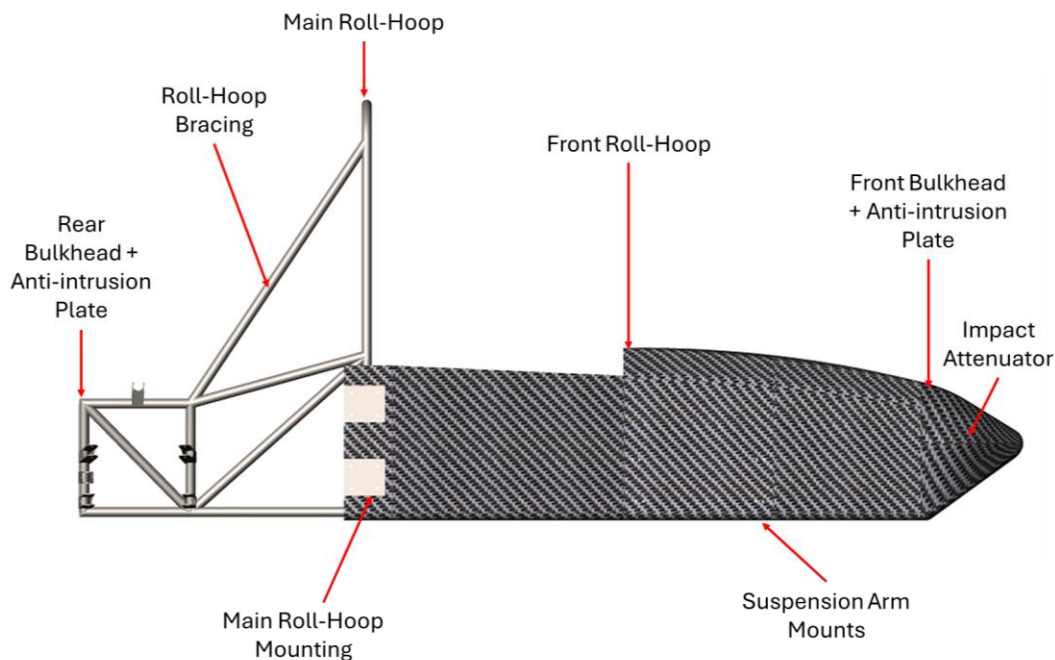
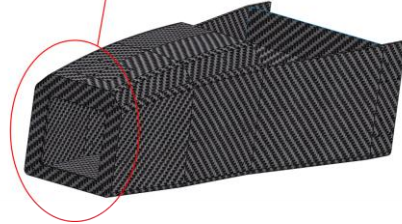


Figure 2: Design Solution - Side View Drawing of Hybrid Frame for representation only.

Anti-Intrusion
Plate and Impact
Attenuator
Mounting Zone



Anti-Intrusion
Plate Mounting
Zone

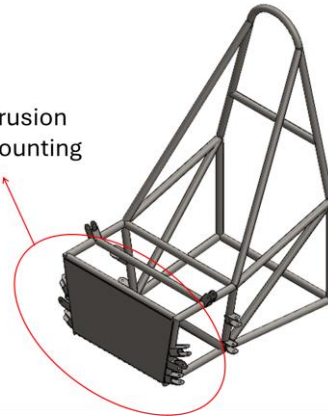


Figure 3: Design Solution – Front and Rear Anti-Intrusion plate mounting zone.

ALL DIMENSIONS IN MM

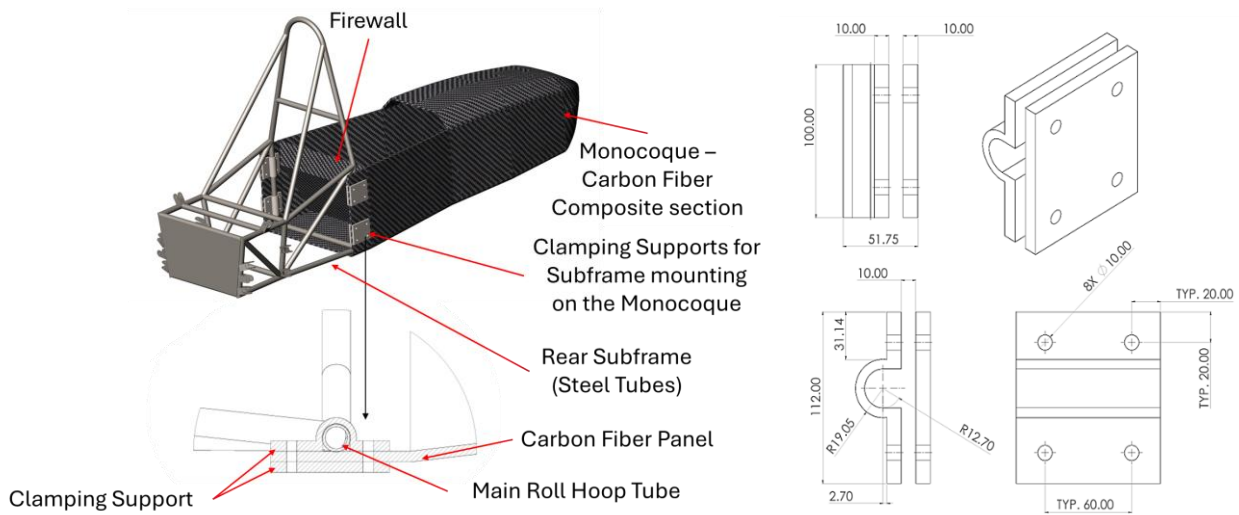


Figure 4: Design Solution - Mounting Attachment between Tubular-Monocoque.

2.1.2 Suspension Design

The selected solution features a double wishbone push-rod suspension connected to inboard-mounted springs and dampers with bell cranks for both the front and rear suspension, as seen in Figure 5. The dampers selected are the Ohlins TTX Mk2 [4] (see Appendix A for datasheet), which are designed specifically for FSAE vehicle and have a good range of adjustability. Vertical wheel motion is transmitted inboard by the push rod, which acts in compression to rotate a bell crank. The bell crank, shown in Figure 7, determines the motion ratio, for adjusting how wheel displacement translates to spring movement. By relocating heavy components inboard, the system reduces unsprung mass and improves handling responsiveness.

The torsion anti-roll bar is mounted inboard near the bottom of the frame, which lowers the suspension system's center of mass. When there is a differential in wheel displacement under cornering, the ARB is twisted, which increase the roll stiffness. The ARB is supported by chassis-mounted bearings and connected

to the bell cranks through adjustable ARB links, allowing fine-tuning of roll stiffness and overall vehicle balance.

All rod connections, including the wishbones, tie rods, push rods, and anti-roll bar links, utilize rod-end Heim joints, providing adjustability in setup changes, precise articulation, and compensation for slight manufacturing misalignments, as shown in Figure 61.

Suspension geometry starts with determining a wheelbase and trackwidth. The wheelbase is restricted by the rules to 1525 mm (V.1.2) [5]. The shorter the wheelbase the better the maneuverability [6], so for our vehicle the minimum wheel base was chosen. For track width, the only requirement is to pass the tilt test. The tilt test requires the car to tilt to 60° sideways without tipping. This is the equivalent of taking a flat corner with a 1.732 *G* lateral acceleration without aerodynamic downforce. The calculation for minimum track width based on the center of gravity height is demonstrated later in the report but the chosen track width was 1.27 m (50 in) which leave a safety margin.

The suspension geometry was optimized using a Multi-Body Dynamics (MBD) model in Altair MotionSolve, ensuring the desired kinematic behavior through the full range of travel. To ensure structural integrity, the tube profiles and ARB were optimized using MATLAB, balancing the required stiffness and strength against mass targets. Suspension mounting design solution is presented in appendix E.2

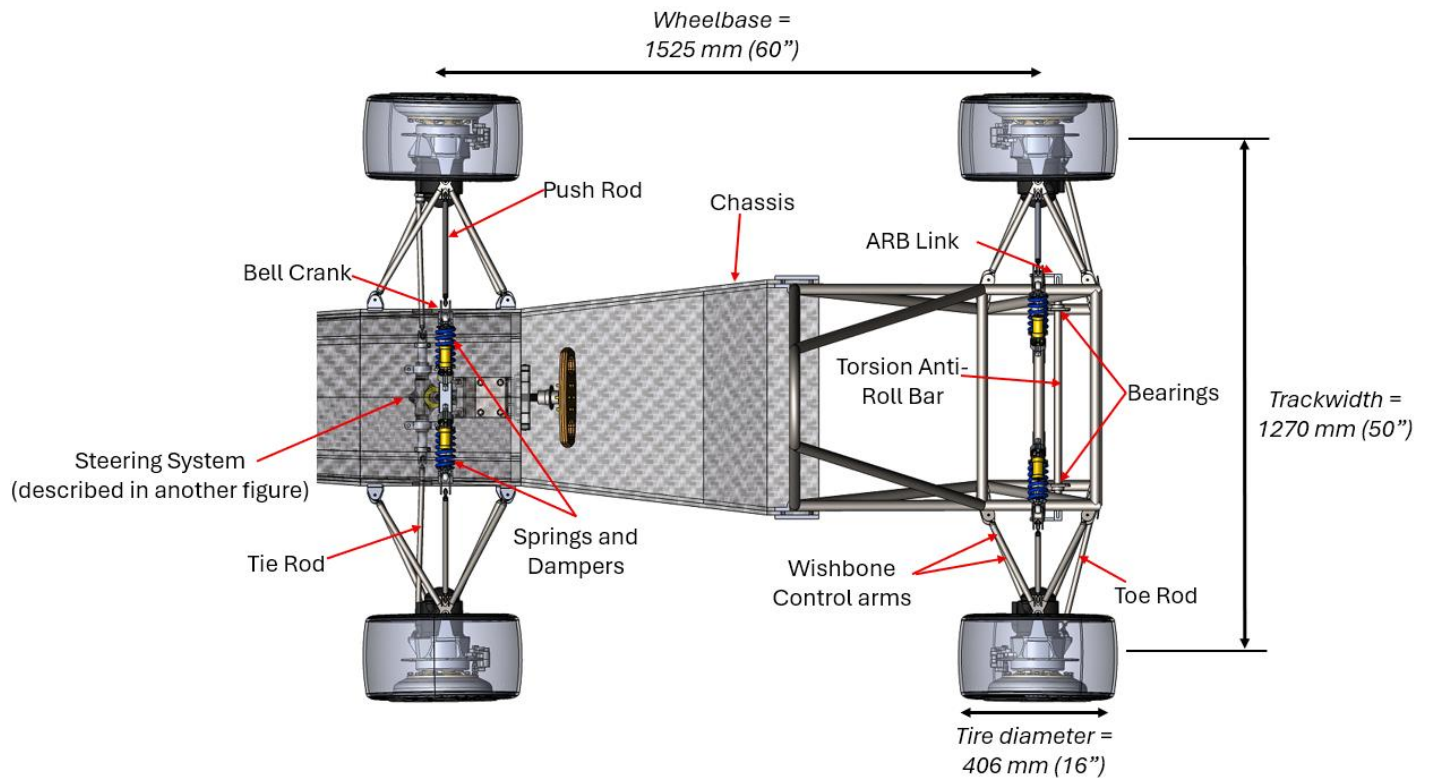


Figure 5: Design Solution - Top View of Overall Suspension System.

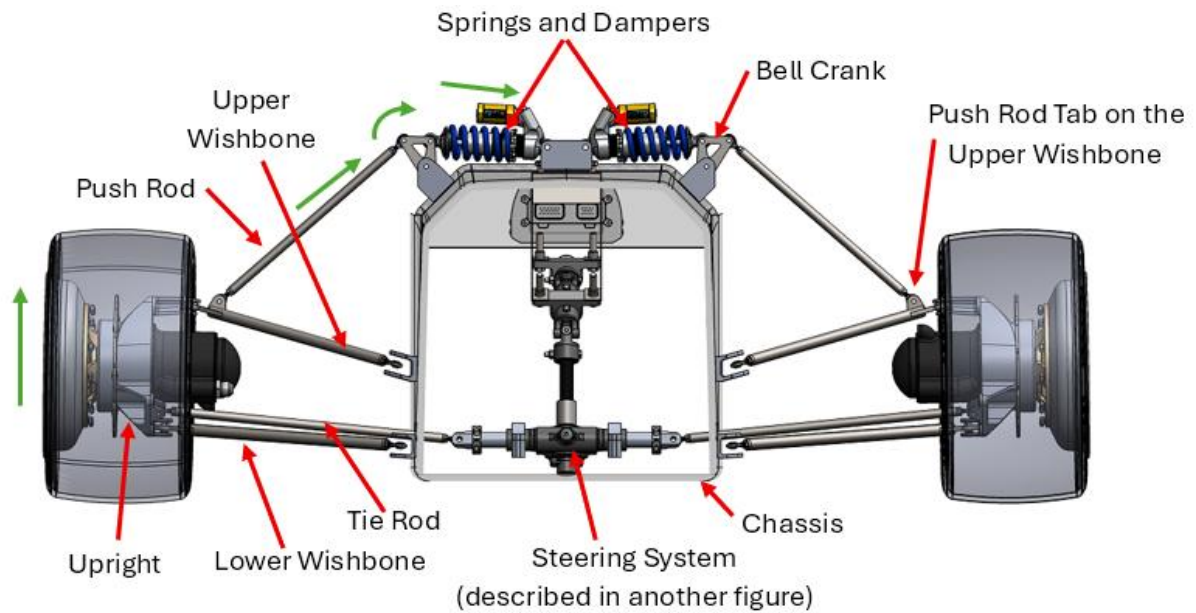


Figure 6: Design Solution – Front View of Suspension System.

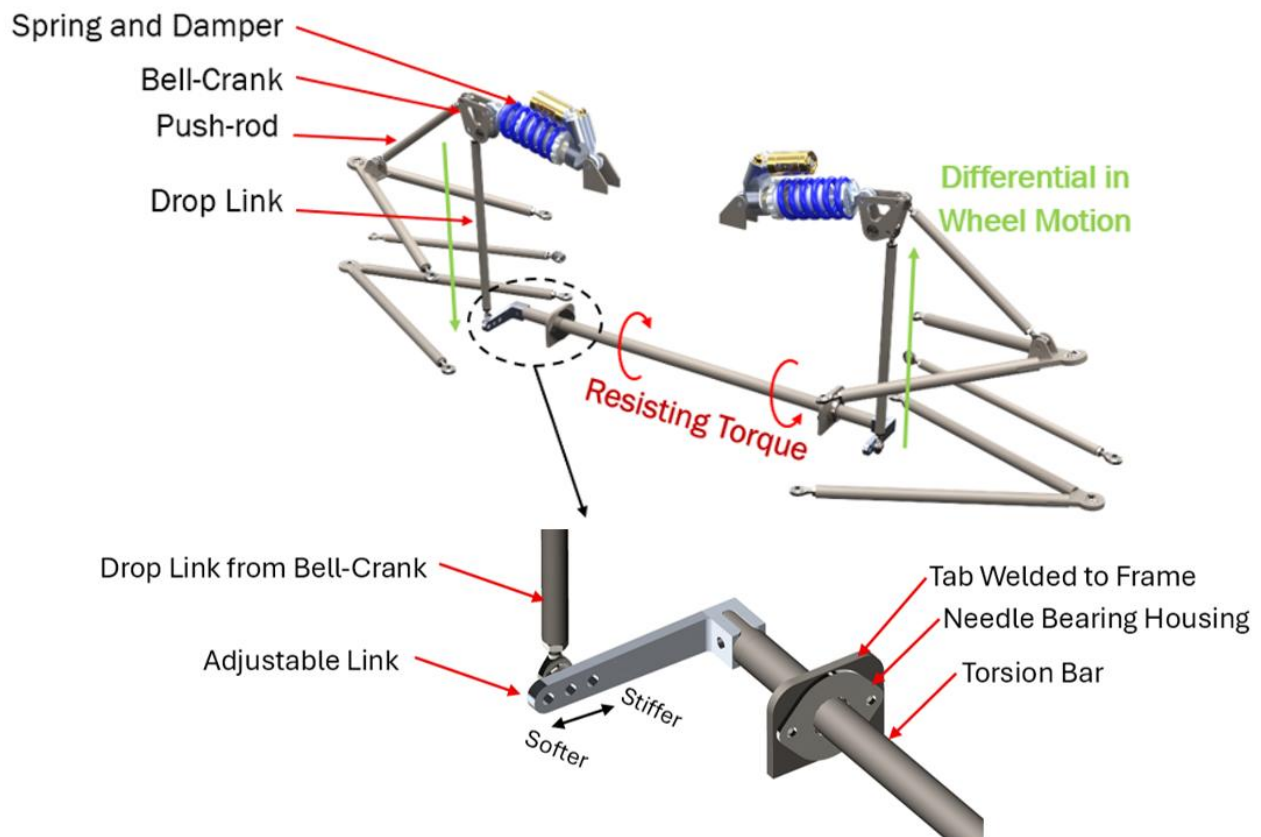


Figure 7: Design Solution - Torsion Anti-roll Bar

2.1.3 Drivetrain Design

For the wheel assembly, an in-hub drivetrain architecture was selected. This means that the car is in an all-wheel drive configuration, with independent wheel control. The two-piece billet uprights act as the housing for the compound planetary gear reducers. The planet gear carriers hold three precision ground axles, which also act as the wheel mounting studs. These carriers rest on angular contact bearing in the upright, and the planetary gearset itself replaces the function of the spindle on regular, undriven wheel assemblies.

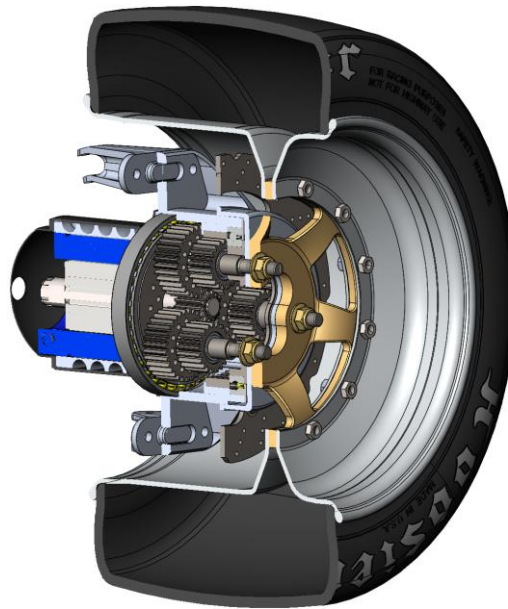


Figure 8: Design Solution - Section view of the wheel assembly.

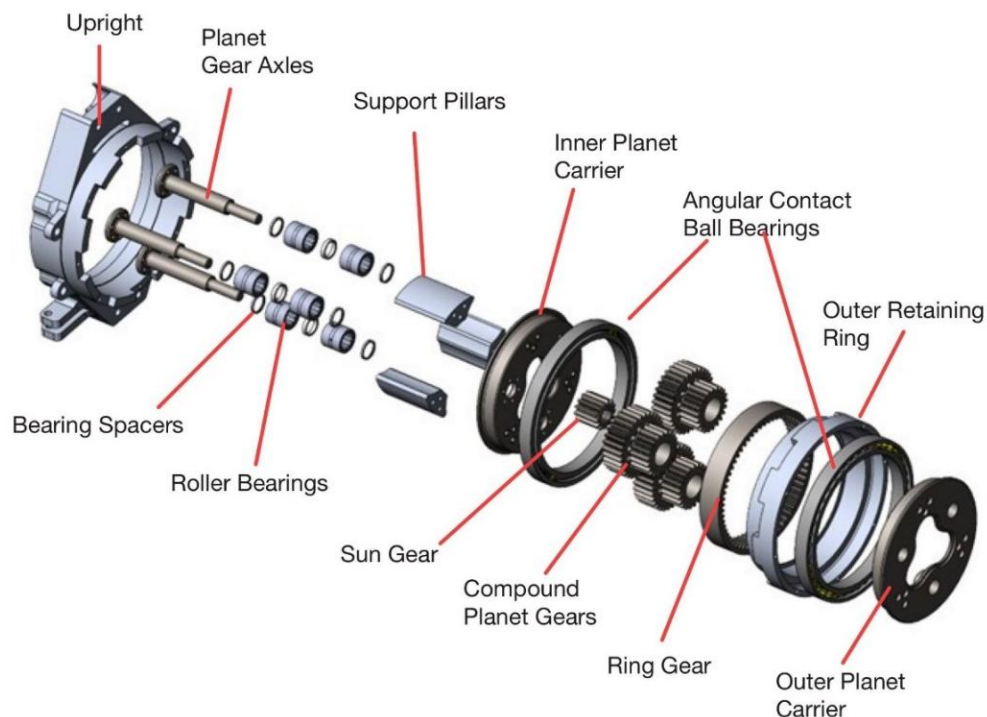


Figure 9: Design Solution - Exploded view of the gearbox housed in the upright.

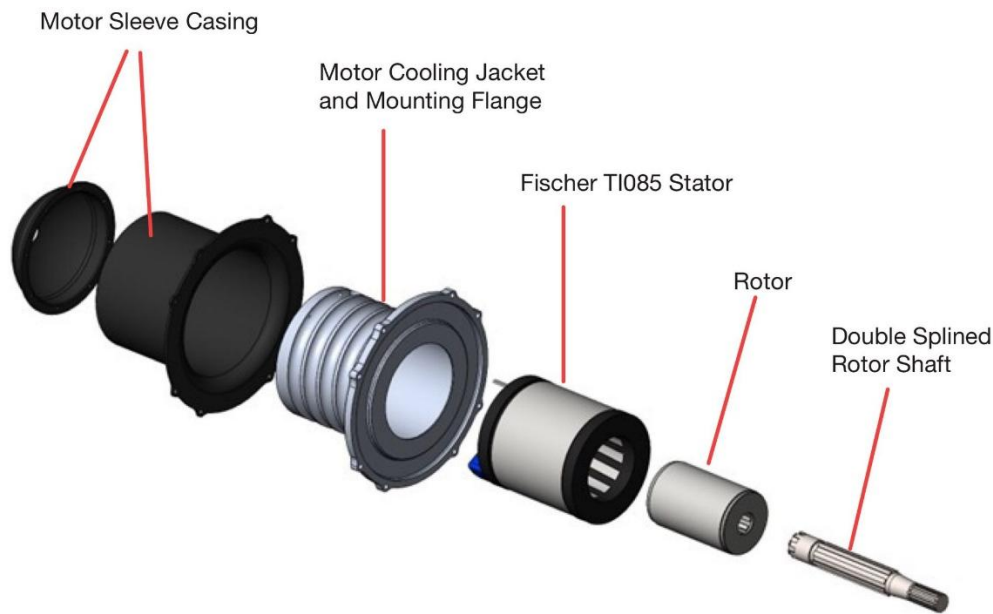


Figure 10: Design Solution - Exploded view of the motor assembly.

2.1.4 Steering Design

This system transmits the steering wheel input through three composite shafts connected in series via U-joints, which allows the column to transmit the torque to follow a downward path through the chassis structure. The first and second shafts of the column are supported by pillow-block bearings bolted to the monocoque section of the hybrid chassis with a sheet metal bracket. This arrangement directs the rotational input towards a rack-and-pinion assembly, where its housing supports the third shaft of the column. The rack-and-pinion converts the torque into a linear movement that pushes and pulls each tie rods. These same tie rods are mounted on the front up right to steer the wheels into the desired direction. The description on how it is mounted to the chassis is described in Appendix E. The datasheet of the following off-the-shelf components described below can be found in the Appendix A.

Firstly, A quick release hub from Fire and Safety System Ltd. is connected to the steering column to enter and exit the vehicle freely (Ingress/Egress). Secondly, all three shafts are made from Carbon Fiber-Reinforced Polymer (CFRP) tubing from Rock West Composite to keep the system light as possible while still maintaining a good weight-to-stiffness ratio under axial torque. Additionally, adhesive 3M Scotch-Weld Epoxy is added on the diameter of the shaft and u-joint interface, which serve as the primary torque transfer. Thirdly, the U-joints from Mc-master Carr are made to remain accurate under repeatable motion, can accommodate up to a 40° working angle and weighs approximately 100g each. Lastly, the rack-and-Pinion comes from the supplier KAZ Technologies, which the total gear assembly is advertised to weighs 3 lbs, with its rack and pinion gears machined out of steel and is housed in an aluminum body. Its geometry gives a steering ratio of 4.71 per

revolution and corresponds to 248° of pinion rotation. Clamp collars are attached to both ends of the center link (rack) to act as lateral end stops, limiting wheel travel when contacting the steering rack mounts.

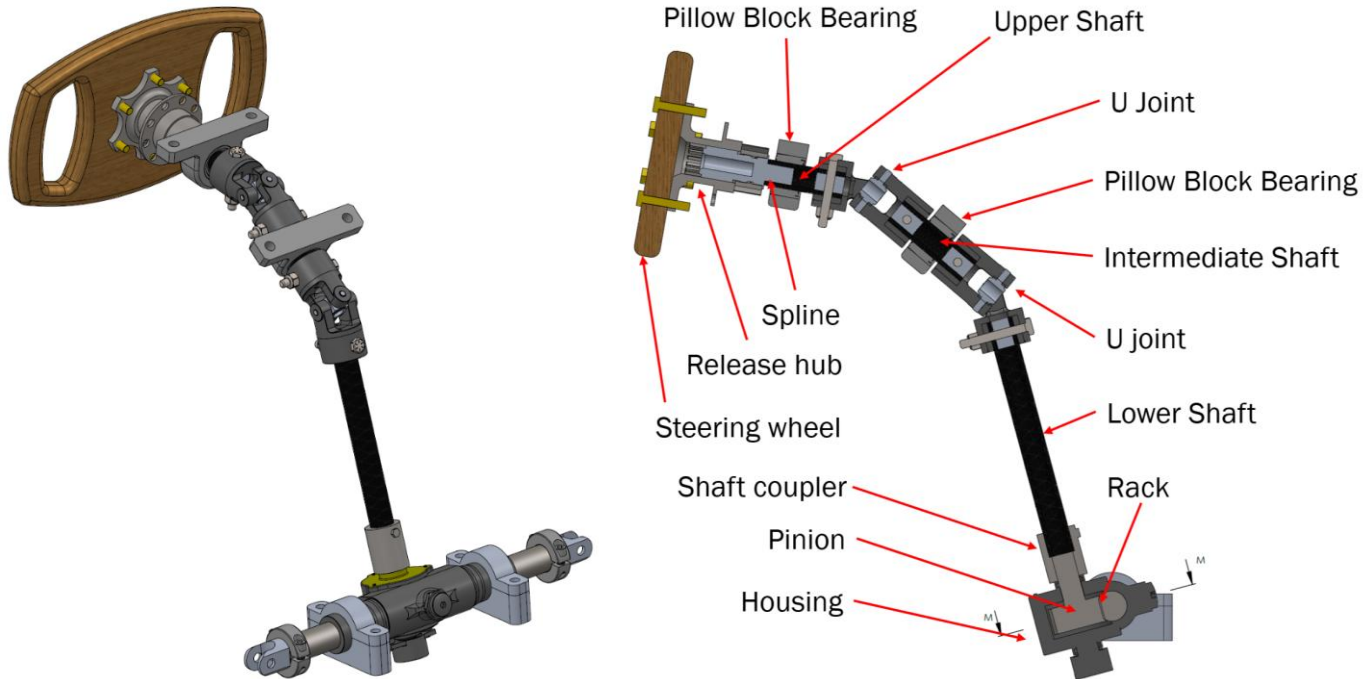


Figure 11: Design Solution – Cross section & Isometric view of Steering Column Assembly

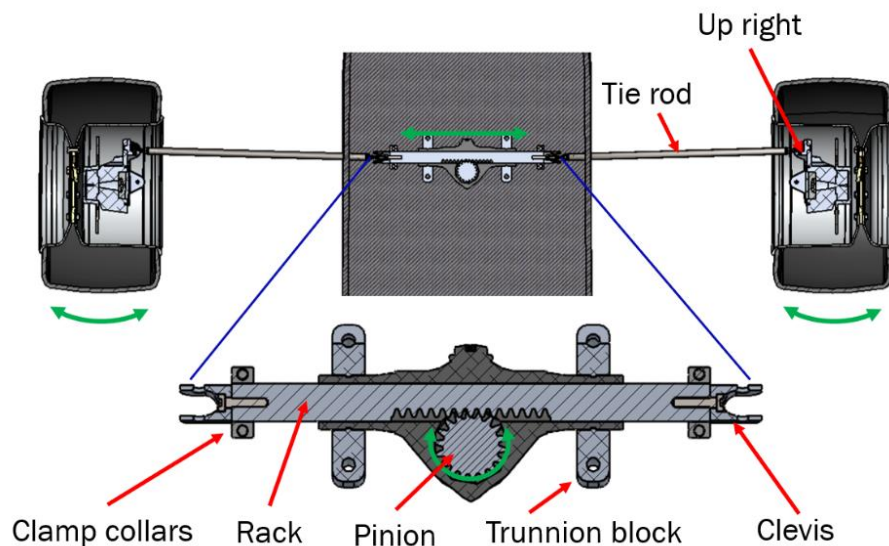


Figure 12: Design Solution - Cross section view of Rack-&-Pinion and tie rod with wheel Assembly

2.1.5 Pedal Box Design

This pedal box employs an adjustable platform guided by a pair of linear rails. The rails are permanently fastened to the chassis and feature longitudinal rows of indexing hole to provide enough multiple positioning options for ergonomic adjustment and driver fitment. The platform is secured to the rails using four quick-release pins (one per corner). The platform is CNC-milled, with carved mounting tabs for the pedals to offer

better structural rigidity and limiting the amount of failure points. Both pedals have shoulder bolt that are used as pivot point and are fastened through the tabs of the platform.

The throttle pedal uses two potentiometers to convert the angular displacement of the pedal into a voltage for the motor to read. After pressing to accelerate, two spring-return are mounted on the pivot to bring back the pedal to the original position. More images on its assembly can be found in the Appendix E. For the brake, two master cylinders are used to convert the mechanical force of the foot into a hydraulic pressure. The balance bar is assumed to provide a 50/50 pressure distribution, which will later be tuned for optimal brake bias. Both cylinders are held with a CNC machined Bracket to align horizontally the transmitted force. The resulting pressure is routed to the calipers of the wheel as described previously via hydraulic line with fittings. In case the hydraulic line fails, the pedal will continue its motion and flip an emergency switch (BOTS) that will electrically turn off the car.

The rotary potentiometer, Safety switch, balance bar, master cylinders, spring return and important fasteners are all off-the-shelf components (see Appendix A), including their electric and hydraulic cables routed behind the system.

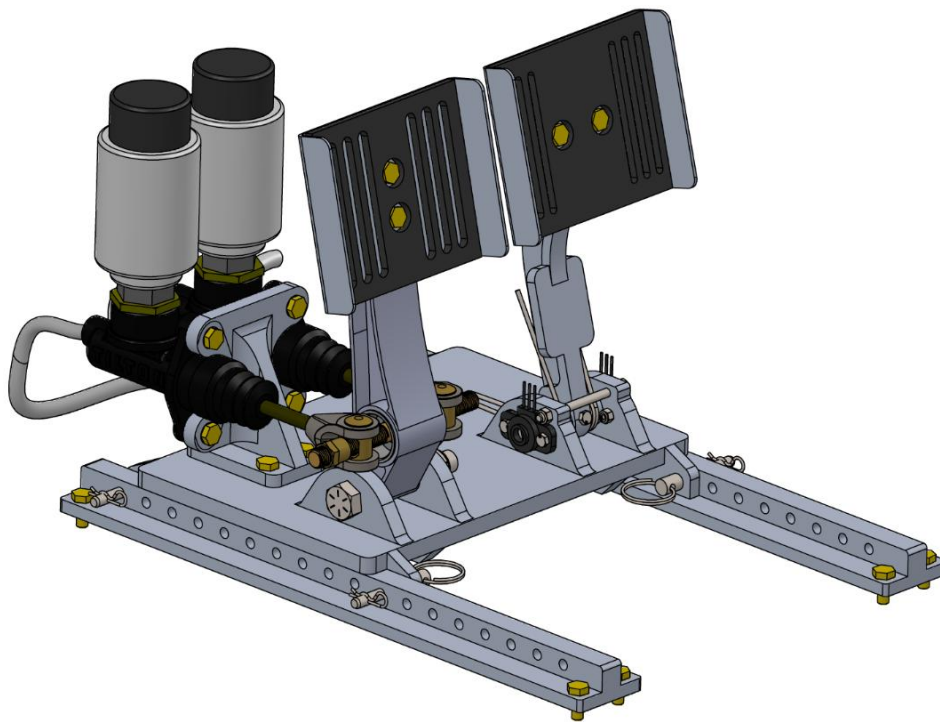


Figure 13: Design Solution – Pedal box Assembly

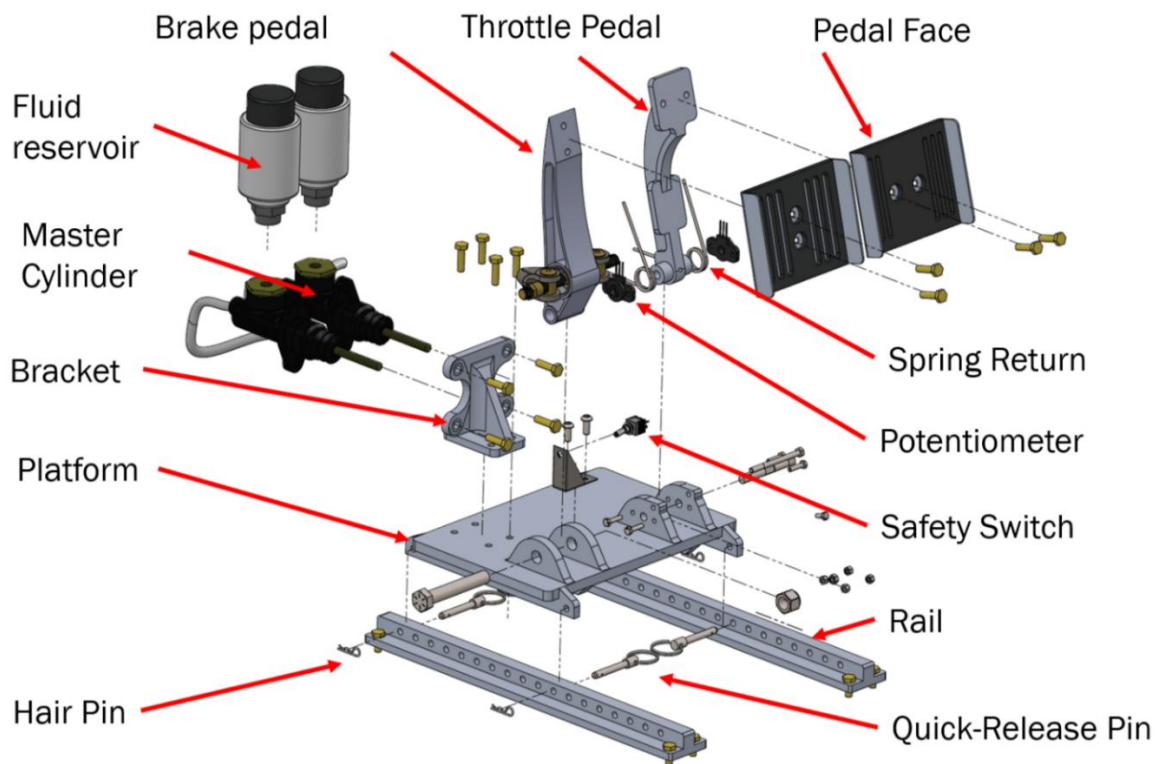


Figure 14: Design Solution – Exploded view of Pedal box Assembly

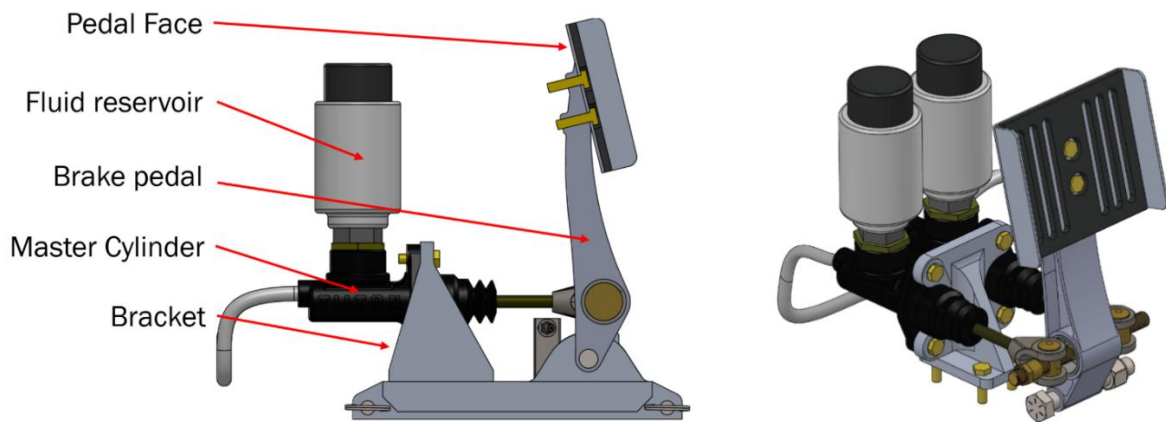


Figure 15: Design Solution - Cross Section View of Brake Assembly

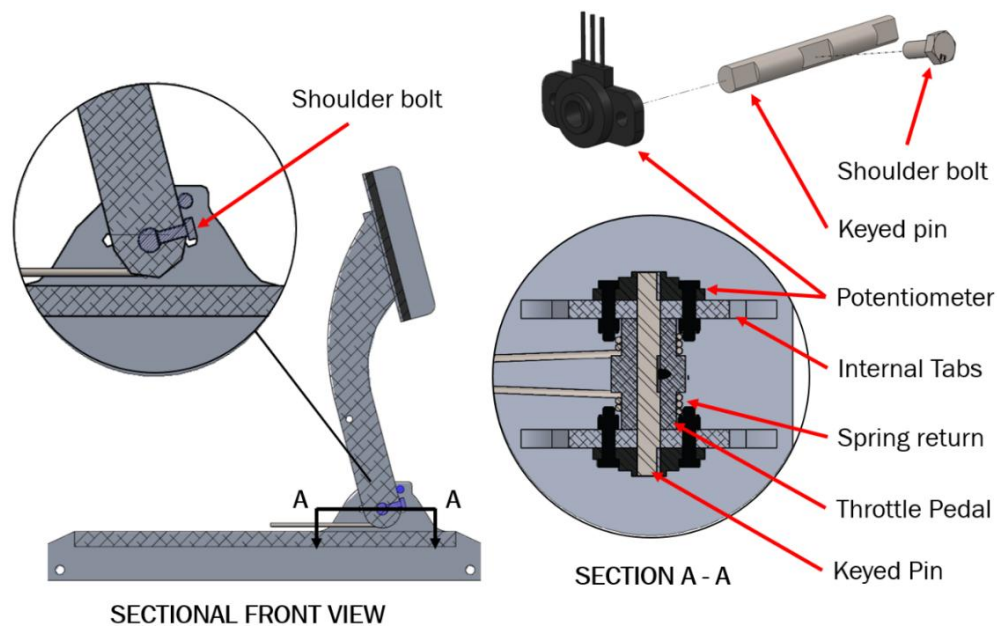


Figure 16: Design Solution - Cross section view of Throttle Assembly

2.1.6 Accumulator Design

The accumulator in this vehicle concept is called a “skateboard pack” as it is very slim and is mounted low to the ground, filling the vehicle floor between the firewall and rear bulkhead. It has a bent and welded sheet metal construction, bolted to the chassis at multiple hardpoints. The battery cell modules are mounted in isolated compartments in a single layer at the bottom, with the high-voltage to low-voltage interface and the power distribution systems on top. The whole unit is air cooled, with designated channels, vents and fans to promote airflow and heat transfer. The baseplate is structural, allowing the container to be mounted at the edges, and the accumulator easily drops out of the bottom of the car when the bolts are removed.

A low and wide battery assists in giving the car a low centre of gravity height. The removable top panel allows all components of the electrical system of the car to be accessed for maintenance or inspection through the firewall. Arranging the battery modules in one layer also simplifies the internal walls of the container. With this setup, sealing and structural design are intricate. Routing critical connections and components near the ground also requires careful consideration placed on system mounting and protection.

The accumulator design process began with defining the total power requirements based on 80 kW system power limit from the FSAE rulebook. Early decisions focused on cell selection, voltage architecture and structural design. Molicel P42A 21700 cells in a 4s1p submodules (see Appendix A) were chosen for their high-power density, proven use in Formula Student applications and wide availability. Since the motors chosen are rated for use at 600 V, the battery module configuration was optimized to approach this voltage. Multiple different configurations were evaluated with the goal of achieving a voltage close to 600V, while keeping cell counts low. A 27s4p (27 in series x 4 in parallel) module configuration was adopted. This created a module

voltage of 97 V, while remaining under the 120 V, 6 MJ and 12 Kg per module requirement. Six modules in series yields a system voltage of 582 V, maximizing motor utilization within the 600 V limit.

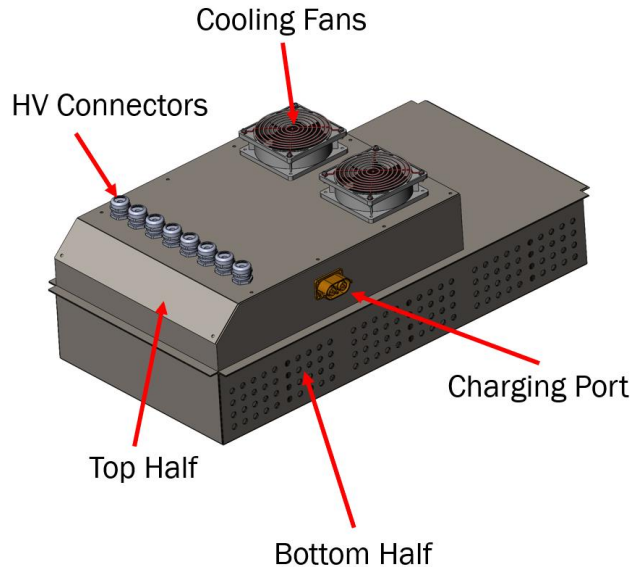


Figure 17: Design Solution - Skateboard pack accumulator.

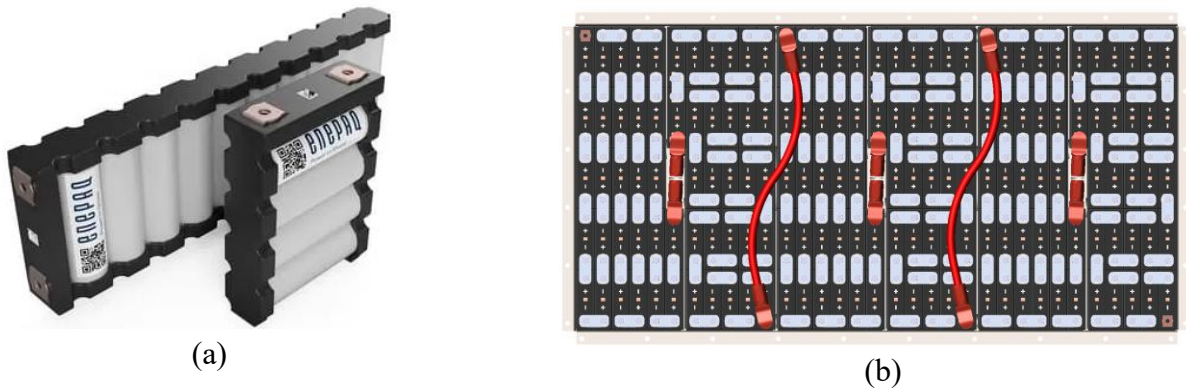


Figure 18: Accumulator submodules (a) and the internal submodule layout of the accumulator (b)

2.1.7 Aerodynamics Design

The aerodynamics of this concept include a standard and static front wing and rear wing. The secondary elements of the front wing only span from the outer face of the wheels to the nosecone while the tertiary element span from the inside of the wheel to the nosecone. Along with these wings, an undertray/diffuser is included with the floor to increase downforce under the car and to direct air to the rear of the car. Lastly, sidepods are mounted around the radiators on the sides of the car to increase the flow rate and create more controlled flow through the radiators. For additional content on mounting, dimensions, and weight of the components, refer to Appendix E.8.

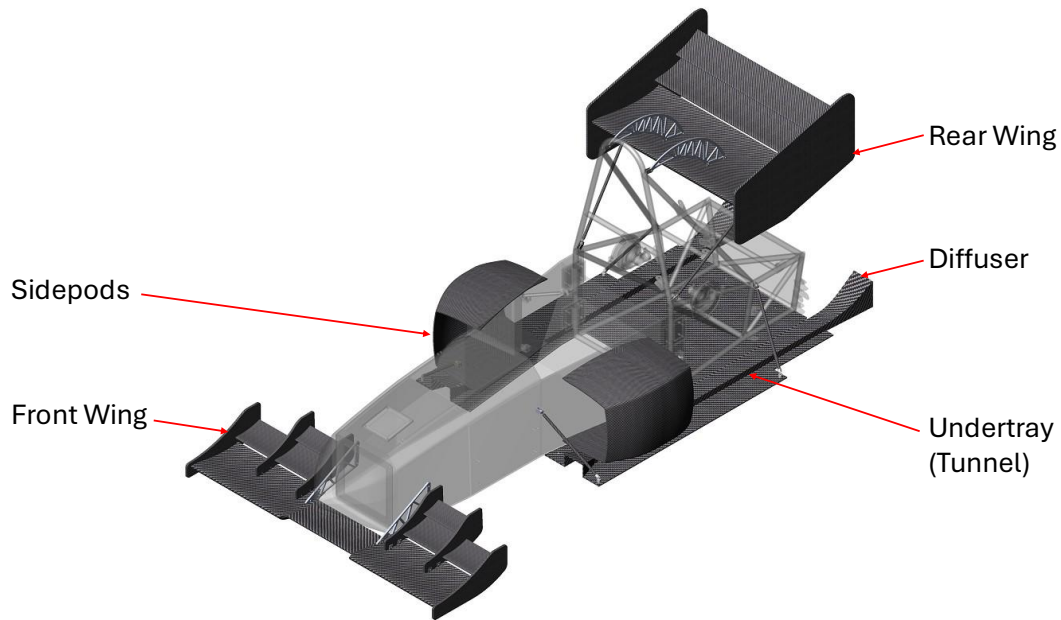


Figure 19: Design Solution - Aerodynamics Package

The front wing design philosophy considers the increased lift from ground effect with the wing proximity to the ground. This phenomenon is observed when the ground clearance of the wing element is below 50% of the chord length in which case the air flow passes through a smaller area under the wing increasing its velocity which increases downforces [7]. The downside to ground effect is the possibility of porpoising which is an oscillatory behaviour in dive and squat caused by unsteady airflow through a varying cross-sectional area.

For the rear wing, the approach was a simplistic effective design revolving around three elements and small endplates. The three wing elements are utilized to generate increased lift with minimal flow separation due to flow recovery between elements. This allows for an efficient rear wing in the low Reynolds number conditions. Also, endplates are utilized to limit flow separation and recirculation at the ends of the wing between the high pressure above and the low pressure below at the ends of the wings.

In FSAE, the modelling of front and rear wings strongly depends on the airfoils. Certain airfoil profiles are most used for their high lift to drag ratios and their performance in low Reynolds number conditions. For this reason, the wings will incorporate S1223 airfoils for their primary elements and E423 for the secondary and tertiary elements. These relative dimensions provide the assumed values for surface areas of the front and rear wing when considering the span and chords as the surface area of these components, $0.575m^2$ for the front wing and $0.616 m^2$ for the rear wing.

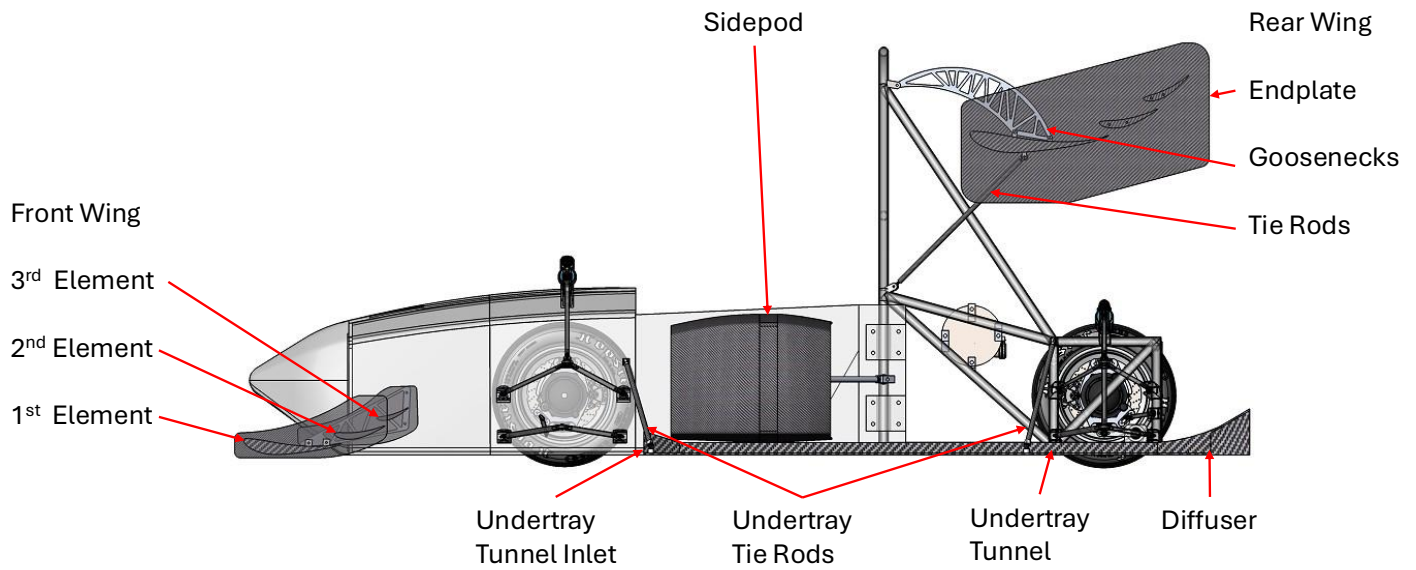


Figure 20: Aerodynamics Components Left View Annotated

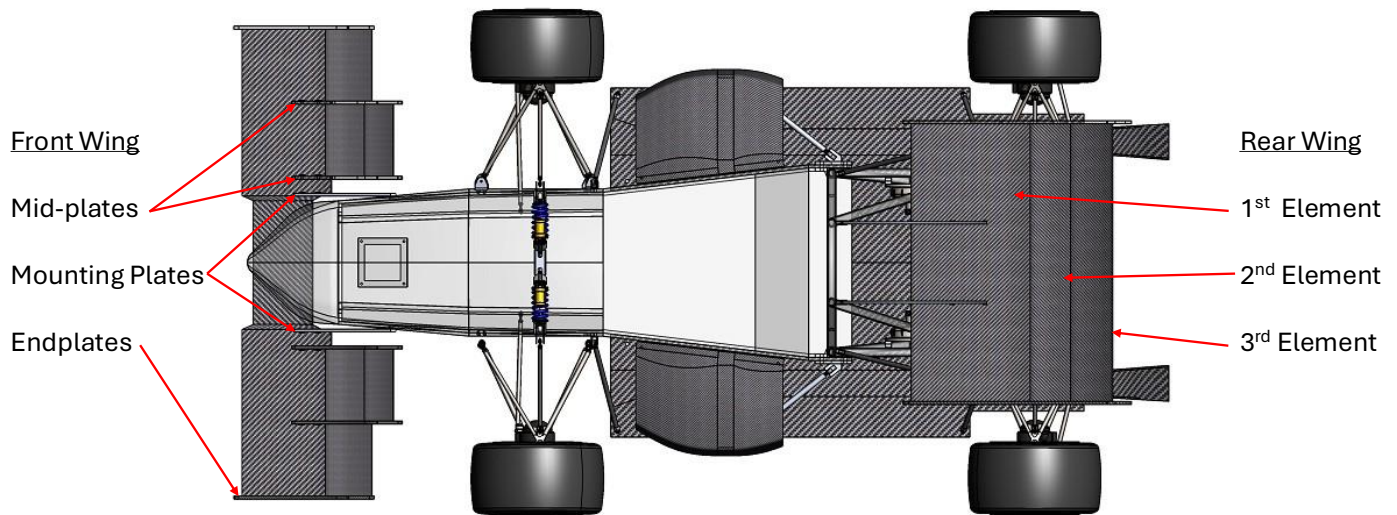


Figure 21: Aerodynamics Components Top View Annotated

The undertray uses venturi tunnels to generate downforce with negligible amounts of drag making it the most efficient component for generating downforce. The goal is to create a tunnel optimized around the flow separation which can be solved using Reynold's number. The geometry used for the tunnel is a rectangular cross-sectional area reducing in size utilizing the venturi effect to generate downforce. At the end of the tunnel, the cross-sectional area expands to allow the flow to reattach to the surrounding air smoothly without creating too much turbulence. This area is known as the diffuser and the angle is determined based on boundary layer calculations to determine the ramp angle. The overall area of the undertray tunnel in this scenario is $0.07m$.

The last aerodynamic component implemented is the sidepods with a geometry that wraps around the radiator to promote more controlled airflow for cooling. The downside of these ducts is the induced drag from the radiator that is acting like a bluff body in line with the flow. However, the forces are minimal compared to the importance of the cooling effect.

2.1.8 Overall Vehicle Mass

To ensure that the center of mass (CG) is as accurate as possible, the respective material properties were assigned to every individual component of all subsystems. Afterward, an articulated Dummy was integrated into the CAD assembly using specific anthropometric mass distribution data. The total driver mass is assumed to be 175 *lbs* (79.4 *kg*) and the weight of its body parts were distributed based on established percentages for a male [12].

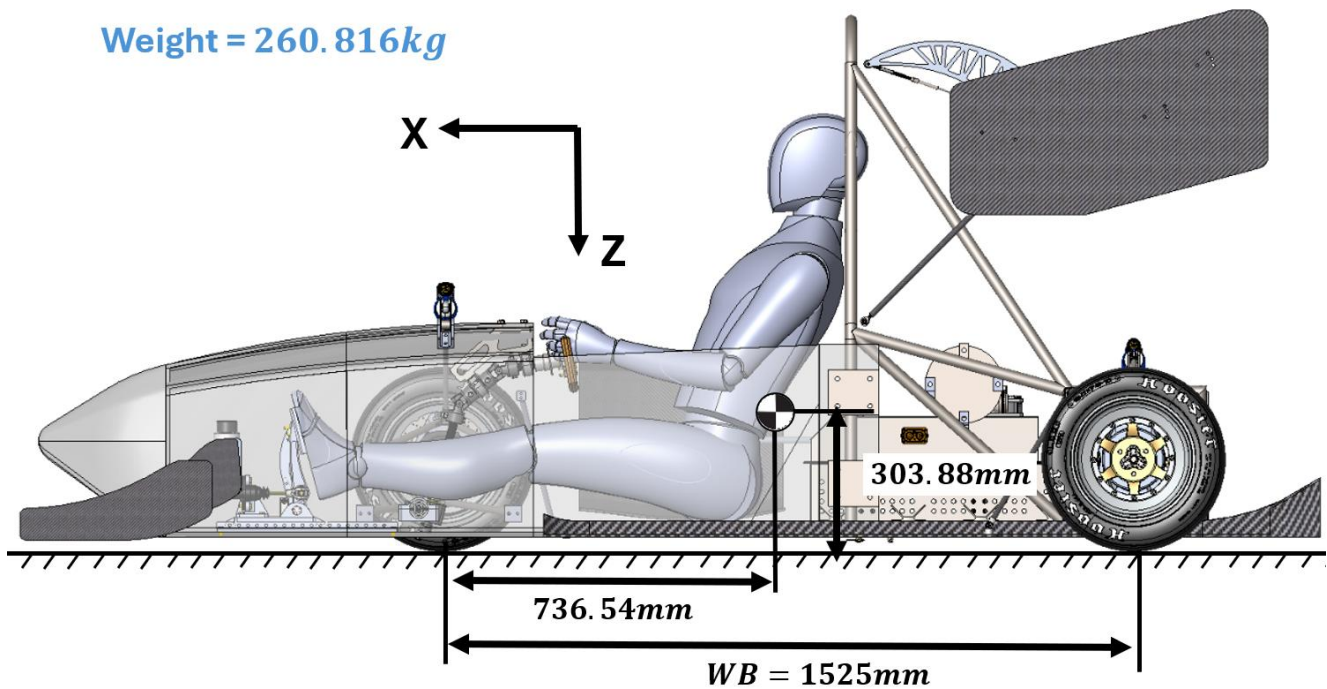


Figure 22: Mass and CG of CAD model

Minimizing vehicle mass is a crucial part of the optimization process while designing an FSAE vehicle. A reduced mass increases the acceleration performance and reduces the inertia in cornering, improving the car's handleability. It is important that the center of mass is as low to the ground as possible and centered within the car to achieve the desired weight balance. The component distribution on the chassis achieved the desired weight distribution of 45% front and 55% rear.

2.2 Discussion on interconnectivity

In the case of a formula car, the components are intentionally designed to have interconnective behaviours to increase each others performance. Typically, the systems are designed around the suspension setup to optimize tire loading and overall balance of the car. This suspension is connected to the rear sub-frame of the chassis with member triangulation at the mounting hardpoint of the A-arms, pushrods (bellcranks) and toe rods. On the other hand, the monocoque is designed around the suspension mounting points with reinforced composite material in this area. On the wheel side of the suspension, the upright design is a complex and tightly packed assembly with the in-hub motors taking up much of space in the rim. The chosen rim affects the suspension mounting and the uprights and spindle packaging. On the uprights the tie rods and toe rods affect the alignment of the tires, and the front tires are connected to the steering rack and pinion. Lastly, the aerodynamic components are major factors affecting the suspension loading relative to car velocity and cornering conditions. To properly tune the suspension for optimal performance, the aerodynamics must be designed to control the center of pressure of the car enhancing the balance and driver feel. When designed together, the suspension, the steering and the aerodynamics of the car will affect the balance in pitch, yaw, and roll for optimal load distribution maintain grip and turning radii.

Another major aspect of the performance of the car that requires multiples systems is the power distribution system. Designing the uprights around the motors and the suspension is a major structural consideration but it is also relative to the motor size which is determined relative to the power outputs permissible by the rules. The accumulator sizing is determined based on the allowable 80 kW limit set by the rules which also plays an effect on the sizing and torque graph of the motors. In turn, this setup affects the power distribution to the wheels through the planetary gear boxes to determine the best performance in acceleration and max speed. Through this power distribution, there is also a high amount of heat generation that is delt with the cooling system. The dual side radiator with independent systems is a system that is designed around the cooling requirements of the power distribution system to maintain a cool running system. This also affects the geometry of the sidepods which are designed relative to the radiator geometry for optimal cooling. The last aspect of interconnectivity of the power distribution system is the mounting of these components with the accumulator and cooling system requiring mounting points on the chassis. This is most important for the accumulator setup that requires rigorous attention to the vibration damping and the high level securing of the part to the chassis.

Lastly, the ergonomics system is a sub-system on the car that require interaction with many other systems. With the brake rotors being mounted to the spindles on the uprights, it is important to design them in cohesion with the rest of the upright components given the tightly packaged assembly in those regions of the car.

3. Optimization

3.1 Outline

The following flowcharts represent every design dependence needed to optimize the desired component. The table afterward presents the range of the optimized dimension of each component depending on the allowable inputs of their respective modeling system and GUI inputs.

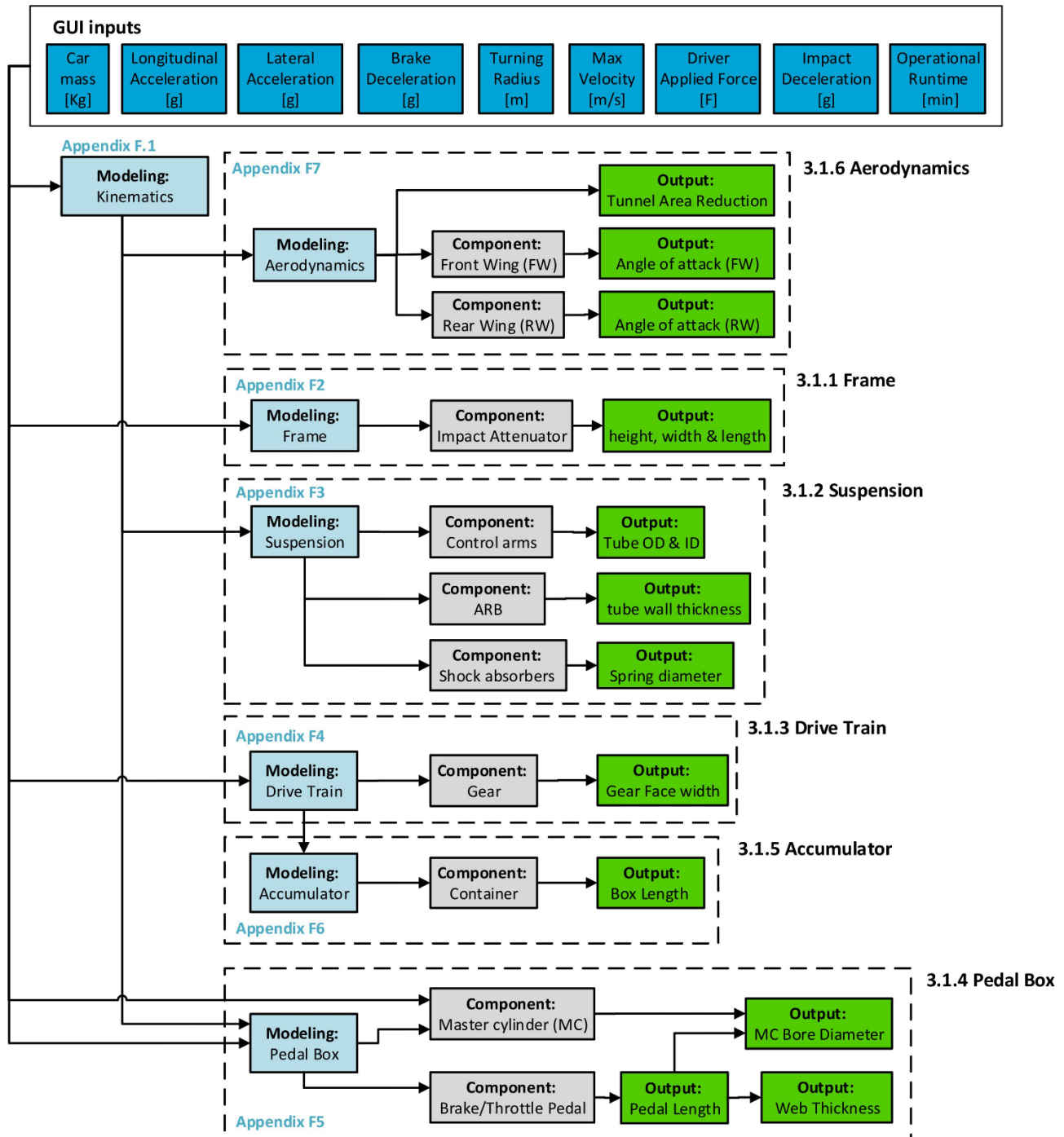


Figure 23: Solution Flowchart

3.1.1 Optimized Frame Components

Because the impact attenuator is primarily responsible for dissipating the kinetic energy of the vehicle during a frontal impact, the optimization focuses on ensuring that the required energy absorption is achieved while minimizing the added mass to the vehicle. Since the honeycomb structure already provides an efficient strength to weight ratio, the optimization is performed on the geometric parameters rather than changing the material, meaning the length and surface area. Therefore, the improvement in performance is obtained by adjusting the cross-section area and the crushing length to allow the attenuator to safely absorb the required impact energy while maintaining a compact and lightweight design that satisfies the FSAE safety requirements.

For the optimization of the impact attenuator, the required crash force is first determined using the vehicle mass and the maximum allowable deceleration to ensure that the driver is not injured. The cross-section area of the honeycomb core is then calculated iteratively in a loop such that the generated crushing force corresponds to the allowable deceleration limit within an acceptable tolerance. At the same time, the required crushing displacement is obtained from the vehicle kinetic energy at the specified impact velocity. Since the absorbed energy depends directly on the crushing distance, the total length of the attenuator is optimized to ensure sufficient energy dissipation while avoiding unnecessary material usage.

It is important to note that the maximum crash velocity range has been changed to minimize the required length. In fact, the impact attenuator only needs to withstand a crash speed of around 7 m/s to pass FSAE standards [9]. This allows a compact impact attenuator that is rule compliant and easy to fit with the aerodynamic package. A reduction factor of 5.5 was used to convert the maximum speed of the car to the maximum crash velocity. This results in a crash velocity ranging from 4.7 m/s to 7.7 m/s .

3.1.2 Optimized Suspension components

All suspension linkages, including the upper and lower control arms, pushrods, tie rods, and toe rods, were modeled as two-force members. Because of their joint configurations, these components primarily experience axial loading. The optimization script evaluated member sizing by analyzing axial stress under tension and buckling stability under compression. Using an iterative loop, the code selected the smallest possible tube outer diameter and wall thickness that maintained a sufficient factor of safety against yielding and buckling under peak load cases. This approach ensured a lightweight design without compromising structural integrity.

The functional optimization focused on the helical coil springs and the torsion anti-roll bar to satisfy vehicle dynamics requirements. The spring rates were sized based on the worst-case kinematic events, including maximum braking, cornering, and launch accelerations. For the coil springs, the model calculated

the ideal wire diameter required to achieve specific spring rates while ensuring the material remains below the shear yield limit at maximum suspension travel.

Simultaneously, the ARB dimensions were optimized using a load-based approach. The script derived the required wall thickness from peak reactive torques transmitted through the pushrod assembly during cornering. By iterating through available material dimensions and comparing calculated torsional stresses against fatigue limits, the code finalized a cross-section that provides necessary roll control while meeting manufacturing constraints. Together, these optimization codes produced a balanced suspension package that integrates structural reliability with precise kinematic performance.

3.1.3 Optimized Drive Train components

The sizing and shape of most of the components in the drivetrain system are designed for specific purposes. Given the extremely tight packaging required to fit inside of the 10-inch rims, most of the component dimensions are dictated entirely to accommodate suspension mounting points, brake rotor and caliper clearance, and motor packaging. The parts are also so small, that any weight saving optimization would drastically increase the manufacturing costs, and provide little performance gain. The main components of the drivetrain which need to be optimized are the gears within the gearbox, as changing the top speed (and therefore the gear ratio) of the car drastically changes the loads seen by the gears, and being such small but highly stressed components, have a high likelihood of failing. The modelling section of the gear optimization starts with the user input of the car's maximum velocity. From there, vehicle inputs like the tire diameter, motor speed and gear module are used in conjunction with a list of functional gear sets to determine system parameters like the diameters of the gears and the interface forces between them all.

The gear ratio selections presented to the user range from top speeds of 93 to 153 kilometres per hour, meaning the diameter of the gears, and therefore the tooth bending and contact stresses change a lot. To be able to keep everything in such a tight package, the best way to optimize the gears is to change the face width. In other words, increasing the surface area of the gear interface reduces the concentrated stress at the surface, and gives the gear teeth more strength to resist bending. With the outputs from the modelling section, and multiple predefined inputs like the target safety factors, material properties and the geometric manufacturing features, some gear factors which influence the strength of the gears are calculated. From there, the optimization enters an iterative loop, where the gear face width begins at 10mm, contact and bending stresses are found, and then are compared with the material properties and safety factors. When the safety factors are below the target values, the gear face width is increased gradually, and the loop repeats until the gears reach an optimal face width.

3.1.4 Optimized Pedal Box components

Given that the assembly is constructed from aluminum and is already inherently lightweight, optimizing the geometry of other components for mass reduction is considered negligible. Therefore, the performance gains from additional weight savings are minimal compared to the ergonomic and structural reliability of the current design. For the optimization of the pedal box, the subsystem will instead prioritize the ergonomic interface to tailor the physical geometry and force response to the driver. The length of both pedals will be optimized based on the size of the driver's foot, which is derived from the anthropometric height-to-weight ratios of the driver. Simultaneously, the required front/rear hydraulic pressures are used to optimize the necessary master cylinder bore size for the driver to accurately stop the vehicle. This will ensure the driver can effectively modulate the brake system while maintaining a comfortable and predictable input force with its foot.

3.1.5 Optimized Accumulator components

The optimization of the accumulator was not structural, but rather functional. The capacity of the accumulator was optimized in a way to keep the required motor voltage and have enough capacity for a user specified run time. The run time slider varied between 30 minutes, which is barely enough to complete the endurance event, and 60 minutes, which gives a buffer for longer races, idle powering devices, potential yellow flags or other delays in the race, etc. The optimization process takes the user defined run time and uses it with inputs like the estimated power draw, cell specifications and rules limits to first find the number of cells in series that could lead to 600V. From there, the number of cells in parallel is iteratively increased until the operational runtime of the accumulator surpasses the user input. The operational runtime calculation is done using a generous approximation of the average power draw over the course of a race. Once the cell count is determined, the container size is computed, which directly translates to the SolidWorks assembly.

3.1.6 Optimized Aerodynamic components

The optimization of aerodynamics components is utilized to analyse the overall balance of the car while maintaining no slip around the track. To find the right balance in different conditions, the calculations performed analyse the coefficient of lift of the front wing, rear wing and undertray relative to the desired conditions by the user. The first thing that is calculated is the undertray cross-sectional area which changes in area reduction relative to the airspeed at the inlet tunnel. This part is not optimized but is analysed relative to different conditions. This is however an important aspect of the optimization of the front and rear wings. These

wings have secondary and tertiary elements with varying angles of attack relative to the target downforce. Overall, this allows the car to increase performance while maintain its general balance for the driver.

3.2 Component optimization

Table 1: Optimization Input with Range of Values

Inputs	Range	Units	Description
Car Mass	230 - 280	kg	Driver mass and component weight.
Longitudinal Acceleration	0.8 – 1.4	g	Launch targets and objectives.
Lateral Acceleration	1.2 – 1.6	g	Cornering speed targets.
Braking Deceleration	1.0 – 1.6	g	Brake sensitivity and peak performance.
Turning radius	8 - 30	m	Performance for track or event layout.
Driver Applied Force	350 - 900	N	Force applied by the driver on the pedal
Impact Deceleration	22 - 25	g	Crash deceleration
Operational Runtime	30 - 50	min	Operation time during the event
Top Speed	153.6 – 93.1	km/h	Maximum velocity of the car

Table 2: Optimization Output with Range of Results

Sub System	Components Optimization	Component
Frame	Impact Attenuator length & Cross section area	$105\text{ mm} < L_{IA} < 195\text{ mm}$
		$14945\text{ mm}^2 < A_{IA} < 21145\text{ mm}^2$
Suspension	Two-Force members Outside diameter & wall thickness	$9.53\text{mm} (0.375\text{in}) < OD_{2FM} < 17.15\text{mm} (0.675\text{in})$
		$0.89\text{ mm} (0.0035\text{in}) < t_{2FM} < 2.54\text{mm} (0.100\text{in})$
	Coil Spring diameter	$7\text{mm} < D_{wire} < 8\text{mm}$
	Anti-roll bar wall thickness	$3.2\text{mm} < t_{ARB} < 4.1\text{mm}$
Drivetrain	1 st stage gear face width	$26.4\text{mm} < f_{1st\ stage} < 43.2\text{mm}$
	2 nd stage gear face width	$10\text{mm} < f_{2nd\ stage} < 22.8\text{mm}$
	Overall gearbox diameter	$130\text{mm} < d_{gearbox} < 138\text{mm}$
Pedal Box	Brake/Throttle pedal length	$170.48\text{mm} < L_{pedal} < 230.48\text{ mm}$
	Master cylinder bore	$15.875\text{mm} (0.625\text{in}) < d_{bore} < 28.575\text{mm} (1.125\text{in})$
	Brake pedal web thickness	$2\text{mm} < t_{web} < 34.9\text{mm}$
Accumulator	Container width	$437.8\text{mm} < w_{pack} < 443.5\text{mm}$
	Container length	$400.6\text{mm} < l_{pack} < 623.8\text{mm}$
	Cell Count	$966 < N_{cell} < 1518$
Aerodynamics	Front Wing angle of attack	$0^\circ < \theta_{FW} < 42.5^\circ$
	Rear Wing angle of attack	$0.5^\circ < \theta_{RW} < 65.5^\circ$

3.3 Discussion on Results

3.3.1 Impact Attenuator Results

The optimization results indicate that the impact attenuator length varies between 105 mm and 195 mm, while the cross-section area ranges between 14945 mm² and 21145 mm². These results show that the required crushing distance remains relatively small while still ensuring that the impact attenuator can dissipate the required kinetic energy. As the required energy absorption increases, the optimization routine increases the attenuator length in order to provide sufficient deformation distance while maintaining a lightweight structure.

For the cross-section area, the obtained range of 14945 mm² to 21145 mm² shows the variation required to generate the appropriate crushing force for different loading cases. A larger area increases the crushing force capacity of the honeycomb, while a smaller area reduces mass but also reduces the energy absorption capability. Therefore, the final optimization reflects a balance between satisfying the crash requirements and limiting unnecessary mass.

Overall, the results show that the impact attenuator design is governed by the relationship between the impact energy and the crushing force generated by the honeycomb structure. As the impact speed or the allowable deceleration increases, the optimization increases both the attenuator length and the cross-section area to ensure that the required amount of energy is absorbed in a stable and progressive crushing mode.

The following pictures illustrate the change of the honeycomb after modifying the inputs in MATLAB. The first result is typical for a low impact acceleration and low speed crash, and the second one represents the case of a higher impact acceleration with a higher impact speed.

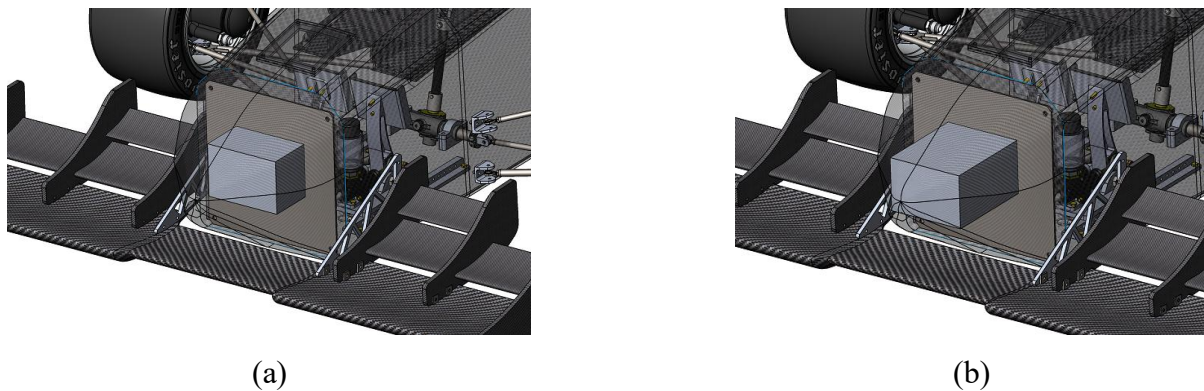


Figure 24: Change of Dimension of the Impact Attenuator between (a) Minimum Length and (b) Maximum Length

3.3.2 Suspension Results

Analysis of the two-force members confirmed that 4130 chromoly steel is well-suited for the suspension load cases due to its high yield strength. Axial stress calculations were performed for the worst-case tensile loading, identified at the front lower control arm under maximum braking, and compressive loading, utilizing

Euler buckling analysis with pinned-pinned boundary conditions. The most critical compressive case was found in the rear lower control arm. To address these load cases, the optimization script determined that the required dimensions for the suspension members range from 0.375 in (9.53 mm) to 0.675 in (17.15 mm) for the outer diameter, with a corresponding wall thickness between 0.0035 in (0.09 mm) and 0.100 in (2.54 mm).

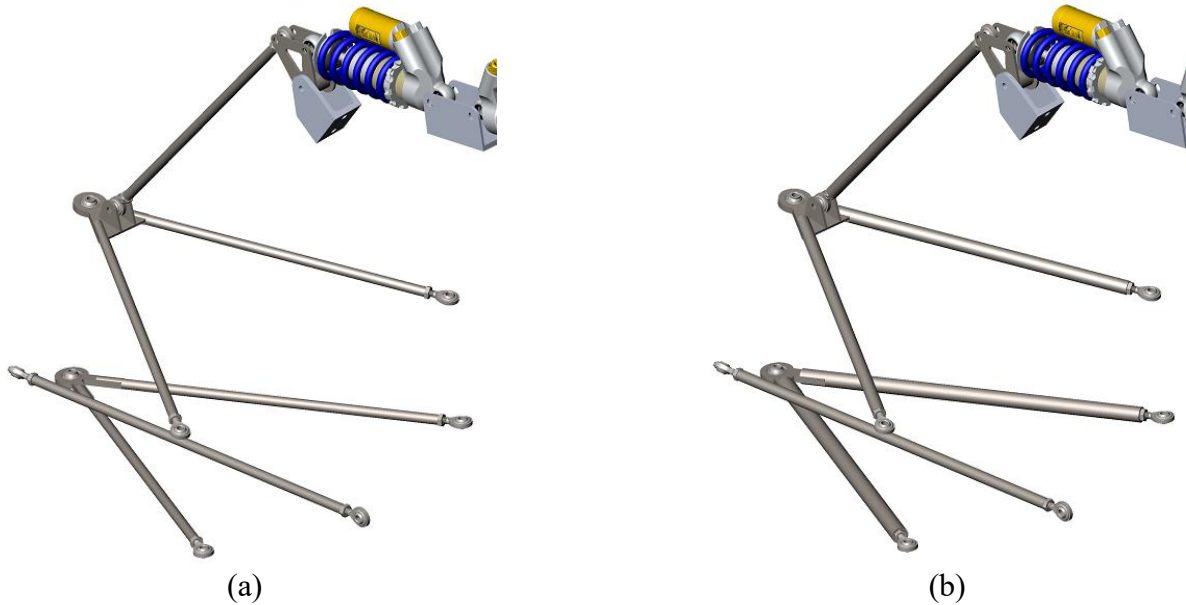


Figure 25: Change of Dimension of the Suspension Members between (a) Minimum Load and (b) Maximum Loading

The design of the coil springs successfully balanced mechanical stiffness with material limits. By solving the helical spring rate equation, the design ensured necessary resistance to suspension travel. While the allowable optimization range was set between 5 mm and 9 mm, using the GUI inputs for the worst kinematic events, the calculated wire diameter resulted a final wire diameter range between 7 mm and 8 mm. Structural integrity checks verified that the maximum operating shear stress remains within safe limits, providing a safety factor above 1.5 during peak dynamic events.



Figure 26: Change of Dimension of the Suspension Springs between (a) 5mm and (b) 9mm

The anti-roll bar was finalized using a load-based approach derived from the peak reactive torque transmitted through the pushrod assembly. This method produced a robust structural result, centering on an ARB with a 19 mm outer diameter and a 3.9 mm wall thickness, which maintains a safety factor of 1.5 against

yielding. Across the full range of GUI input variables, the required ARB wall thickness varied between 3.2 mm and 4.1 mm. This range ensures the suspension package provides the necessary roll control and manufacturability for the competition vehicle.



Figure 27: Change of Dimension of the Suspension Members between (a) 1.5mm and (b) 5mm

3.3.3 Drive Train Results

The gears in the planetary gearbox are the most stressed part of the drivetrain, and so it is important that they are optimized properly. To get a wide range of vehicle top speeds for different tracks and events, the gear were optimized for speeds between 93 and 153 km/h. While realistically the car wouldn't reach 153km/h due to aerodynamic drag and power limitations, this speed is meant to represent a low gear ratio, and wheel speed under slippage can still reach such velocities. To get the best results, the first and second stage of the gearbox were optimized separately. The first stage, which is typically under greater stress than the second stage, had results varying between 26.4 and 43.2 millimetres, with the smallest dimension coming from the fastest case (least gear reduction). For the second stage, the gear face width varied from the minimum of 10mm to 22.8mm. The change in gear sets also brought along a slight change in the overall diameter of the case, which only varied between 130 and 138 millimetres. The safety factors considered for this optimization were 1.3 for gear tooth bending and 1.2 for surface contact stress. However, this optimization code required a few simplifications, as the number of factors and modifiers in gear calculations would overcomplicate the code. These simplifications to the analysis equations will only give worse load cases though, ensuring that the gears are designed with an even higher safety factor than considered directly in the code. This optimization ensures the integrity of the gears, but some components like the planet carriers and the upright itself may need some software validation such as FEA to make sure they are strong enough.



Figure 28: Change in gear sizes between (a) the fastest (153.7 km/h) and (b) the slowest (93.1 km/h) configurations.

3.3.4 Pedal Box Results

With the higher range of the GUI inputs, the fixed “motorsport athlete” BMI assumption produced unrealistic anthropometric outputs and produced an abnormal 2.5 *m* stature. To maintain realistic cockpit dimensions, the MATLAB code caps the driver stature at 1.9 *m* (95th percentile male). Using Topinard’s formula, this cap gives a pedal length of 230.48 *mm*, for total vehicle mass of 255 *kg*. It’s important to note that the excessive total car masses between 255 *kg* and 280 *kg* are still utilized for the structural analysis of other subsystems. The lower range of the GUI inputs for a driver mass of 50 *kg* (5th percentile female) gives a pedal length of 170.48 *mm*. The largest pedal length (230.48 *mm*) with the strongest assumed possible driver-applied force (900 *N*) during low deceleration (1.2 *G*) reaches the maximum Tilton bore size of 1.125 *in*. The opposite range gives range reaches the minimum bore size of 0.625 *in*. Every GUI input was able to cover all the commercially available diameter provided by Tilton. For any GUI input value, the thickness consistently defaulted to the 2 *mm* minimum threshold. This indicates that even at the maximum pedal length, this minimal thickness provides sufficient structural contribution to satisfy safety requirements without necessitating further reinforcement.

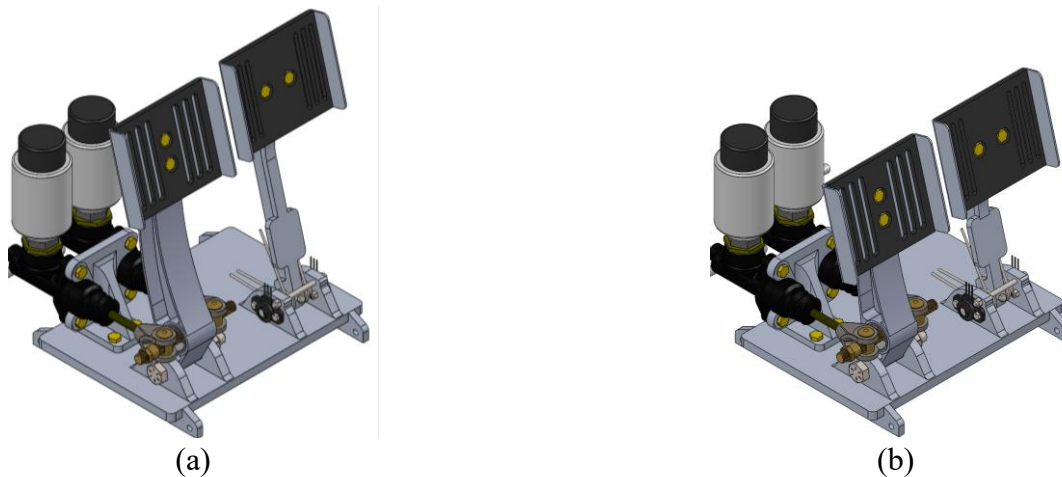


Figure 29: Change of the brake/throttle pedal length between (a) 170.45mm and (b) 230.45mm

3.3.5 Accumulator Results

The results of the optimization gave cell counts varying between 966 and 1518 cells. This number is higher than the number calculated in the analysis, and this is because of the major simplifications needed to be able to implement the design to a MATLAB code. The initial design took lots of trial-and-error calculations, and different case studies, something that is not possible to automate with a simple code. This higher value, though, will net a battery with a decent safety margin on the energy requirements. One effect of the higher cell count is that the container became too large for a single layer. This meant that the accumulator is now more of a box than a tray, with two stacked layers of cells and modules. The case dimensions varied between 400.6 mm and 632.8 mm in length and the width stayed constant at 437.2 mm, which still fits comfortably in the rear subframe.

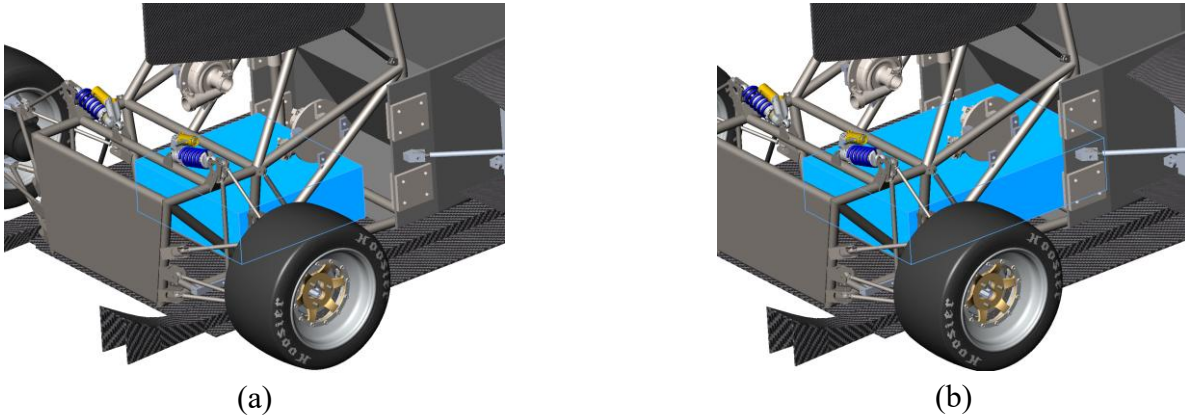


Figure 30: Change in accumulator (in blue) size between (a) the shortest and (b) the longest run times.

3.3.6 Aerodynamic Results

The optimization of the aerodynamics revolves around the variation of the undertray tunnel to calculate the coefficient of lift required out of the front and rear wing. The undertray tunnel inlet size is what changes the quantity of downforce generated through the tunnel. This area reduction ratio varies from x1 to x2.25 of the through section generating downforce.



Figure 31: Undertray Inlet Tunnel Area Ratio at (a) Lowest Downforce (x1.006) and (b) Highest Downforce (x2.25)

With the calculated amount of downforce generated from this tunnel, the front and rear wing coefficients of lift are calculated. The front wing has a coefficient of lift that varies from -1.1 to -3.7 depending on the quantity of downforce required. Similarly, the rear wing varies from -0.6 to -4.0 causing the aerodynamic system to work in cohesion for stability and downforce. The lower absolute values are utilized for event like acceleration where there is no downforce requirement since the car is not slip limited, it is power limited. The higher absolute values are used for events like skid pad or autocross where downforce is required to increase lateral forces in cornering depending on the target G-forces.

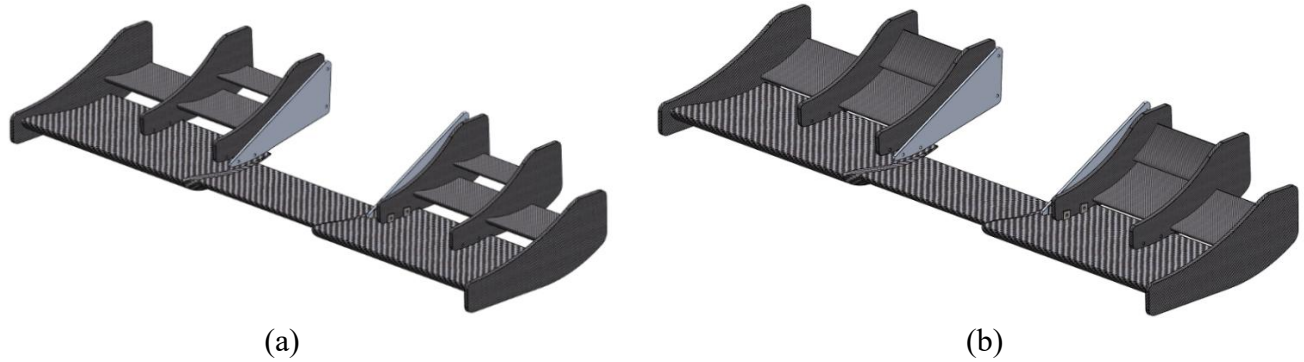


Figure 32: Front Wing Angle of Attack at (a) Lowest Downforce, 0°, and (b) Highest Downforce, 42.5°

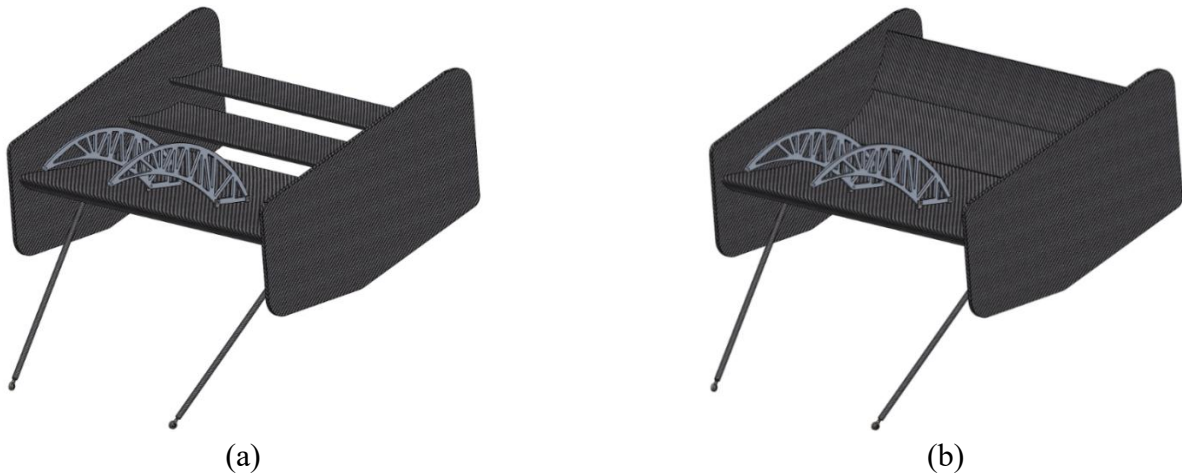


Figure 33: Rear Wing Angle of Attack at (a) Lowest Downforce, 0.5°, and (b) Highest Downforce, 65.5°

4. Ethics case study

4.1 Impact Attenuator Case Study

The component selected for this case study was the car's impact attenuator. When tasked with generating an improved attenuator design, the AI produced an ambitious proposal that is worth examining both for its ingenuity and its risks. This report summarizes the AI-generated concept, identifies its critical shortcomings, and proposes a practical, ethics-grounded remediation plan.

The AI proposed a functionally graded auxetic metamaterial hybrid attenuator. Rather than using traditional aluminum honeycomb, it called for a 3D-printed, variable-density Scalmalloy [10] lattice core that collapses inward to resist buckling, all encased in a carbon fiber-reinforced PEEK shell. The logic is very compelling, pairing the plastic deformation of the auxetic core with the controlled fragmentation of the composite outer layer is meant to maximize specific energy absorption and produce a smooth, stable deceleration pulse. The full AI response is included in Appendix E.

However, blindly trusting an AI-generated design for a safety-critical system would be a serious mistake. The auxetic hybrid carries some major risks. First, accurately simulating its crash behavior demands non-linear dynamic FEA software like LS-DYNA, a level of modeling complexity well beyond conventional structural analysis. Second, unlike conventional honeycomb structures, auxetic metamaterials have no long track record of predictable, linear performance, making their real-world crash behavior genuinely uncertain. Third, manufacturing defects such as micro-porosity in the 3D-printed metal core or poor interlaminar bonding in the CF-PEEK shell could trigger sudden brittle failure instead of the gradual, controlled crush the design depends on [11]. Finally, the design is especially vulnerable to off-axis glancing impacts, which could shear the outer shell and completely bypass the core energy-absorbing mechanism.

To move forward safely, the design needs simpler materials and empirical validation. On the materials side, swapping the metal lattice core for 3D-printed TPU or polycarbonate removes the brittleness risk, and replacing automated CF-PEEK winding with a standard wet layup vacuum bag process brings costs down to roughly \$220 per unit while eliminating a key failure mode. For validation, rather than relying solely on simulation, a physical drop-tower rig should be built and a number of prototypes tested using a crash-rated accelerometer and a MATLAB analysis script. This brings an estimated cost of \$1,380 to physically confirm the structure meets the required 20g average and 40g peak deceleration targets.

From a utilitarian standpoint, shifting to predictable materials and physical testing drastically lowers the probability of catastrophic driver injury, which is a benefit that far outweighs any gain in weight savings or design novelty. From a virtue ethics perspective, the plan reflects engineering at its best, which means being

prudent, honest, and clear-eyed about what the team can and cannot reliably build. The PEO Code of Ethics [12] reinforces this directly, requiring that a practitioner's first duty be to public welfare and that engineers only take on work they are genuinely competent to perform. Replacing an unverifiable, simulation-dependent concept with a locally fabricated, physically validated structure is the fulfillment of that professional obligation.

The AI-generated design is a creative starting point but is far from a finished solution. By choosing simpler materials and fabrication, and using empirical testing, we can build an attenuator that is competitive, trustworthy, and puts safety first.

4.2 Cost and Budget Analysis

For the first step of the remediation plan, the cost of the new impact attenuators is itemized below.

4.2.1 Itemized Cost Estimate for the Impact Attenuator (per unit):

- TPU/PC Filament (Core): ~\$81 (Assuming 1.5 spools of high-quality engineering filament) [13]
- Carbon Fiber Twill Fabric: ~\$110 [14]
- Structural Epoxy Resin & Hardener: ~\$65 [15]
- Consumables (Vacuum bagging, peel ply, breather): ~\$55 [16]
- **Estimated Total:** \$311 per attenuator

For the second step of the plan, multiple attenuators need to be manufactured, as well as the actual physical test rig. This is a very rough estimate, a more detailed cost for this would require a comprehensive design of the apparatus.

4.2.2 Itemized Cost Estimate for the Testing Setup:

- Raw Materials for 4 Test Prototypes: ~\$1244 (Based on Remediation 1 costs)
- Crash-Rated Accelerometer (+/- 100g minimum): ~\$150 [17]
- Drop Rig Materials (Assuming largely scrap/in-house steel tubing): ~\$500 [18]
- **Estimated Total:** \$1,894

Though this price may seem excessive and difficult to justify for a component which does not improve the performance of the car in a meaningful way, it is absolutely a crucial expenditure. Driver safety cannot be compromised in any way when designing a high-performance machine capable of reaching upwards of 120 kilometres an hour.

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Appendix A. Datasheet

Ohlins Dampers for suspension



Ohlins TTX25 MKII FSAE Damper set

Part #: 23.TTX25KIT

Brand: Ohlins

[Reviews and Success Stories:](#) Be the first to review this product

Set of 4 low friction twin tube FSAE dampers

[Show All vehicles this product fits](#)

\$2,860.00

Starting at \$100/mo or 0% APR. [See if you can pay with affirm.](#)

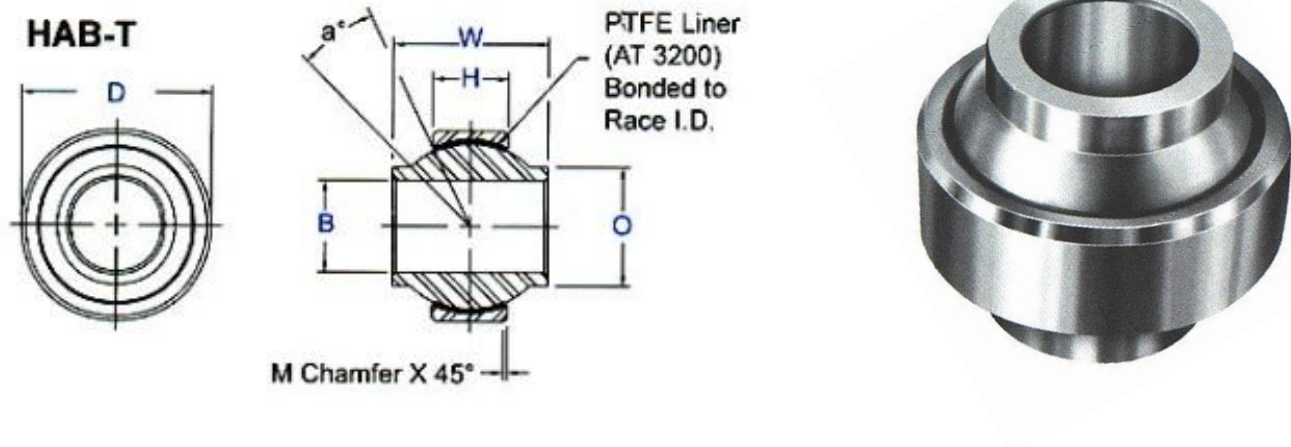
6 In Stock: Ready to Ship | [Add to Wishlist](#)

The Öhlins TTX 25 FSAE, the Formula SAE version of the Cane Creek Double Barrel damper has been very successful among the collegiate design teams throughout the world. So to continue development of the product, we have updated the damper for the 2011 season. The TTX25-MKII FSAE will remain a twin tube damper with four completely independent external adjusters and solid main piston, only now the new damper has lower friction and allows for an even softer valving option and finer adjustment. The damper graphs have been updated below and spare parts, revalve kits, rebuild kits and rebuild tools are still available.

- 200mm free length
- 57mm stroke
- Weight approx 394g each (without spring)
- 8mm ID bearings
- 4way adjustable with independent high and low speed adjustments for compression and rebound
- Solid main piston
- Small diameter, twin tube design
- Nitrogen gas charge in external reservoir
- Adjustable spring preload
- Spring perch locking collar
- Full range of spring rates available ([Click here](#) for Hyperco options)
- Part Number: 23.TTX25MKII.FSAE

Figure 34: Ohlins TTX25 MKII Damper [4]

HAB4TG, Aurora Bearing

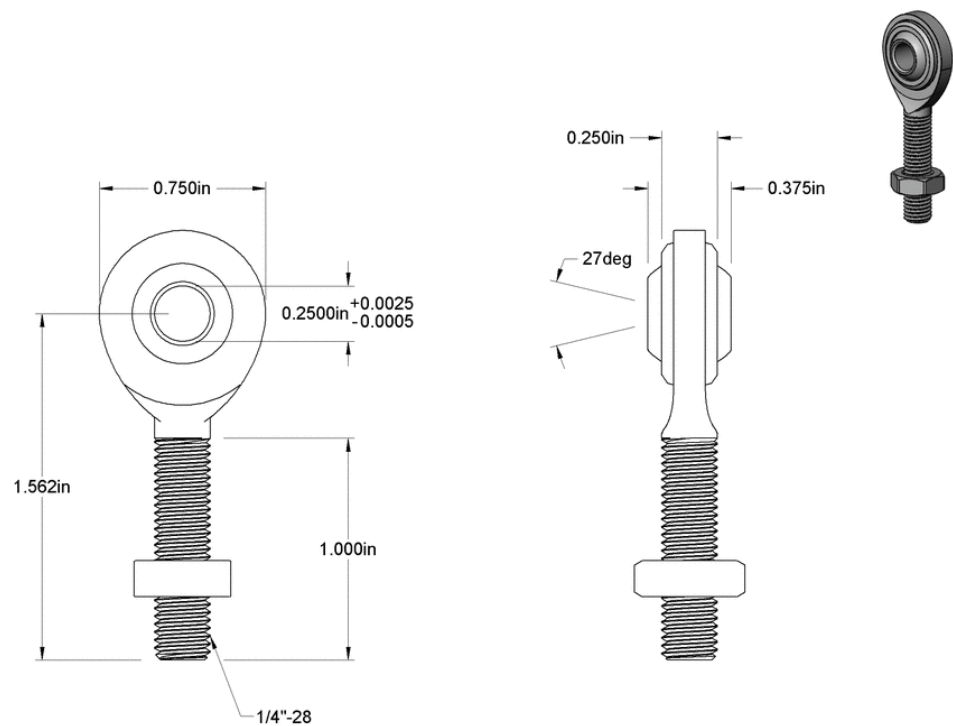


- Ball - Alloy steel - heat treated - hard chrome plated.
- Race - 17-4 PH stainless steel - heat treated.
- PTFE liner - (AT 3200) Permanently bonded to race inside diameter. Qualified to SAE-AS81820 Type A.
- Staking Groove is available. When a staking groove is required, specify by adding suffix "G". Ex.: HAB-4TG.

Specifications	
Part Number	HAB-4TG
Bearing Type	Grooved
B - Ball Bore	.2500 in
D - Outside Diameter	.7400 in
W - Ball Width	.593 in
H - Race Width	.255 in
M - Chamfer	.020 in
O - Ball Shoulder Diameter	.390 in
P - Groove Depth	.025 in
G - Groove Pitch Diameter	.674 in
R - Grooved Radius	.010 in
Ball Diameter	.593 in
a - Misalignment Angle	24°
Radial Static Limit Load Capacity	7,560 lbs
Approximate Brg. Weight	.04 lb

Figure 35: HAB4TG Specification sheet [19]

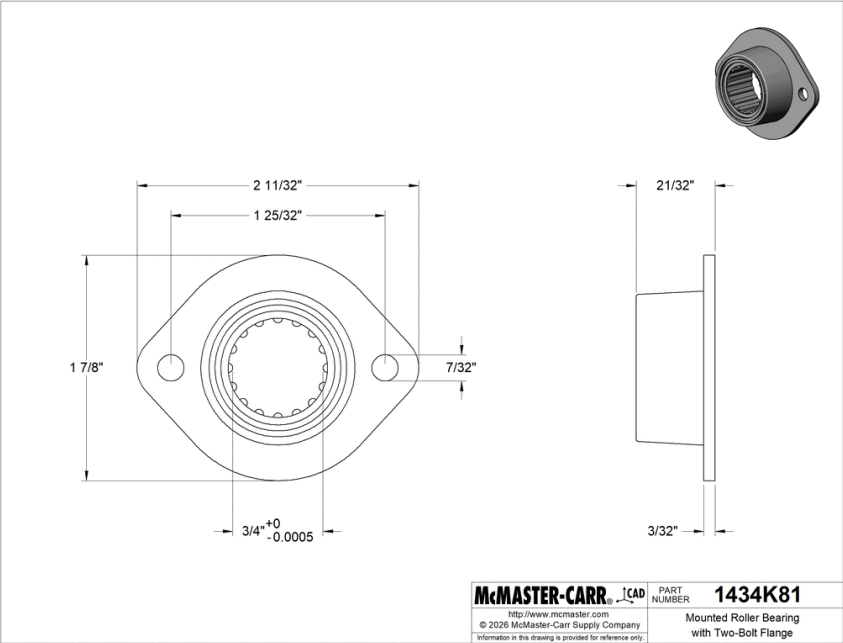
Rod-End Bearing with Nut



Shank Thread Size	1/4"-28
ID	1/4"
ID Tolerance	-0.0005" to 0.0025"
Maximum Ball Swivel	27°
Overall Width	3/4"
Overall Thickness	3/8"
Shank Center Length	1 9/16"
Shank Thread Length	1"
Static Radial Load Capacity	2,800 lbs.
Material	Zinc-Plated Carbon Steel
Ball Material	Chrome-Plated Bearing Steel
Lubrication	Required
Relubrication	Required
Rod End Type	Ball Joint
Shank Threading	Fully Threaded
Shank Gender	Male
Includes	Nut
RoHS	RoHS 3 (2015/863/EU) Compliant
REACH	REACH (EC 1907/2006) (01/17/2023, 233 SVHC) Compliant
DFARS	Specialty Metals COTS-Exempt
Country of Origin	Peoples Republic of China
Schedule B	848330.8070
ECCN	EAR99

Figure 36: Rod End Bearing Specification Sheet [20]

Mounted Needle-Roller Bearing with Two-Bolt Flange for Anti-Roll Bar



Bearing Type	Roller
Roller Bearing Type	Needle
Housing Material	Steel
Shaft Mount Type	Press Fit
For Shaft Diameter	3/4"
Overall	
Height	1 7/8"
Length	2 11/32"
Width	2 1/32"
Flange Thickness	3/32"
Dynamic Radial Load Capacity	2,200 lb.
Maximum Rotation Speed	14,700 rpm
Bearing Material	Steel
Bearing Seal Type	Open
Alignment Style	Fixed
For Load Direction	Radial
For Shaft Surface Smoothness, Ra	16 µin
For Shaft Type	Round
Grease Fitting Included	No
ID	0.75"
ID Tolerance	-0.0005" to 0"
Lubricant	Grease with Lithium Thickener
Lubrication	Lubricated
Mounting Hole	
Hole Diameter	7/32"
Hole Center-to-Center	1 25/32"
Fasteners Included	No
Static Radial Load Capacity	2,800 lb.
System of Measurement	Inch
Temperature	-20° F to 210° F
Weight	0.2 lb.
Country of Origin	United States
DFARS Compliance	Specialty Metals COTS-Exempt
Export Control Classification Number (ECCN)	EAR99
REACH Compliance	REACH (EC 1907/2006) (06/25/2025, 250 SVHC) Compliant
RoHS Compliance	RoHS 3 (2015/863/EU) Compliant
Schedule B Number	848320.0050
U.S.-Mexico-Canada Agreement (USMCA) Qualifying	Yes

Figure 37: Mounted Needle-Roller Bearing Specification [21]

Rack and Pinion from KAZ Technology

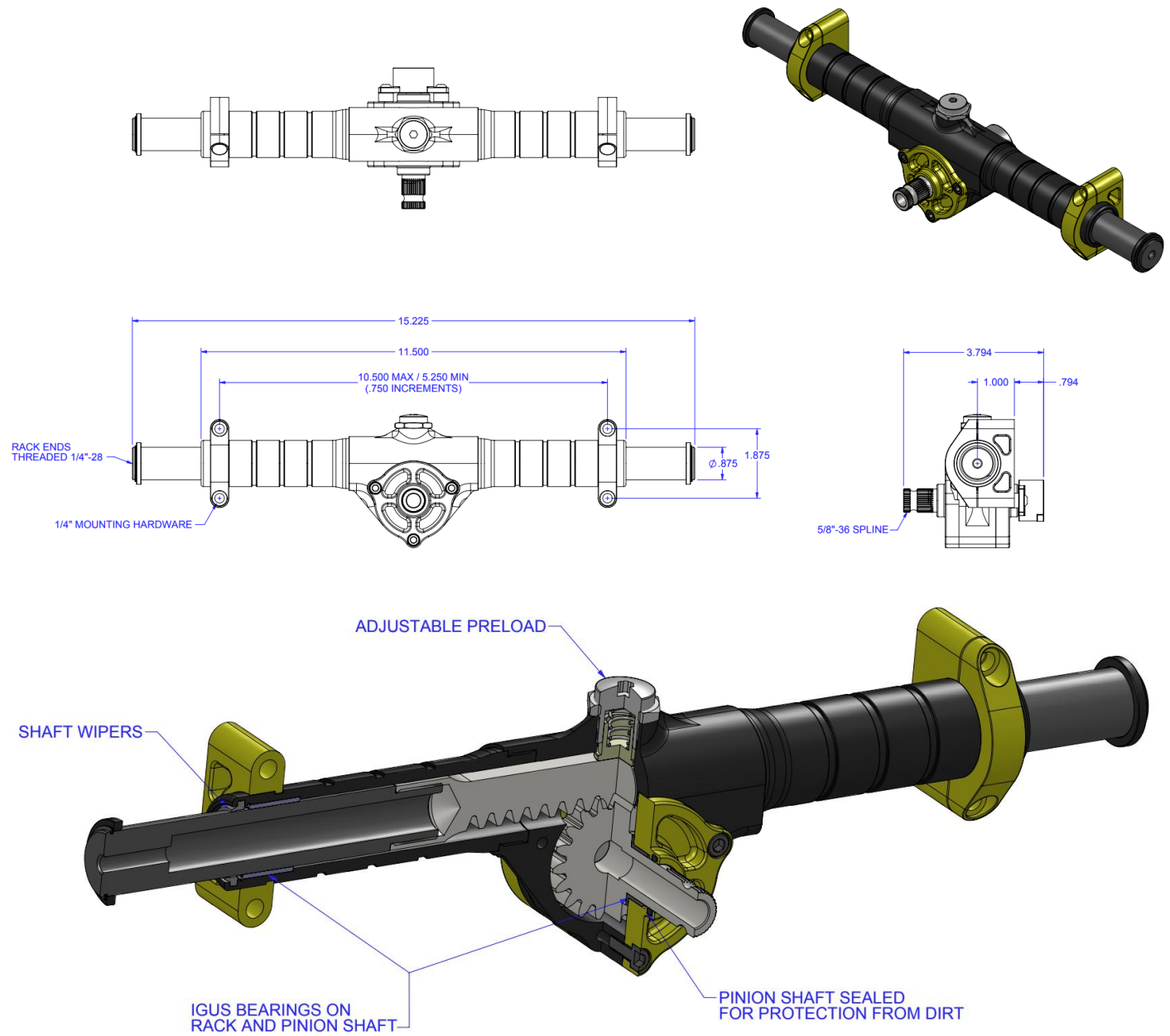


Figure 38: KAZ Technologies Steering Rack Assembly [22]

Steering Column U-joints from Mc master Carr



Pin-and-Block Joint—Pin-and-block U-joints work well in most applications since they handle higher torque at lower speeds than joints with ball bearings.

Acetal—Acetal U-joints weigh less than metal U-joints. They protect components in your system from stray electrical currents by blocking the flow of electricity between your shafts.

Spring-Pin Shaft Mount—U-joints that connect with spring pins require you to drill holes through both the U-joint and your shafts. [Spring pins](#) are not included.

Lubrication Not Required—U-joints that don’t require lubrication are often used with optical devices and other sensitive equipment because they're less likely to attract dust and debris compared to lubricated U-joints. They’re also easier to maintain.

Figure 39: Precision Single U-Joint from Mc-Master Carr [23]

Carbon Fiber Reinforced Polymer Tubing from Rock West Composites

- Tube - Filament Wound - Sanded - 0.500 x 0.750 x 12 Inch
- SKU : 35142-S-12



Avg EX Young's Modulus	1.11e+7 psi
Avg EY Young's Modulus	5.54e+6 psi
GXY Shear Modulus	3.22e+6 psi
X Tensile Yield Strength	6.66e+4 psi
X Compressive Yield Strength	7.61e+4 psi
Y Tensile Yield Strength	3.41e+4 psi
Y Compressive Yield Strength	4.23e+4 psi
XY Shear Strength	3.10e+4 psi
CTEX	-5.20e-8 in/in-°F
CTEY	1.89e-6 in/in-°F
Poisson's Ratio νXY	0.382
Poisson's Ratio νYX	0.190

Figure 40: Carbon Fiber Tubing from Rock West Composites [24]

Adhesive Epoxy from 3M Canada



Figure 41: Adhesive Epoxy from 3M Canada [25]

Clevis From KAZ Technology

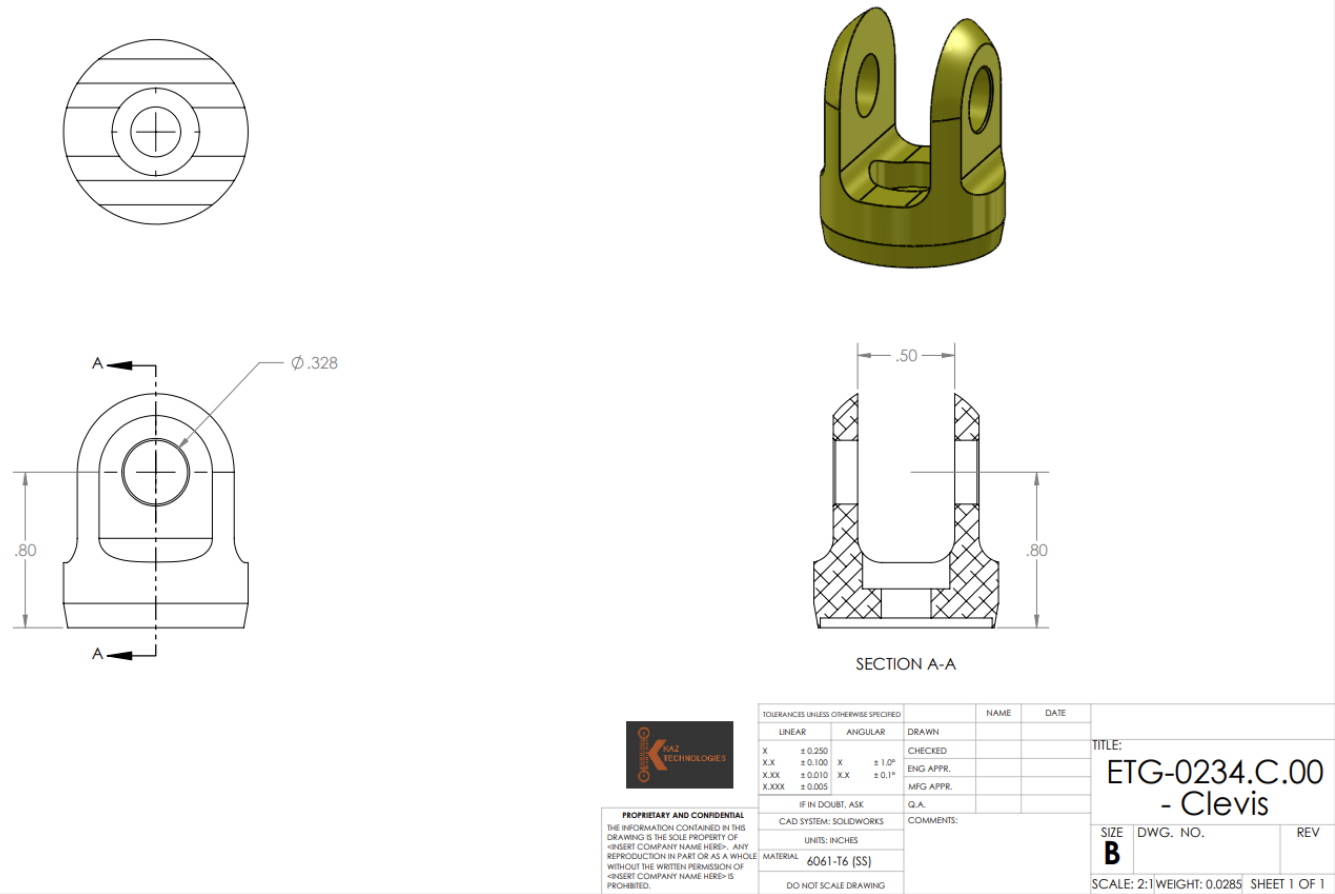
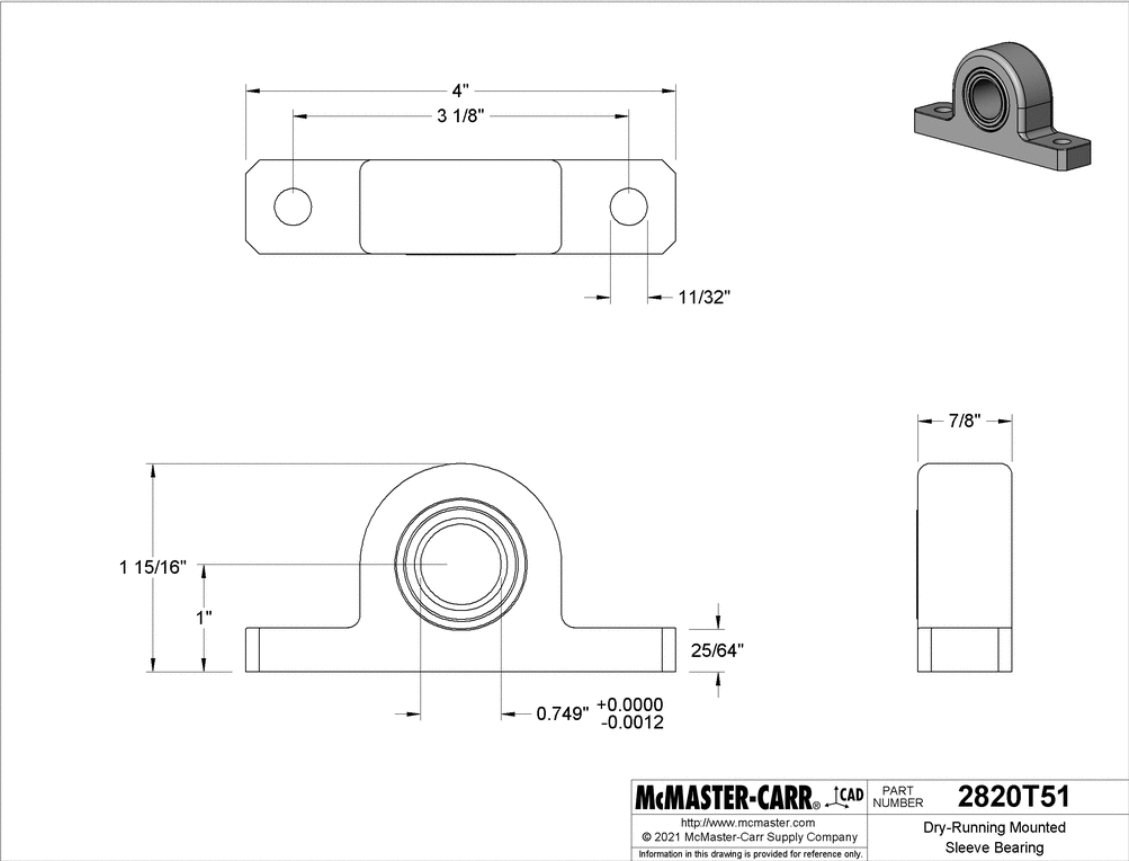


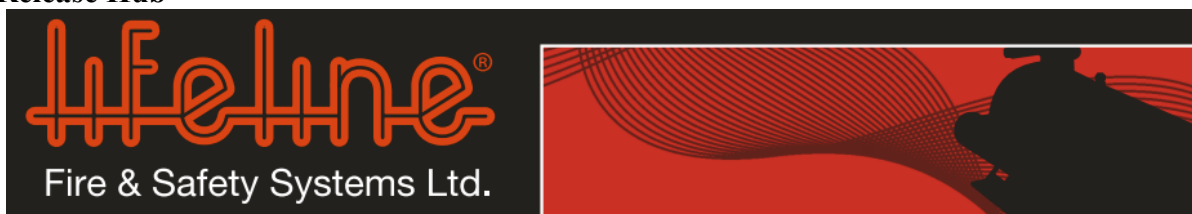
Figure 42: clevis from KAZ Technologies [22]

Mounted Sleeve Bearing



Mounted Bearing Type	Base Mount
Bearing Type	Plain
Plain Bearing Type	Sleeve
Housing Material	Aluminum
For Shaft Diameter	3/4"
Center Height	1"
Overall	
Height	1 15/16"
Length	4"
Width	7/8"
Dynamic Radial Load Capacity @ Speed	2,750 lb. @ 60 rpm
Misalignment Capability	5°
Alignment Style	Self-Aligning
Backing Material	Bronze
Bearing Material	PTFE
Bearing Width	7/8"
For Load Direction	Radial
For Shaft Type	Round
Lubrication	Not Required
Mounting Hole	
Hole Diameter	11/32"
Hole Center-to-Center	3 1/8"
Number of Holes	2
Fasteners Included	No
P Max	8,000 psi

Figure 43: Mounted Sleeve Bearing from Mc-Master Carr [26]



Technical Bulletin 027 – Quick Release Steering Bosses

Rev1 22/12/2015

The Lifeline **Quick Release Steering Boss** is CNC manufactured using aircraft quality aluminium, hard anodised for strength and durability. Both male and female splines are precision gear cut incorporating a master spline for accurate alignment. The Lifeline **Quick Release Steering Boss** is available in four different formats:



Formula Car – 3 x 50.8 PCD

Touring Car – 6 x 70 PCD steering wheel fitment e.g. Sparco

Group N/Tuning – allows you to use the quick release mechanism in conjunction with any aftermarket steering wheel boss that has a 6 x 70 PCD steering wheel fitment e.g. Sparco

Classic 9 Hole – 9 x 101.6 PCD

Lifeline steering quick release bosses can be supplied with a prewired 8 way electrical connector to allow the use of steering wheel mounted instruments or switch gear, and in 2,3,4,5,6,7,10, 14, 16 & 22 electrical connector arrangements (non-prewired) to suit individual and constructor's needs, details upon request. All connectors are manufactured to IP50 specification.

Lifeline can manufacture bespoke steering wheel quick releases to suit the requirements of any constructor including different diameter splines, shafts and whole steering column assemblies.

Diameter of the standard weld-in male spigot is 20mm, other customer sizes also available. Lifeline Quick Release Bosses are anodised gold to meet FIA requirements except for the Tuning option in black for road car use only.

The male spline is manufactured from EN3B steel as standard and can be optionally manufactured from SAE 4130 or titanium on request.

Part No.	Description	Weight	Overall Dims
410-100-001	Formula Car Quick Release	0.260kg	80 x Ø63mm
410-100-002	Touring/Rally Car Quick release	0.290kg	80 x Ø81mm
410-100-003	Group N Quick Release	0.410kg	66 x Ø81mm
410-100-004	Group N Tuning – Black (Tuning Market Only)	0.410kg	66 x Ø81mm
410-100-005	Classic 9 Hole – Fit Moto Lita	0.830kg	69 x Ø115mm
410-100-007	Classic 9 Hole – Fit Moto Lita – Weld on Spline	0.374kg	87 x Ø115mm
410-200-001	Formula Car – Electric – 8 Pin – Prewired	0.286kg	83 x Ø63mm
410-200-002	Touring Car – Electric – 8 Pin – Prewired	0.308kg	83 x Ø81mm



Lifeline operates a policy of continual improvement and reserves the right to amend this Technical Bulletin without notice. For latest advice contact Lifeline Technical Department on +44 (0)24 7671 2999

Figure 44: quick-release-steering-bosses from Fire and Safety System Ltd [27]

Shaft Coupler from Borgeson



Steel Steering Coupler, 5/8-36 X 3/4, 2" Length

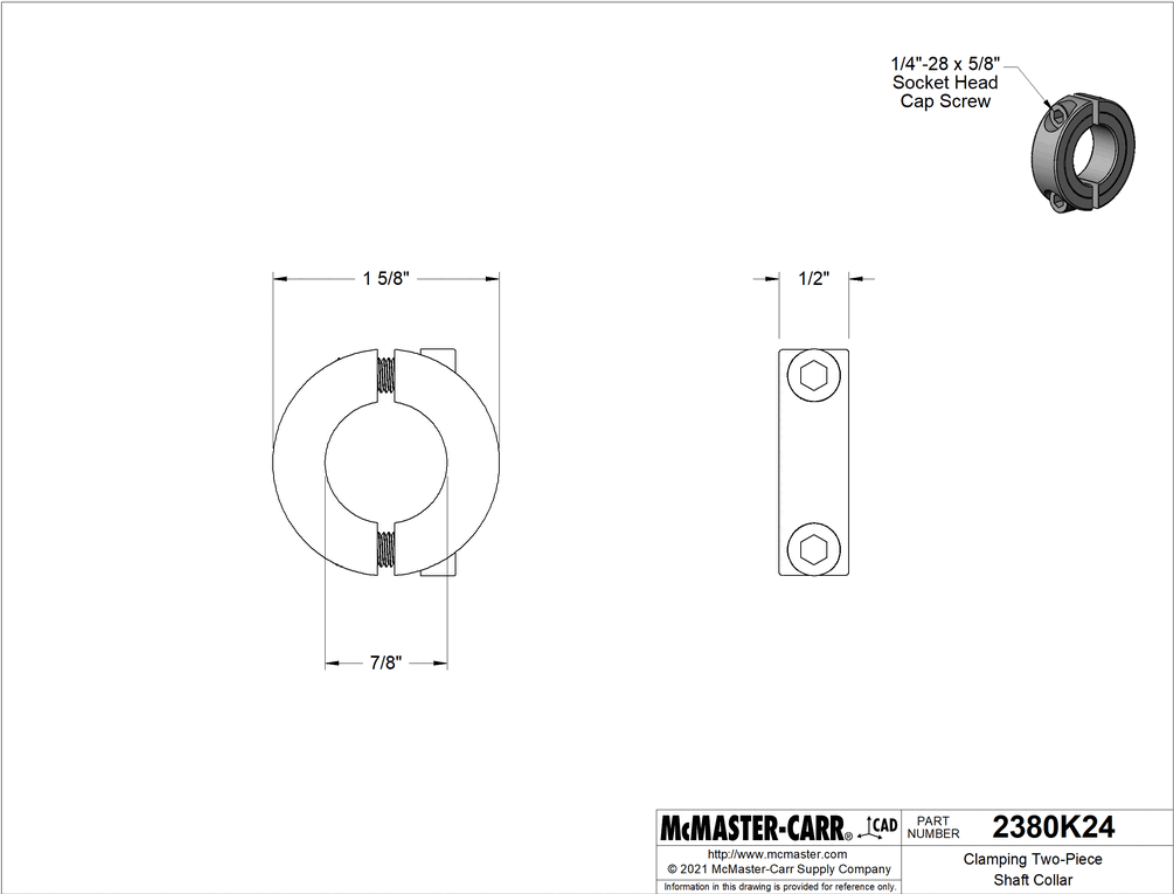
Part #	312100
Weight	0.28 lbs
Manufacturer	borgeson
CA Proposition 65	WARNING
WARNING:	Cancer and reproductive harm - http://www.p65warnings.ca.gov
Coupler Type	Straight Steel Coupler, 2" Long
Universal Joint Material	Natural Steel
Shaft End 1	5/8"-36 Spline Chrysler
Shaft End 2	3/4" Smooth Bore

Applications

Fitment Type	Regular Product
--------------	-----------------

Figure 45: Steering Shaft Coupler from Borgeson [28]

Clamping Two-Piece Shaft Collar



Material	Zinc-Plated 1215 Carbon Steel
For Shaft Diameter	7/8"
OD	1 5/8"
Width	1/2"
Clamping Screw	
Type	Socket Head Screw
Material	18-8 Stainless Steel
Number Included	2
For Shaft Type	Round
Shaft Mount Type	Clamp On
Construction	Two Piece
RoHS	RoHS 3 (2015/863/EU) Compliant
REACH	REACH (EC 1907/2006) (01/21/2025, 247 SVHC) Compliant
DFARS	Specialty Metals COTS-Exempt
Country of Origin	United States
USMCA Qualifying	Yes
Schedule B	848790.0080
ECCN	EAR99

These metal collars are stronger than plastic shaft collars. The two-piece design allows you to install them anywhere on a shaft without removing components or having access to the ends of the shaft. Collars clamp evenly around the shaft to create a strong, mar-free hold. Tighten the clamping screws to secure.

Zinc-plated steel and chrome-plated steel collars have moderate corrosion resistance with a bright finish.

Figure 46: Rack-&-Pinion collar from Mc-Master Carr [29]

Tilton Master cylinder

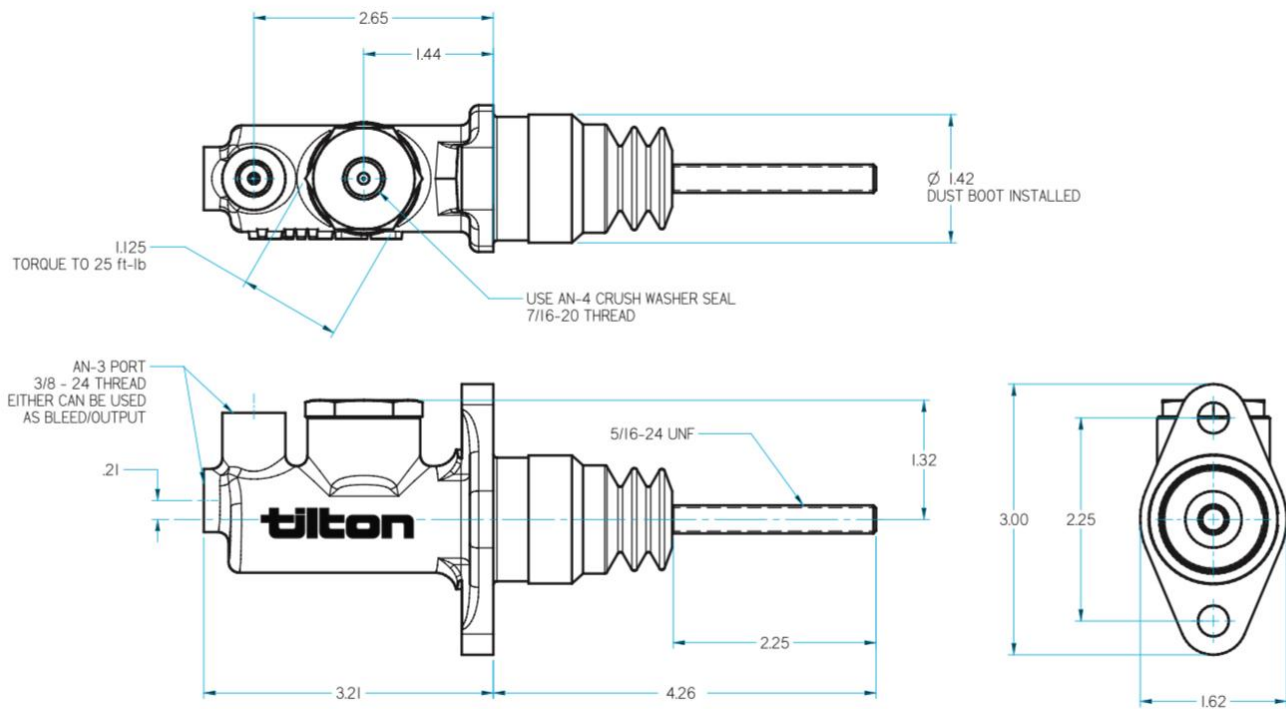
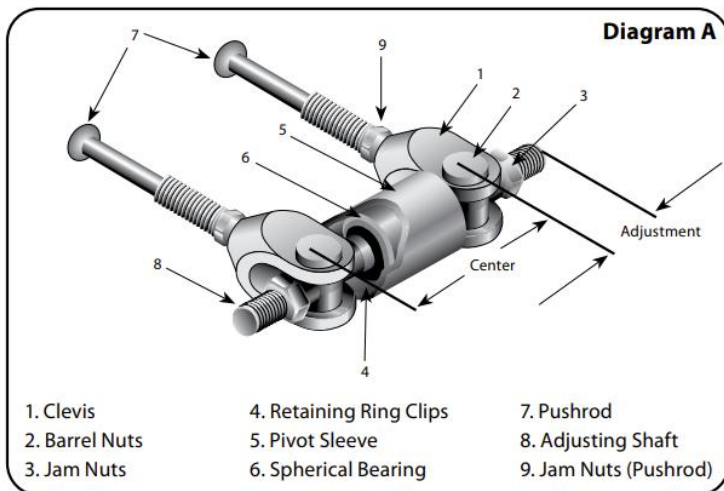


Figure 47: Master Cylinder 74 series from Tilton [30]

Tilton balance bar 72-260 .4375



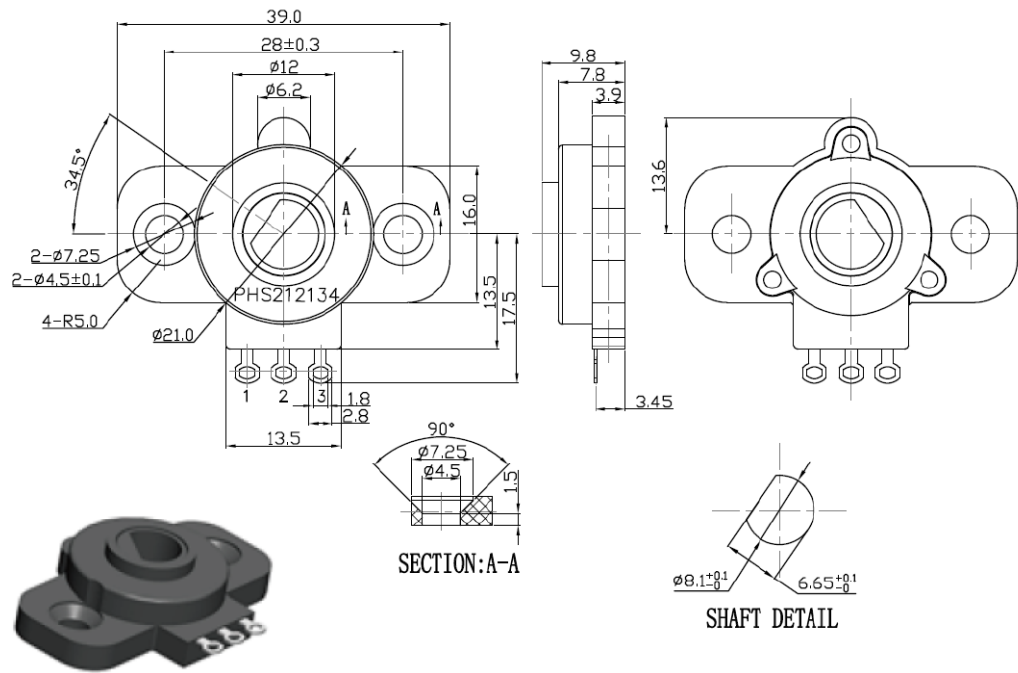
Force Distribution (2.5" Clevis Center-to-Center)		
Spherical Joint Position	Left Clevis	Right Clevis
3/8" left-of-center	65%	35%
1/4" left-of-center	60%	40%
1/8" left-of-center	55%	45%
Centered	50%	50%
1/8" right-of-center	45%	55%
1/4" right-of-center	40%	60%
3/8" right-of-center	35%	65%

Figure 48: Balance Bar from Tilton [31]



PHS21 Series

Outline Drawing



Mechanical Characteristics

$T_A = 25^{\circ}\text{C}$ unless otherwise noted

Total Rotational Travel	360°
Rotational Torque	20 - 150 gf.cm
Rotational Life	50,000 cycles

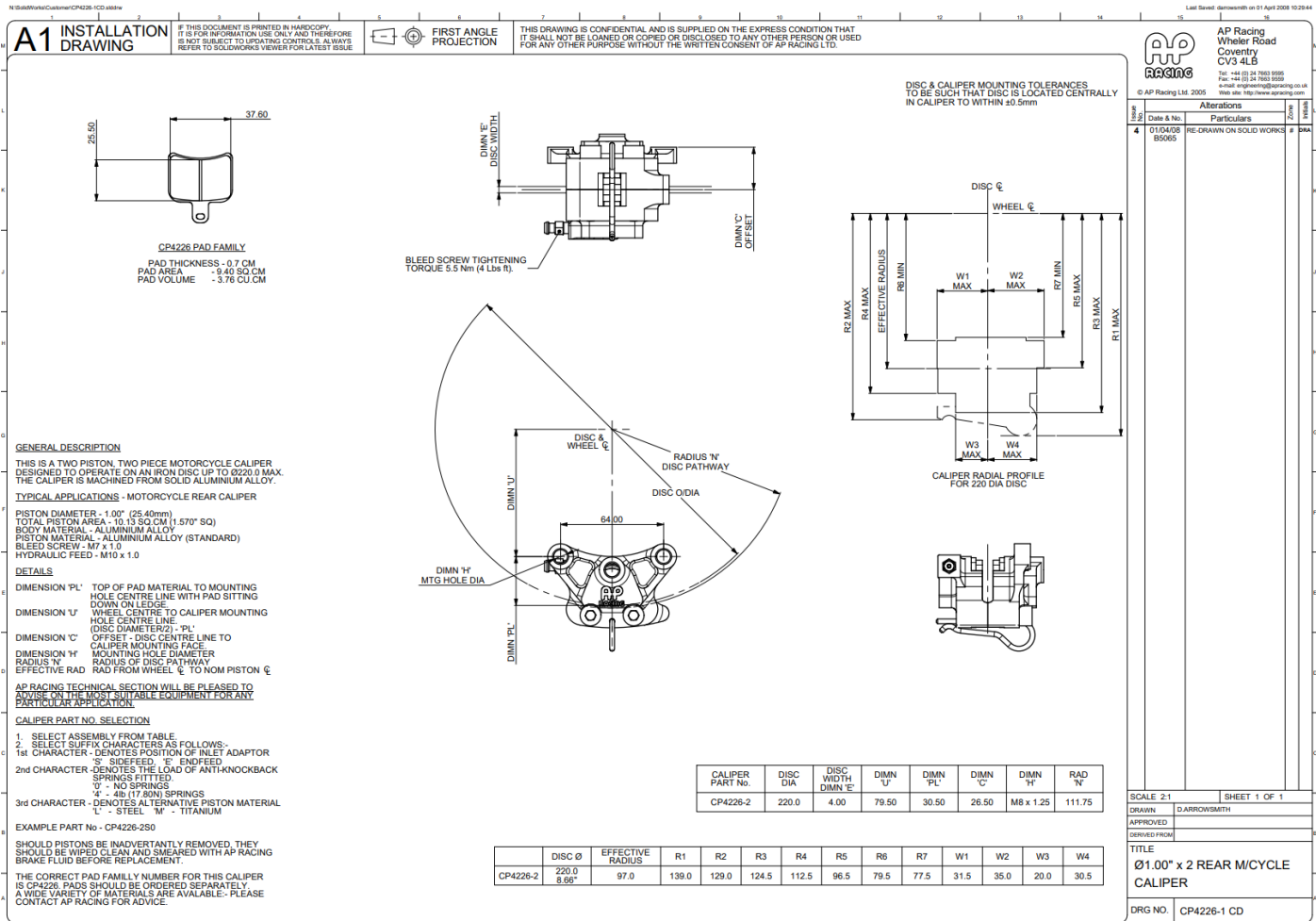
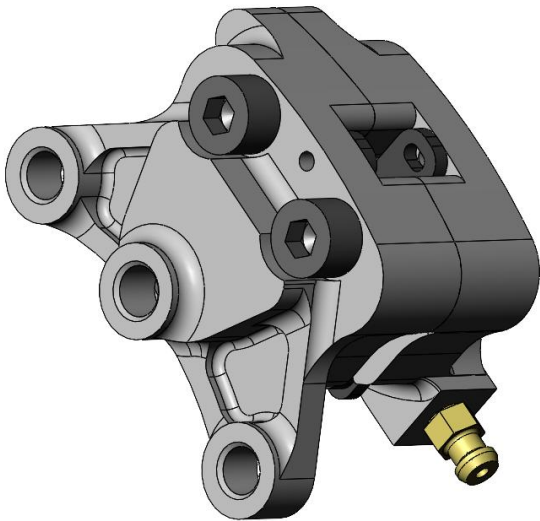
Environmental Characteristics

$T_A = 25^{\circ}\text{C}$ unless otherwise noted

Operating Temperature	-20°C to +105°C
IP Rating	IP 40
Soldering Condition	Wave Soldering 260°C maximum for 1-3 seconds Hand Soldering 300°C maximum for 3 seconds
RoHS	Please refer to TT Electronics website
REACH	Please refer to TT Electronics website

Figure 49: rotary potentiometer from TT Electronics [32]

brake caliper AP Racing CP4226



Motor Datasheet

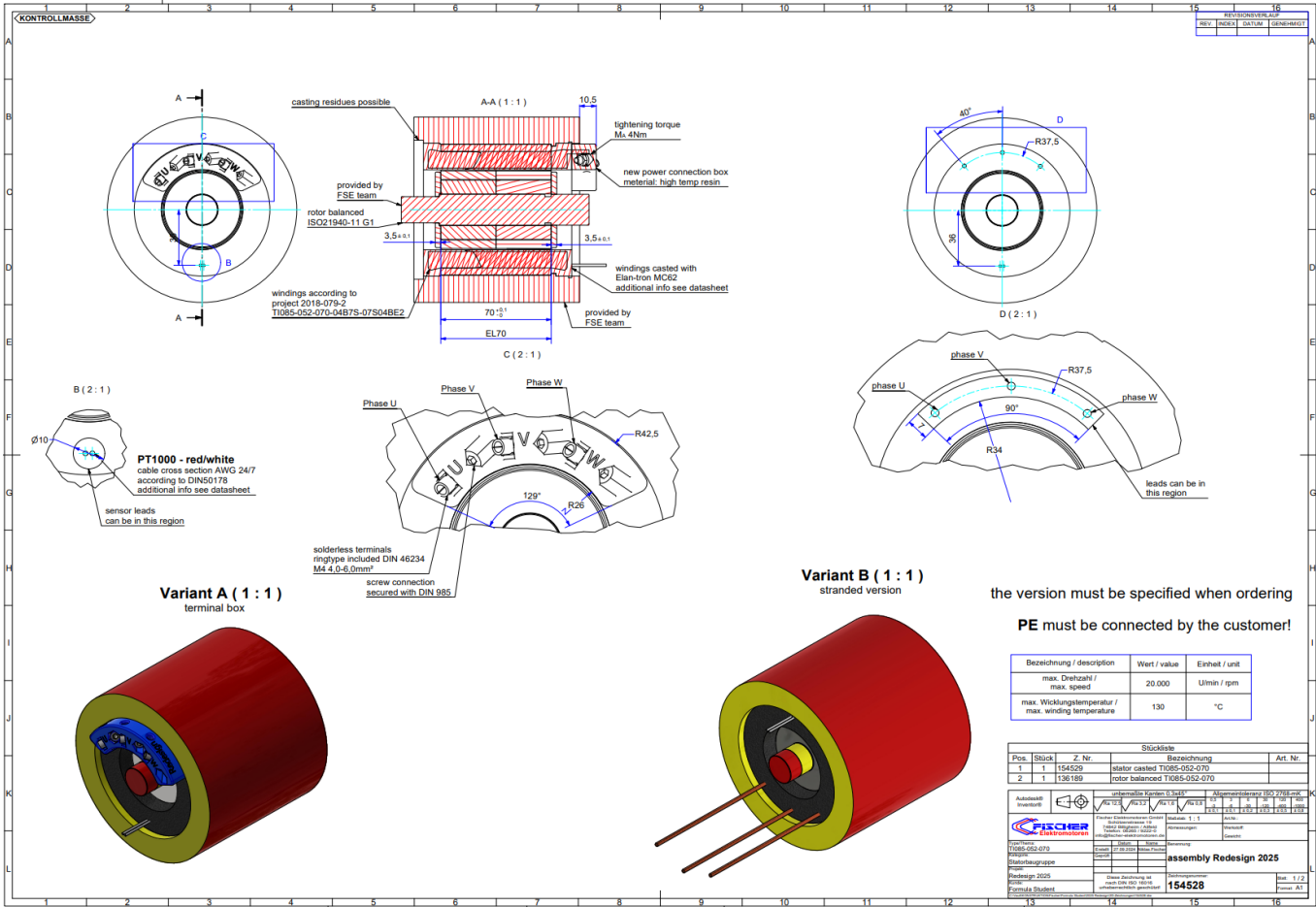


Figure 51: Datasheet for the Fischer TI085 electric motor (1 of 4) [33]

Project-No.:

	Symbol	Unit	Value	
Rated Data Water cooled ($\varphi = 0^\circ$)				
Nominal Torque	T _{NomWC}	Nm	11,1	
Nominal Current	I _{NomWC}	Arms	22,6	
Nominal Speed	n _{NomWC}	rpm	13250	
Nominal Power	P _{NomWC}	W	15404	
Winding Losses ¹ / Total Losses ^{1,2}	P _{DWC}	W	254	617
Holding Torque	T _{HWC}	Nm	7,9	
Holding Current	I _{HWC}	Arms	16	
Rated Data S6 duty ($\varphi = -10^\circ$)				
Torque	T _{S6}	Nm	24,6	
Current	I _{S6}	Arms	51	
Speed	n _{S6}	rpm	12100	
Power	P _{S6}	W	31199	
Winding Losses ¹ / Total Losses ^{1,2}	P _{DS6}	W	1282	1619
Holding Torque	T _{HS6}	Nm	17,4	
Holding Current	I _{HS6}	Arms	35,9	
Peak Data ($\varphi = -10^\circ$)				
Peak Torque	T _{Peak}	Nm	29,1	
Peak Current	I _{Peak}	Arms	61	
Speed at Peak Torque	n _{Peak}	rpm	11600	
Peak Power	P _{Peak}	W	35366	
Winding Losses ¹ / Total Losses ^{1,2}	P _{DPeak}	W	1843	2167
Data				
Torque Constant	k _t	Nm/Arms	0,492	
BEMF Constant (Phase - Phase)	k _e	V _{rms} /(rad/s)	0,296	
		V _{rms} /rpm	0,031	
Motor Constant	k _m	Nm/√W	0,447	
Idle Speed	n _{idle}	rpm	13650	
max. Speed (Fieldweaking)	n _{max}	rpm	20000	
max. Frequency (Idle/Fieldweaking)	f _{max}	Hz	910	1333
DC Bus Voltage	U _{DC}	V _{DC}	600	
Ø Resistance per Phase (Winding only)	R _{Ph20}	Ω	0,126	
Ø Inductance per Phase (Winding only)	L _{Ph}	mH	0,393	
electr. Time Constant $\tau=L/R$	τ _{el}	ms	3,11	
Number of Polepairs	n		4	
Winding Connection			Star	

Figure 52: Datasheet for the Fischer TI085 electric motor (2 of 4)[33]

	Symbol	Unit	Value
Data Watercooling			
Inlet Temperature of Coolant	T_{in}	°C	10 ... 40
Max. Temperature rise of Coolant	T_{max}	K	5
Min. required Coolant flow	Q_{min}	l/min	---
Volume of cooling channel	V_{cool}	l	---
thermal Time Constant	τ_{th}	min	---

Data Mechanics			
Rotor Inertia (assembly set)	J	kgm ²	$0,33 \cdot 10^{-3}$
Weight of Motor w/o Housing	m	kg	2,8
Outer Stator Diameter w/o Housing	dA	mm	85
Inner Stator Diameter	dAg	mm	51,6
Length of Stator	l	mm	70

Annotations - Losses

¹ Winding Losses are referred to a Coil Temperature of 100°C.

² The total Losses are made up of: Winding Losses; Stator Iron Losses; Rotor Losses;

Calculation of total Losses:

Winding Losses + Stator Iron Losses (at speed X) + Rotor Losses (at speed X)

Annotations - general

Ensure that your servo drive can handle the Nominal- and Peakcurrent of the Motor.

An adjustment of the Speed and DC Bus Voltage can be done after consultation.

The nominal data in this datasheet are based on an ambient/coolant temperature of 20°C

The stated nominal Torques are without consideration of friction losses through Bearings or Sealings.

Annotations - thermal monitoring system

Because the exact duty type depends also on the thermal connection of the motor, the embedded thermal monitoring system has to be analysed and attended. However, attention has to be payed that the temperature sensors do not show the exact temperature of the winding and this could be up to 20 K higher due to thermal capacities. Despite an electrical insulation towards the winding, you are only allowed to connect the sensors to your controller by using a galvanic separation in between.

Figure 53: Datasheet for the Fischer TI085 electric motor (3 of 4)[33]

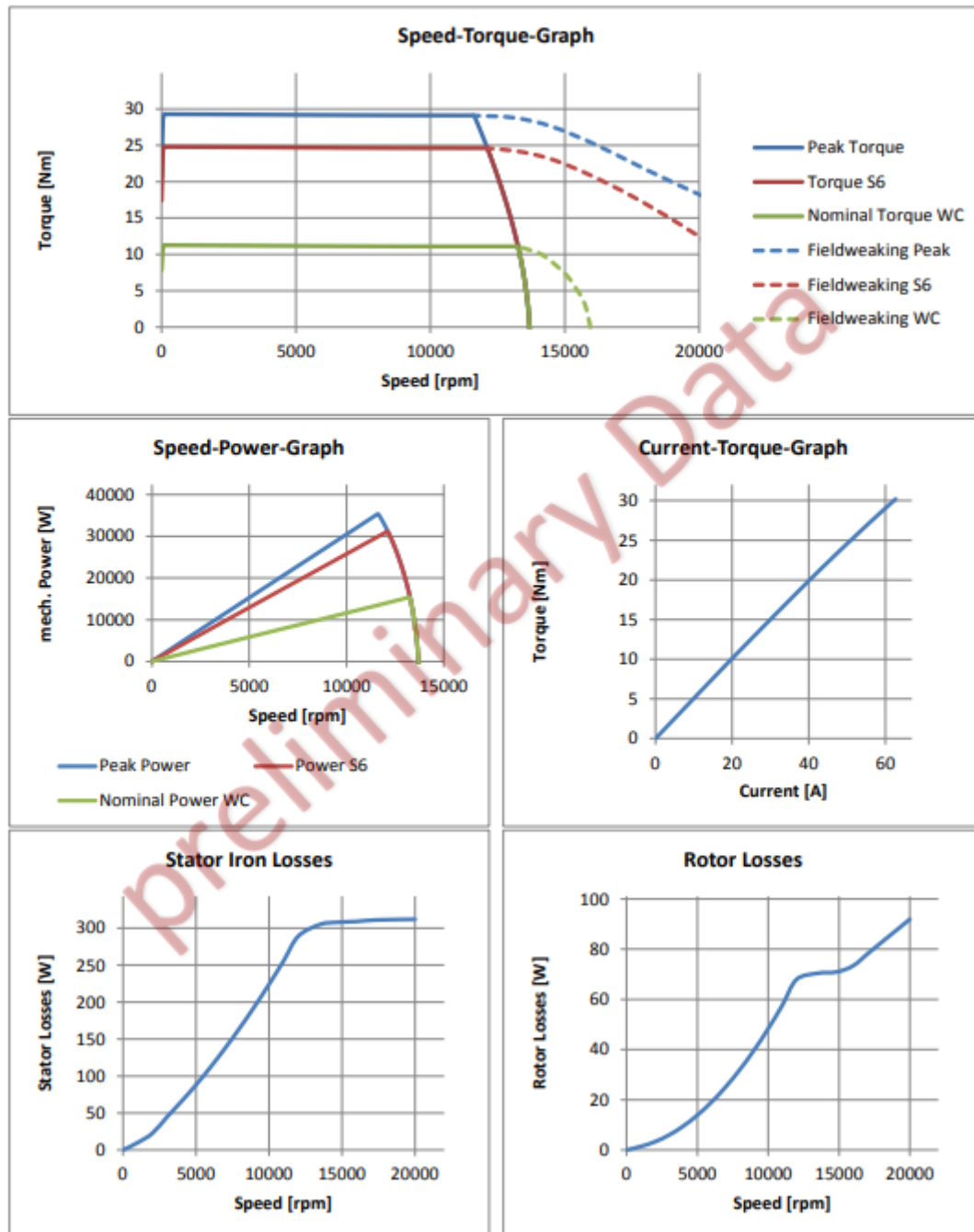


Figure 54: Datasheet for the Fischer Ti085 electric motor (4 of 4)[33]

MECHANICAL DATA

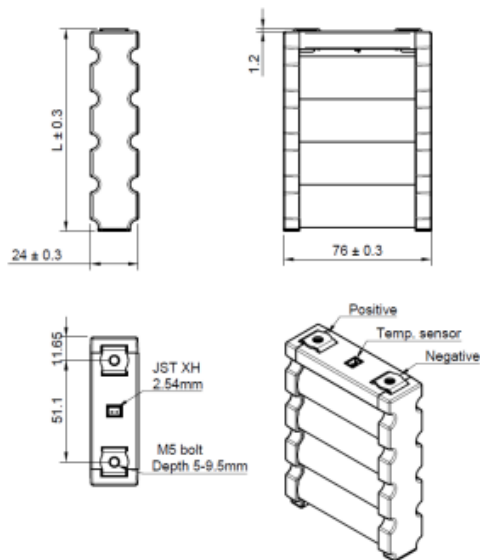


Figure 7: Mechanical dimensions of 1xNp modules

A simplified 3D STP model is available upon request.
1xNp notes:

- 1. Inner M5 nut is stainless steel
- 2. Tightening torque: 6 Nm
- 3. Self-locking washers recommended
- 4. Modules should be mounted in a firm enclosure to avoid mechanical damage.
- 5. Modules should be protected from direct water ingress.
- 6. Temperature sensor connector: JST XH series.

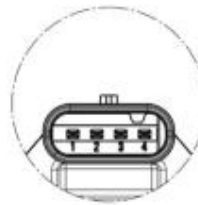
Module	L (mm) ±0.3	Weight* (g)
1x1	37.2	95
1x2	58.7	167
1x3	80.2	239
1x4	101.7	311
1x5	123.2	383
1x6	144.7	455
1x7	166.2	527
1x8	187.7	599
1x9	209.2	671
1x10	230	743

*-Without fasteners

Figure 55: Datasheet for Enepaq 1s4p module with Molicel P42A cells[34]

Pierburg „CWA100“ (CWA100-2)

Pierburg's electric water pump in the medium capacity range.



- 1 - Power (GND)
- 2 - Signal (GND)
- 3 - Signal (PWM)
- 4 - Power (12V)

Pierburg CWA100

The CWA100 is a performance enhanced version of the CWA50. The dimensions and the weight is nearly the same.

This closes the gap between the CWA50 and the CWA200. (Sometimes also called CWA100-2)

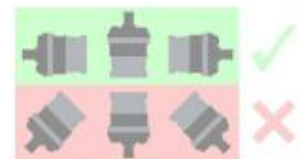
Suitable for instance for intercooling in combustion engine vehicles and cooling of various systems in electrical vehicles.

Specifications:

- Name: "Pierburg CWA100-2"
- Operation voltage: 8 to 16 V (Nominal: 12.5 V) (Full hydraulic power @ 12.5 to 16 V)
- Weight: approx. 1.0 kg
- Current consumption: 8.5A max. / (approx. 0.2mA in standby mode)
- Nominal diff. pressure: ≥ 0.75 bar *
- Flow rate: approx. 30-35 l/min @ 0.75bar / 40-45 l/min @ 0.50bar
- Speed: approx. 24 to 7000 rpm
- Temperature range: -40°C – 110°C (water) / -40°C – 120°C (ambient)
- Protection: IP 67 + IP 69 K
- Inlet / outlet nozzle removable / Four possible positions
- Part numbers: Pierburg: 7.06754.05.0 // Mercedes: A0005000486

Notes:

- Power (speed) reduction below 0°C and/or below 12.5V.
- Works with water, water/glycol mixtures and "other liquids" (according to Pierburg)
- The PWM input is equipped with a 10 kOhm pull-down resistor.
- Flow diagram @ 80°C, 12.5V, Water/Glycol 50/50



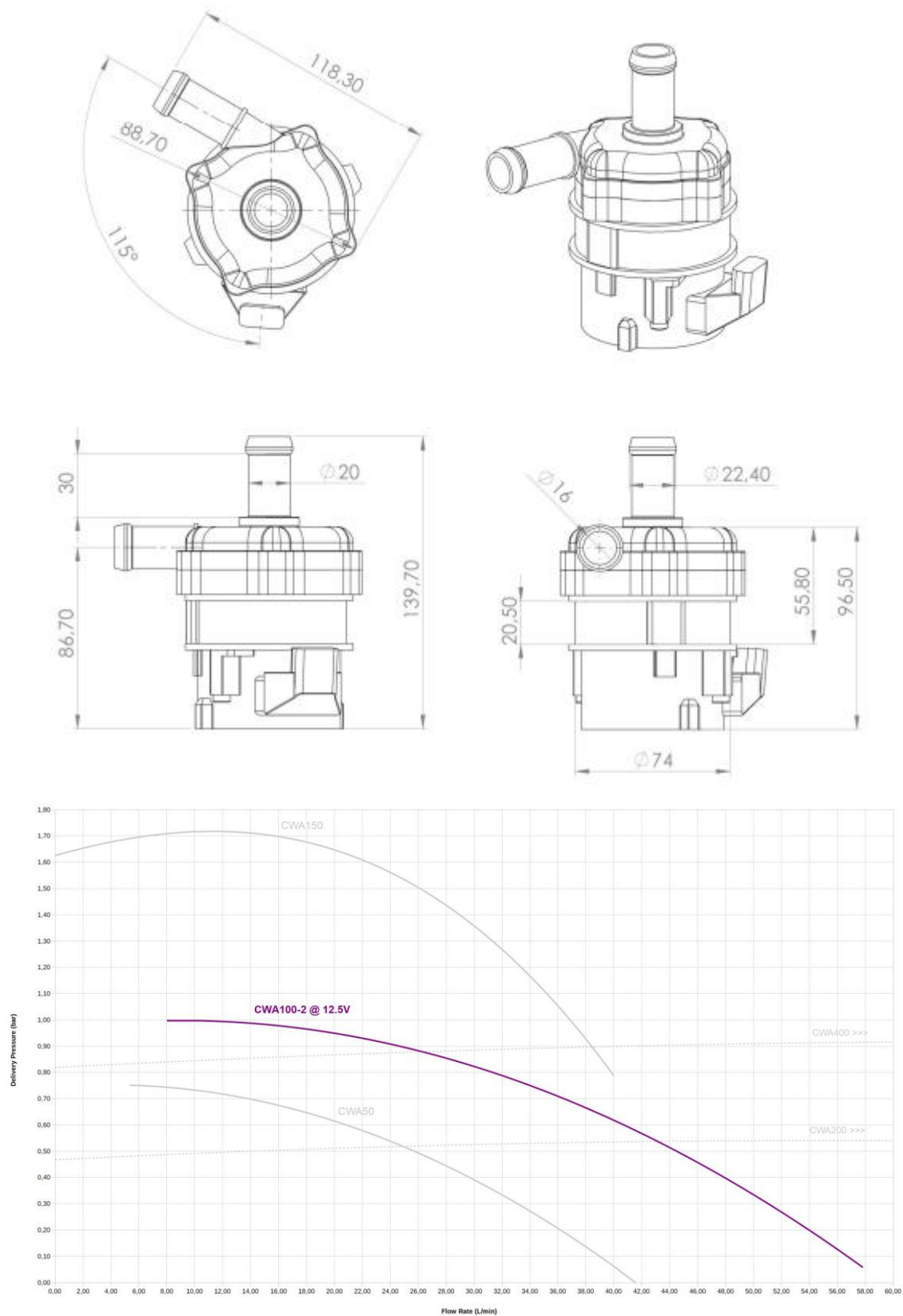
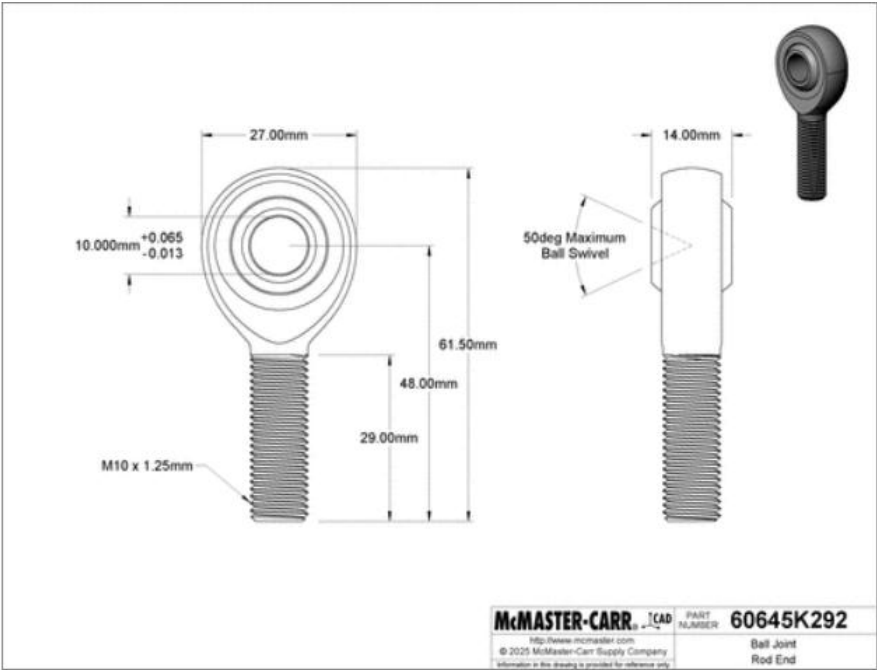


Figure 56: Water Pump Pierburg CWA100 Data Sheet[35]

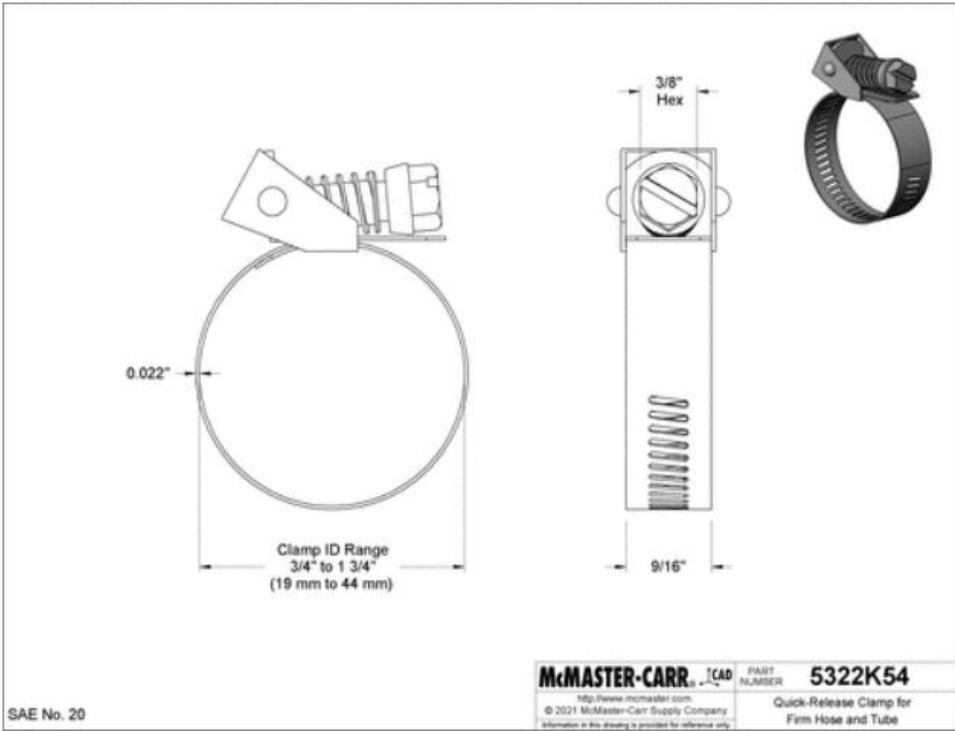
Cooling – Rod-end



Shank Thread Size	M10 × 1.25 mm
ID	10mm
ID Tolerance	-0.013mm to 0.065mm
Maximum Ball Swivel	50°
Overall Width	27mm
Overall Thickness	14mm
Shank Center Length	48mm
Shank Thread Length	29mm
Static Radial Load Capacity	4,600 lbs.
Material	Zinc-Plated Carbon Steel
Ball Material	Chrome-Plated Alloy Steel
Lubrication	Required
Relubrication	Required
Rod End Type	Ball Joint
Shank Threading	Fully Threaded
Shank Gender	Male
Shank Thread Direction	Left Hand
RoHS	Not Compliant
REACH	REACH (EC 1907/2006) (11/07/2024, 242 SVHC) Compliant
DFARS	Specialty Metals COTS-Exempt
Country of Origin	United States
USMCA Qualifying	Yes
Schedule B	848330.8055
ECCN	EAR99

Figure 57: Radiator Support Rod-end [36]

Hose Clamp



For Use With	Firm Hose, Firm Tube
Clamp Type	Quick Release
Clamp ID Range	
Inch	3/4" to 1 3/4"
mm	19 to 44
Band	
Width	9/16"
Thickness	0.022"
SAE Number	20
Temperature Range	-40° to 800° F
Maximum Torque	30 in.-lbs.
Drive Style	Slotted, External Hex
Hex Size	3/8"
Reusable	Yes
Material	201 Stainless Steel
Housing Material	201 Stainless Steel
Screw Material	410 Stainless Steel
RoHS	RoHS 3 (2015/863/EU) Compliant
REACH	REACH (EC 1907/2006) (01/21/2025, 247 SVHC) Compliant
DFARS	Specialty Metals COTS-Exempt
Country of Origin	Mexico
USMCA Qualifying	Yes
Schedule B	732690.8695
ECCN	EAR99

Figure 58: Cooling hose clamps [36]

Appendix B. Additional Requirements

The requirements for the car are the same as those of any modern race car. It must be designed to safely and rapidly complete competitive events, while complying with the series ruleset. For an FSAE EV, the design of the car must satisfy all technical regulations and safety compliance as per the FSAE rulebook [5].

B.1 General Requirements

The harsh nature of motorsport events means that the car must operate safely under all loads and environmental conditions. The car must also pass several physical inspections to ensure vehicle integrity, prove rule compliance and demonstrate effective safety features. The first of these is the mechanical technical inspection, where a checklist inspection of the main regulations is performed on the car. The next inspection is the tilt test, which requires the vehicle to be tilted up to 45 degrees on its side, where it must not produce any leaks, and then to 60 degrees, where all four wheels must remain in contact with the platform to demonstrate a low center of gravity. The next test is the rain test, where the car is artificially rained on, meaning it must be waterproof. The last test before being fully eligible to race is the brake test, where the car must be able to accelerate and lock up all four wheels to prove the full operation of the braking system without assistance from the electric motors.

B.2 Frame Requirements

The chassis structure requirements revolve around safety and component packaging. Heavy components such as the motors, suspension or battery container must not be mounted to the upper portions of the main hoop, main hoop braces or the shoulder harness structure (F.5.11.4). Only limited bends in the roll hoop bracing members are allowed (F.5.6.3). Continuous and wide core load paths around the roll hoop mounts and a core splice between adjacent core sections are required for monocoque construction (F.7.1.7). The chassis must ensure that the hips of the driver are always below the top member of the upper side impact structure and their legs below the top of the frame ahead of the front hoop (T.1.3.1). It is important to note that when choosing a monocoque construction, a full structural equivalency study must be performed to prove that the chassis meets the same requirements stated for the space frame type chassis.

B.3 Suspension Requirements

Section V.3 of the Formula SAE rule book covers the rules and requirements to be respected for designing the suspension. The suspension system must be fully operational on both the front and rear of the vehicle,

incorporating shock absorbers at all four corners to provide compliant and controlled wheel movement (V.3.1.1). It must deliver a minimum of 50 mm of usable wheel travel at each wheel with the driver seated to ensure sufficient compliance and performance under dynamic conditions (V.3.1.1). Additionally, the suspension must enable safe, predictable, and appropriate handling suitable for an autocross-style circuit, as vehicles demonstrating unsafe or unsuitable handling characteristics may be disqualified (V.3.1.2).

B.4 Steering Requirements

The steering system must provide a fully mechanical connection between the steering wheel and the vehicle's front wheels, using components capable of transmitting both tension and compression without reliance on electronic or hydraulic actuation (V.3.2). Positive steering stops must be incorporated to prevent the steering mechanism from exceeding its designed travel or from causing any interference between the wheels, suspension, or bodywork (V.3.2.4). Rear wheel steering can be electrically operated only if the design contains a mechanical stop that limits the range of angular movements (V.3.2.10). Finally, the steering wheel must leave a view to the dashboard and must be attached to the steering column with a quick release disconnect that can be operated with gloves for rapid driver egress (V.3.3).

B.5 Pedal Box Requirements

Braking should be applied to all four wheels using the brake pedal as a single control and must be capable of locking all four wheels (T.3.1.2). As a redundancy, the hydraulic system must include two independent circuits with reservoirs so that any leak allows braking on at least two wheels (T.3.1.3) (T.3.1.4). The Brake Over-Travel Switch (BOTS) is a closed mechanical switch that must be used to electrically shut down the vehicle if pedal travel exceeds its normal range (T.3.3). The throttle pedal must use two independent sensors and return springs (T.4.2.1) (T.4.2.2). All bolts or pins in the pedal assembly are classified as critical fasteners (T.8.2). All sensors and shutdown-related components must be analog and non-programmable (EV.4.5) (EV.7.7).

B.6 Drivetrain Requirements

The drivetrain of the car needs to be fully enclosed and protected from debris and moisture (EV.4.8.6). It must also be securely and mechanically mounted to the chassis (T.3.30). Components such as gear reducers, shafts, and differentials must be capable of withstanding the expected torque and speed without failure. The

entire system must also integrate with the tractive system and maintain electrical isolation from any conductive chassis elements (EV.3.2.6).

B.7 Accumulator Requirements

The accumulator must have adequate energy capacity and discharge rate to complete all endurance requirements without exceeding allowable temperature or voltage limits. Being a critical part of the car, the battery pack is highly regulated. The accumulator must safely store and deliver energy to the tractive system, while protecting the driver in the case of a failure. It must be housed in a rigid, fire-resistant container which is securely mounted to the chassis (EV5.3.1) (EV5.4.2). It must include electrical isolation (EV3.2.6), fusing and insulation (EV5.5.1), and a cooling system to maintain safe operating temperatures. Accessibility is another requirement for inspection and service of the pack, with clearly labeled connections and removable modules (EV5.4.3). The design must also allow for easy disconnection of the high-voltage system during an emergency shutdown.

B.8 Cooling Requirements

Thermal safety and component integrity are the main requirements for this system. The tractive system accumulator container must incorporate measures to prevent the internal temperature from exceeding the specified limits defined by the manufacturer or stated in the data sheet of the relevant component (EV.3.2.4) (EV.3.2.6). All fluid carrying components, including hoses and pumps, associated with the cooling system of the motor must be rated for at least 120°C or the highest temperature the fluid is expected to reach (T.7.2.7).

B.9 Aerodynamics and Bodywork Requirements

It is important to note that the aerodynamic system is not a required system for the car. However, if it is included, it must comply with the applicable rules from the rulebook. First, the components for both aerodynamics and bodywork must be mounted such that they provide sufficient rigidity in static conditions, and do not oscillate in dynamic conditions (T.7.1.3).

The bodywork must follow the aerodynamic requirements if it is designed to have an aerodynamic effect on the car (T.2.2).

Appendix C. Additional Constraints

The project operates within the FSAE Electric rulebook, a defined set of technical, organizational, and regulatory limitations that guide design feasibility and decision-making. All systems must comply with FSAE EV rules, including accumulator design, mechanical protection, aerodynamic limitations and much more.

C.1 General Constraints

Physical constraints outlined in the rulebook are that the vehicle must have a minimum wheelbase of 1525 mm (V.1.2). There is no minimum or maximum track width for the car, but the stagger between the front and rear must be no more than 25% (V.1.3.2). The car must also have a maximum ground clearance of 75mm from the lowest point in the chassis, and a usable minimum wheel travel of 50mm (V.1.4.2) (V.3.1.1). The cockpit of the car must accommodate drivers ranging from 5th percentile female up to 95th percentile male and needs to allow for a minimum field of vision of 100 degrees on each side (V.2.1.1) (V.2.2). Welded inserts must be smaller than the outer diameter of the tube and multi-piece members must be mitered for the structural joints (F.5.2.4).

C.2 Frame Constraints

The chassis tubes need to have a minimum wall thickness of 2 mm, or an equivalent for aluminum for the main hoops. A minimum 50 mm clearance is needed between the driver's helmet and the roll structure. The main hoop should be at least 980 mm high (F.6.1.4). The upper side impact member should be between 240 mm and 320 mm off the chassis floor (F.6.4.4).

C.3 Suspension Constraints

The suspension must be designed such that all suspension mounting points are visible during Technical Inspection, either through direct line of sight or by removing simple covers (V.3.1.3). All suspension fasteners are classified as Critical Fasteners and must meet the safety and retention requirements defined in rule T.8.2 (V.3.1.4). Spherical rod ends and spherical bearings used within the suspension system must either be mounted in double shear or mechanically captured with a bolt head or washer whose diameter exceeds the bearing housing's internal diameter (V.3.1.5). Suspension mounting points may not be attached to the span of the Main Hoop Braces or to the Main Hoop above any tube node or composite attachment (F.5.11.4).

C.4 Steering Constraints

Here are the limitations regarding the steering system. Measured from the steering wheel, the steering system free play must be less than 7° total (V.3.2.5). If rear-wheel steering is used, it must include mechanical stops that limit angular wheel movement to a maximum of 6° (V.3.2.10). Rear-wheel steering may be electrically operated, but only if it complies with the above angular restriction and verification requirements (V.3.2.10).

C.5 Pedal Box Constraints

Both pedals must be made of steel, aluminum, or titanium (T.3.2.1). The brake pedal and its mounting structure must withstand a 2000 N pedal force without failure or permanent deformation (T.3.2.2). No part of the pedal box or any of the brake components may extend below the lowest chassis tube or the undertray (T.3.1.9). Brake-by-wire actuation is prohibited (T.3.1.6).

C.6 Drivetrain Constraints

The drivetrain and tractive system must not exceed 600V DC (EV.3.3.2) and must draw no more than 80kW of energy at any moment (EV.3.3.1). Any size and number of motors are permitted (EV.4.1.1). Any cooling and lubrication systems need proper overflow containment (catch cans) (T.5.6).

C.7 Battery Box Constraints

Being such a critical component of the car, the accumulator has its own separate technical inspection, called the tractive battery pack inspection. Key design constraints mention that insulation resistance must be greater than 500 Ω/V between the tractive system and chassis (EV.7.6.3). The accumulator container must withstand 40 G longitudinal, 40 G lateral, and 20 G vertical loads without cell movement (F.10.3.1). All active components must be isolated from the chassis with non-conductive materials, and venting provisions are required to safely release gases in case of failure (F.10.4). The total energy content must comply with limits defined in EV.3.3.2, and the accumulator must include accessible fuses and shutoff devices meeting the spacing and accessibility requirements of EV.7.

C.8 Cooling Constraints

The coolant fluid can only be plain water with no additives or oil (T.5.4.2). Liquid coolant cannot directly touch the cells in the Tractive Battery. Any catch can on the cooling system must vent through a hose with a

minimal internal diameter of 3 mm down to the bottom levels of the chassis (T.5.6.5). Cooling fans should not be inside the Tractive Battery Container (EV.6.5.4).

C.9 Aerodynamics and Bodywork Constraints

The aerodynamic constraints are summarized in the rulebook with Figure 59 demonstrating the allowable areas for these components (T.7). The location of these components is also prohibited above the tires (T.7.4). Furthermore, no power device can be used to remove air from under the car (T.7.1). Another important constraint for aerodynamic components is the forward-facing edges that could contact a pedestrian require a minimum radius of 5 mm in the horizontal direction and 3mm in the vertical direction (T.7.1.4). Lastly, if parts are mounted to the outside of the Main Hoop Braces, they must be longitudinally offset [37] to prevent point loading in a rollover (F.5.11.5).

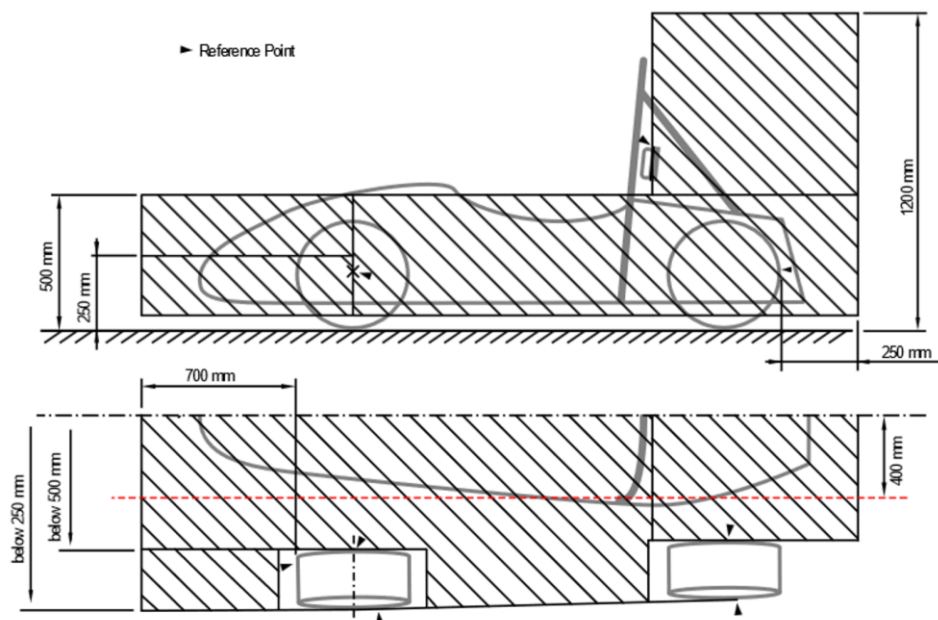


Figure 59: Aerodynamic Component Area Constraints

The constraints for the bodywork unrelated to aerodynamics are that the minimum radius for the front nosecone must be 38 mm with a radius that extends at least 45° relative to the forward direction, along the top, sides, and bottom of all affected edges. (T.7.2.4). Also, the bodywork must not have any openings from the front of the vehicle to the main hoop and firewall except for the cockpit and small openings around the front suspension (T.7.2.3).

Appendix D. Additional Criteria

The vehicle must be designed to achieve competitive performance while maintaining safety, reliability, and manufacturability. These design criteria outline key performance specifications that should be optimized through design and analysis. Every subsystem should be designed to minimize weight, without sacrificing structural integrity or rigidity.

D.1 Mass and Center of Gravity

Minimizing weight improves the car's acceleration, handling, and efficiency. Lightweight materials, topology optimization and compact packaging are all common ways of achieving this. Components on the car should be positioned to tune the Centre of Gravity (CG). Everything should be mounted as low as possible, since a low CG height can drastically improve the stability and cornering performance of the car. With a driver in the car, the front-rear balance changes the dynamic behavior of the car. Typically, a 45-55 front-to-rear balance is desirable. The target of a 50-50 left-right balance ensures consistency in cornering performance. A well-balanced vehicle should be capable of rapid, controllable acceleration, cornering and braking. In other words, maximizing the car's potential within the limit of grip of the tires.

However, when applying the weight and CG criteria for the aerodynamic components, it is important to consider them separately since the CG of these components is not only controlled by the weight. The components affect the CG in different ways depending on their geometry and not the weight.

D.2 Durability

All components must be designed to withstand high cyclic loads and intense vibrations. This includes both individual racing events and extensive testing periods, which amounts to roughly 5000+ km. Structural fatigue, wear resistance, and thermal management are critical to ensure reliability and performance longevity.

D.3 Manufacturability

Designs should consider manufacturing methods and their associated costs when choosing materials and geometries. Not only is reducing the vehicle cost important, but designing parts which are easy to manufacture decreases the time required to source replacement parts if something breaks. The use of highly specialized machinery should be avoided unless necessary.

D.4 Packaging

Packaging assesses the volume and spatial integration of components within the vehicle envelope while still ensuring mechanical compatibility and accessibility. It involves optimizing the arrangement of sub-assemblies to achieve a compact subsystem with proper clearances.

D.5 Safety

Safety evaluates the system's ability to protect the driver by ensuring the chassis and all critical systems can withstand the different dynamic event conditions without failure. This includes appropriate materials and load paths that maintain structural integrity during impacts, rollovers, or heavy braking. Designs must respect the rules governing driver protection, load path integrity, and fail-safe operation.

D.6 Assembly

Assembly evaluates how practical and efficient the subsystem is to construct. It focuses on minimizing time, part variety, and alignment complexity. Parts should be designed for straightforward joining, common fasteners, and traditional mounting styles. The assembly sequence must avoid interference between components and allow for clear visual and physical access.

D.7 Repairability, Serviceability and Maintainability

This criterion evaluates how easily the subsystem can be accessed, repaired, or replaced with minimal disassembly or downtime. Having a high maintainability ensures the parts can be quickly swapped during competitions to reduce the time spent in the pit garage. Components should have standardized fasteners and accessible mounting points. Proper layout of all mechanical interfaces is essential to facilitate troubleshooting and maintenance.

D.8 System Stiffness (Stability)

System stiffness ensures that forces from each sub-system are transmitted accurately without excessive flex or distortion. A rigid structure should maintain its geometry and provide consistent driver feedback. Designs should minimize deflection through proper material choice and load-path continuity to ensure predictable performance under load.

D.9 Ergonomics

An ergonomic design ensures the driver can operate the vehicle comfortably, safely, and efficiently. Every driver control must transmit inputs with minimal fatigue. The layout should accommodate different driver sizes while maintaining easy access to critical systems during operation. The cockpit of the car needs to allow easy driver ingress and egress in the case of a fire or accident.

D.10 Reliability

Reliability ensures consistent performance throughout the vehicle's operation. Components must function as intended under repeated loads, vibrations, and environmental stresses. Designs should prioritize proven mechanisms to minimize the risk of failure during testing and competition.

D.11 Precision and Backlash

Precision ensures that mechanical inputs produce an accurate and consistent motion without delay or deviation. Minimizing backlash in joints, linkages, and interfaces improves control response and overall system accuracy. Designs should aim for tight tolerances and proper alignment.

D.12 Tunability

For most mechanical systems, operating parameters and performance characteristics must be designed into their components. Tunability is the system's ability to allow rapid and precise adjustments to optimize vehicle behavior across different track layouts and dynamic events. This ensures the car can be quickly adapted for grip, stability, and responsiveness under varying conditions. Certain systems can be integrated electronically to allow "on-the-fly" changes to the way the car behaves, while others require partial or complete disassembly to adjust. Systems like the drivetrain and brakes can be tuned by the driver with simple dashboard controls, or even automatically through software. Examples of good control flexibility are brake biasing and torque vectoring.

D.13 Efficiency

Given that every car has the same energy consumption limitations, being able to maximize the amount of power that makes it to the wheels is critical to having a competitive car. Not only does the vehicle's performance improve with an efficient tractive system, but a more efficient system generates less heat, noise and vibrations. The efficiency event at the competitions is also worth 100 points in total, which is enough to require proper consideration in designs.

D.14 Thermal Management Difficulty

Heat management is a critical part of the car that ensures both safety and performance. When many heat generating components are used, such as the case with four motors, the complexity of the cooling system becomes much higher. The more components that require cooling, the higher the cooling capacity needs to be, and the number of hoses, fittings and pumps increases.

D.15 Dynamic Event Criteria

D.15.1. Acceleration (75m sprint)

The objective of the acceleration event is to evaluate the vehicle's straight-line acceleration. This event requires high instantaneous torque delivery, and so the car's drivetrain must be able to handle peak loads without chain slippage, gear failure or excessive voltage drop. The intense acceleration and high speed introduce a high thermal load, so the motor(s) and accumulator must stay within safe operating limits over the course of the event.

D.15.2. Skidpad (figure-eight circuit)

The Skidpad event measures the steady-state cornering ability, balance and stability of the car. For a good result, consistent and predictable torque application and wheel slip are important. The high and prolonged lateral force on the car can introduce cavitation in the coolant pump, or oil starvation in gearboxes, so these systems may need to be designed to account for this.

D.15.3. Autocross (technical sprint)

Contrary to the Skidpad event, autocross is intended to evaluate transient performance and agility. Autocross emphasizes the importance of torque responsiveness, predictable handling and fast energy recovery between short bursts. The numerous quick turns and slaloms the car experiences bring high load cycles on every part of the car, highlighting the importance of designing highly stressed parts for fatigue.

D.15.4. Endurance (22km race)

The endurance event is the biggest event of the competition, and it is used to demonstrate the long-term reliability and thermal stability of the car. The accumulator, motors and overall cooling system must maintain operational temperatures over high, sustained loads. Reliability is a major design consideration for this event, given that any failure results in a disqualification. Mechanical components must resist wear and fatigue throughout the event.

D.15.5. Efficiency (evaluated during endurance)

Given the innovation-focused nature of the competition, energy efficiency is becoming a prevalent part of FSAE. The efficiency event encourages teams to minimize the total energy consumption for the event distance. The drivetrain mechanical efficiency and the optimization of reduction ratios are imperative to maintain efficiency.

Appendix E. Additional Designs

E.1 Chassis Design – Monocoque

This section shows a detailed drawing of the potted inserts that are used for mounting the different systems to the front monocoque.

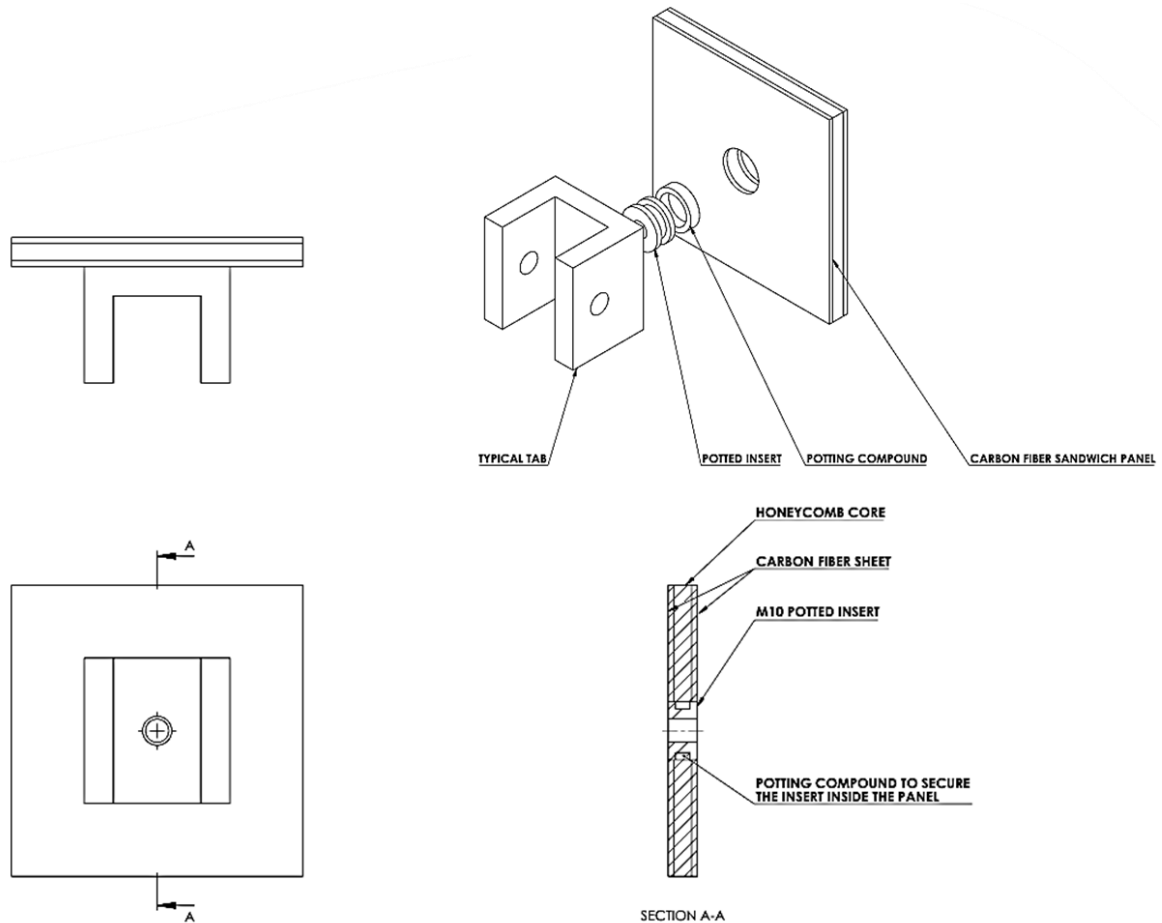


Figure 60: Design Solution - Composition of Monocoque Panel.

E.2 Suspension Design - Mounting

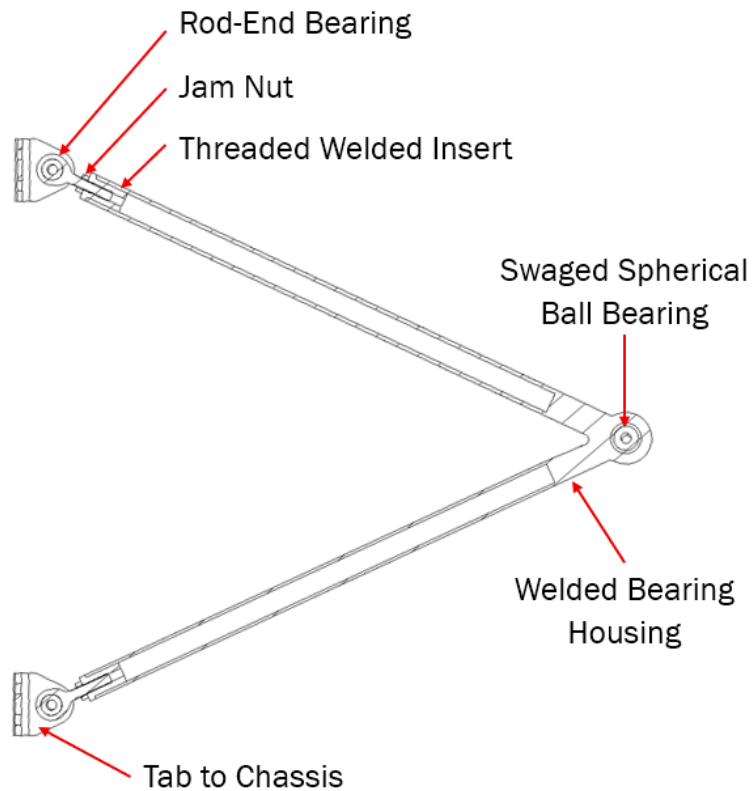


Figure 61: Design Solution – Suspension Mounting

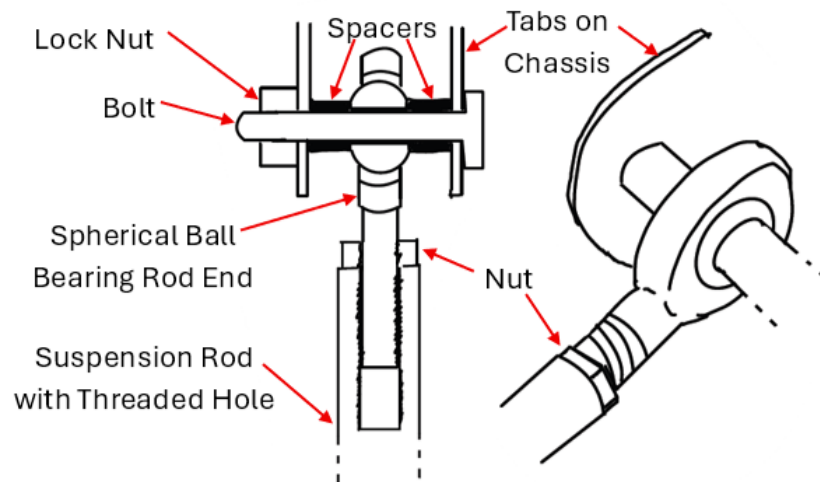


Figure 62: Design Solution - Mounting Attachment of Suspension using Heim Joints

E.3 Drivetrain Design – Additional exploded views



Figure 63: Design Solution - Exploded view of the entire wheel assembly.

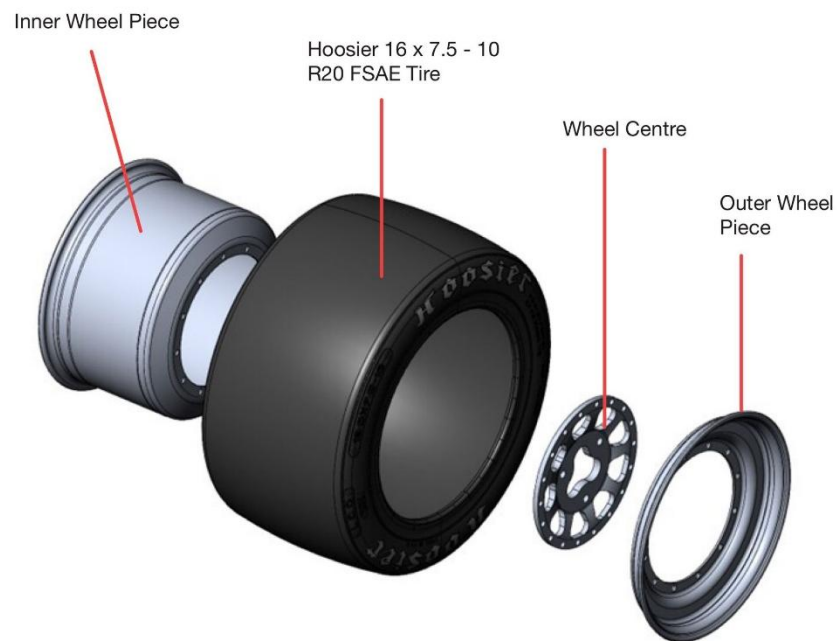


Figure 64: Design Solution - Exploded view of the wheel.

E.4 Steering Design – Mounting & Column Assembly

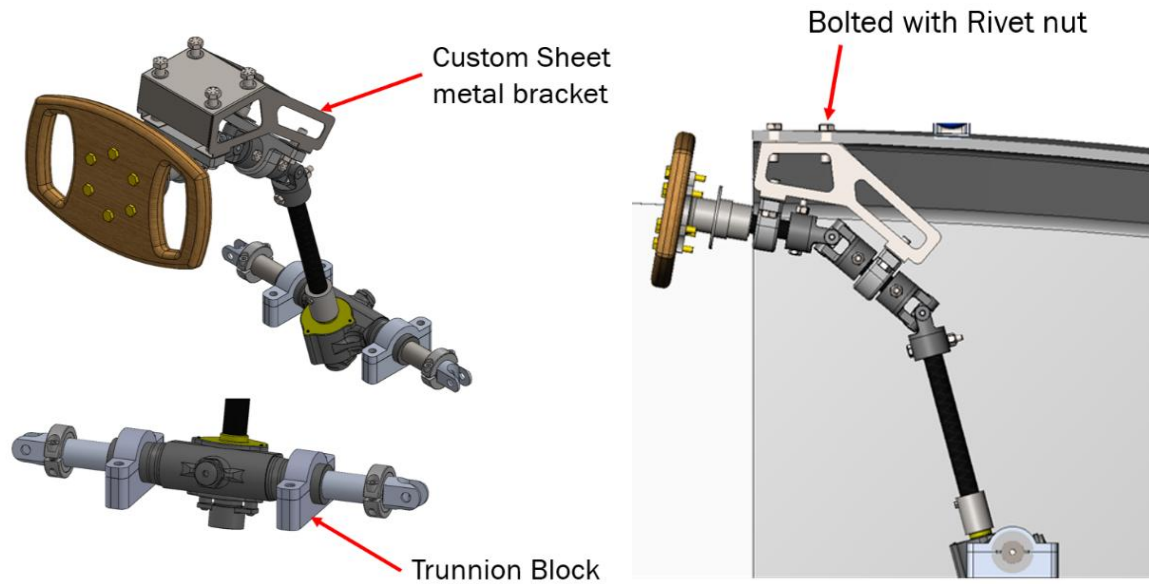


Figure 65: Design Solution – Steering Assembly mounted in Chassis

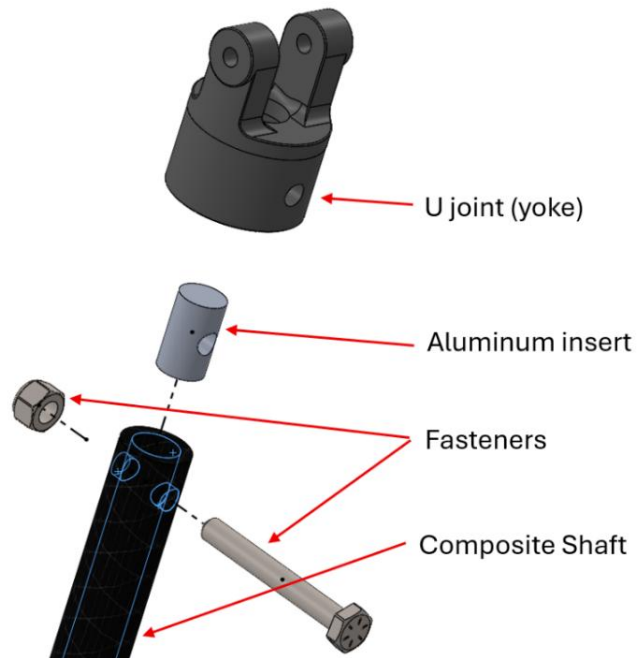


Figure 66: Design Solution - steering column joint assembly

E.5 Pedal Box Design - Throttle Pedal

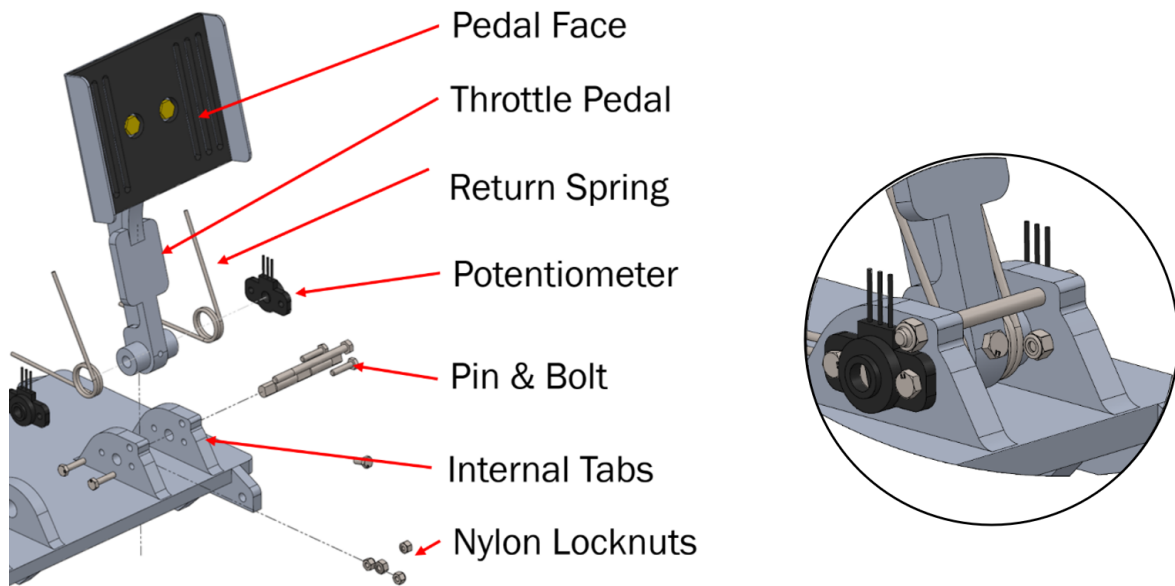


Figure 67: Design Solution – Exploded view of Throttle Assembly

E.6 Pedal Box Design - Platform

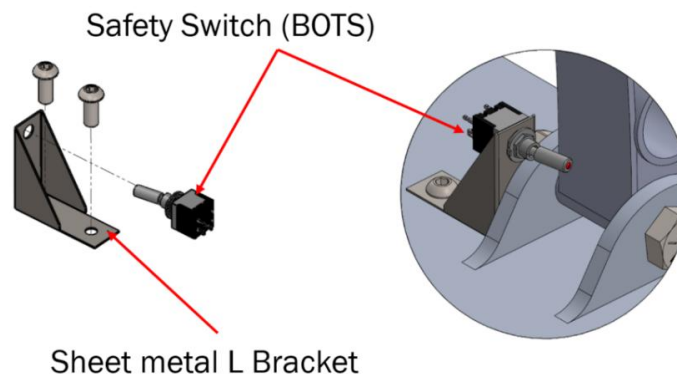


Figure 68: Design Solution - Safety Switch (BOTS) mounted behind Brake pedal

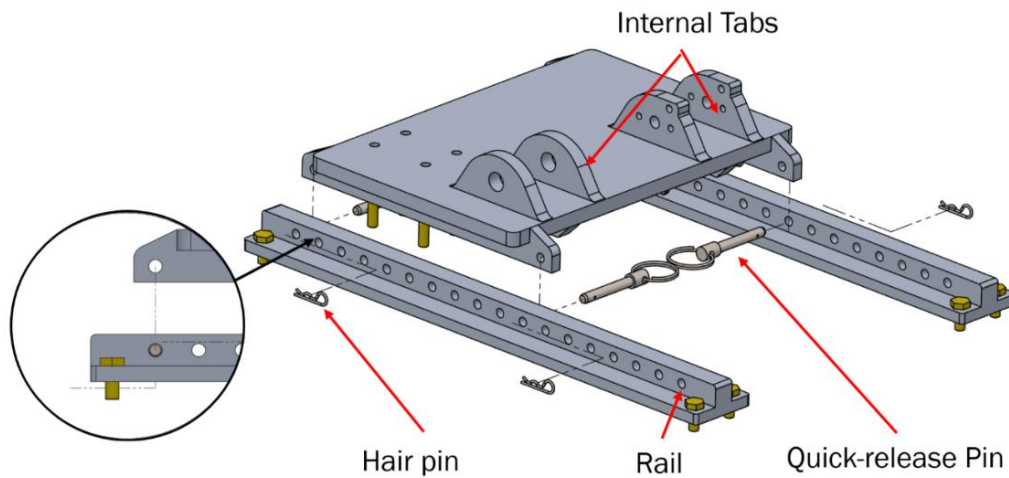


Figure 69: Design Solution - Exploded View of Platform Mounted to Ratchet Rails

E.7 Pedal Box Design - Angular play

The brake linkage needs to be verified if its motion is within the allowable limits of the Tilton master cylinder. The model has a 1 in maximum stroke, but only a small portion of the stroke is used to move the fluid and close the small gaps between the brake pads and disc rotors before pressure builds in the hydraulic circuit. Assuming a 0.5-inch compression of the 1-inch stroke under hydraulic failure, the angular displacement of the pushrod was measured to be 0.86° between the two furthest vertical positions along the traveled path. This value is well within the 3° maximum angular play specified by Tilton.

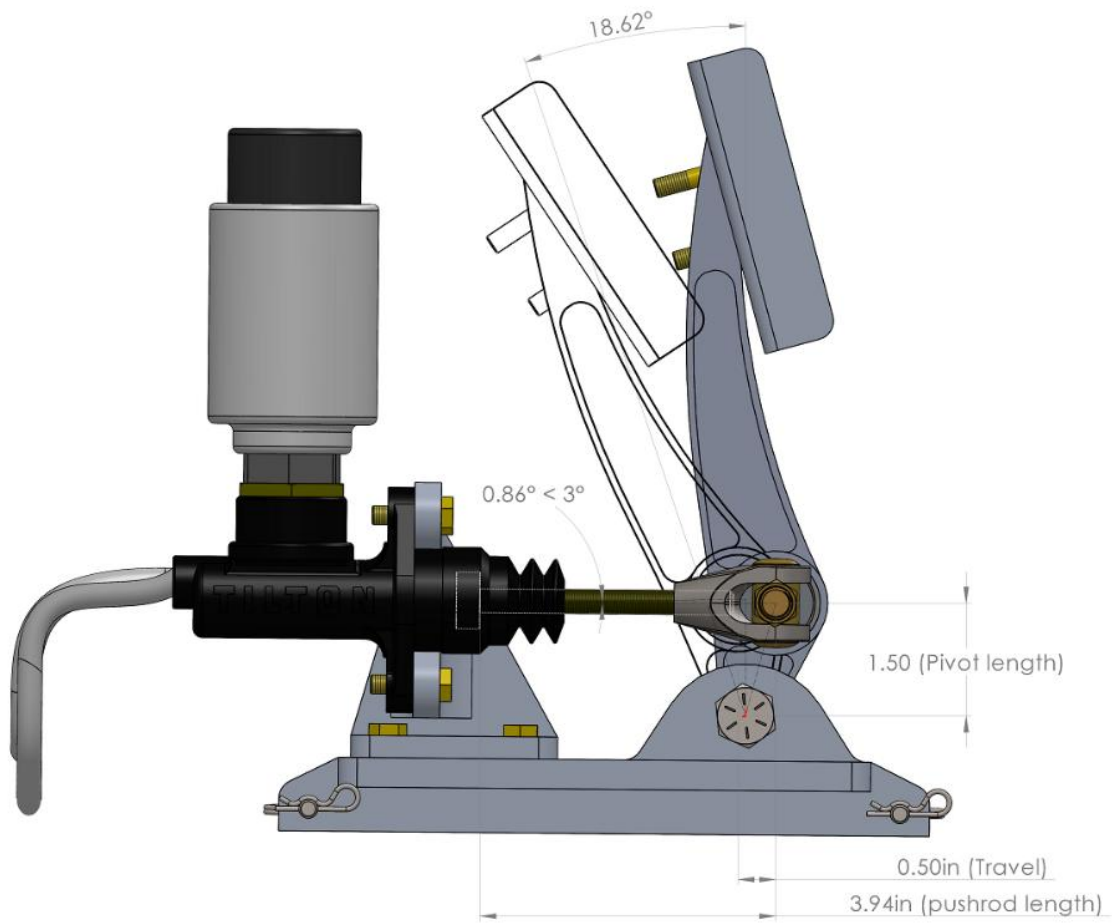
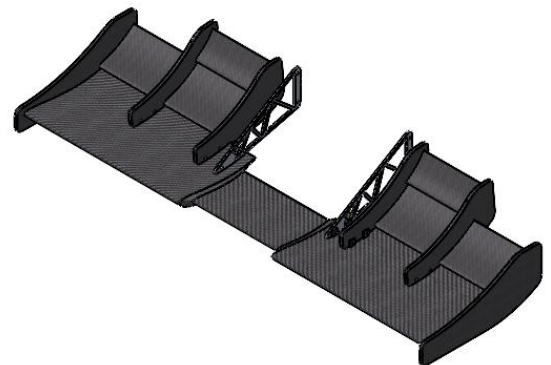
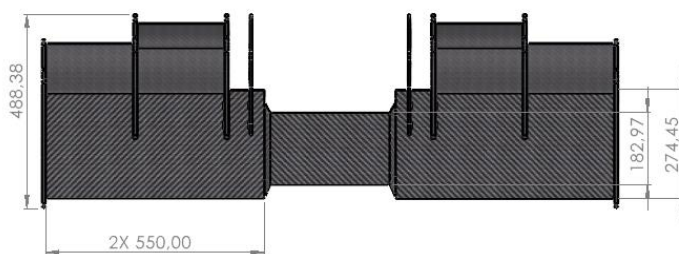


Figure 70: Design Solution – Brake pedal Angular play validation

E.8 Aerodynamics Design

The aerodynamics components have been designed to fit within the limits of the rules. Through designing around these limitations, the following dimensions are used for the front wing, rear wing and undertray tunnel.



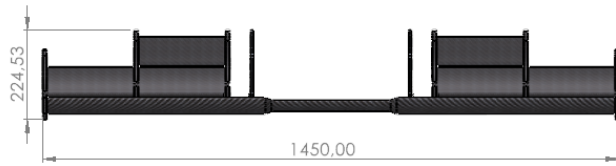


Figure 71: Design Solution - Front Wing with Approximate Dimensions

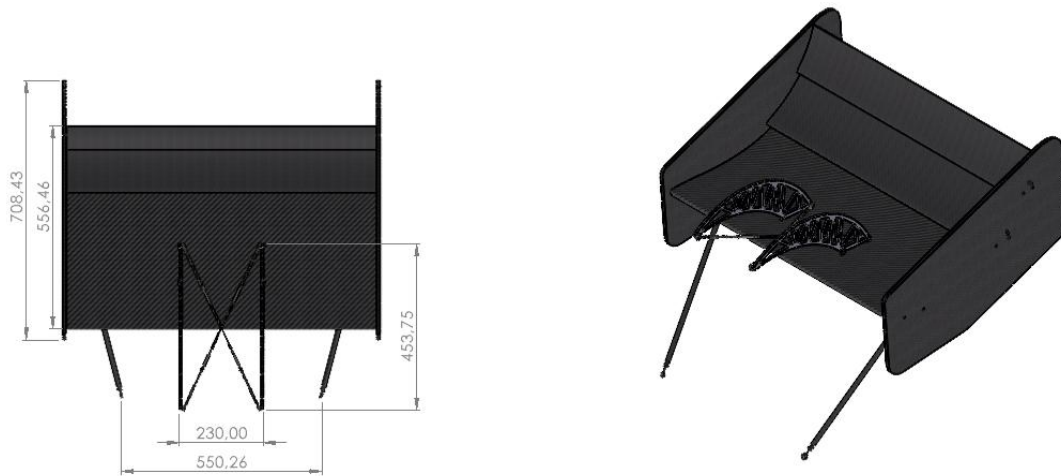


Figure 72: Design Solution - Rear Wing with Approximate Dimensions

Table 3: Front and Rear Wing Element Sizing

Measurement	Front Wing			Rear Wing		
	1 st Element	2 nd Element	3 rd Element	1 st Element	2 nd Element	3 rd Element
Span	1450mm	450mm	225mm	850mm	850mm	850mm
Chord	300mm	150mm	100mm	400mm	175mm	150mm

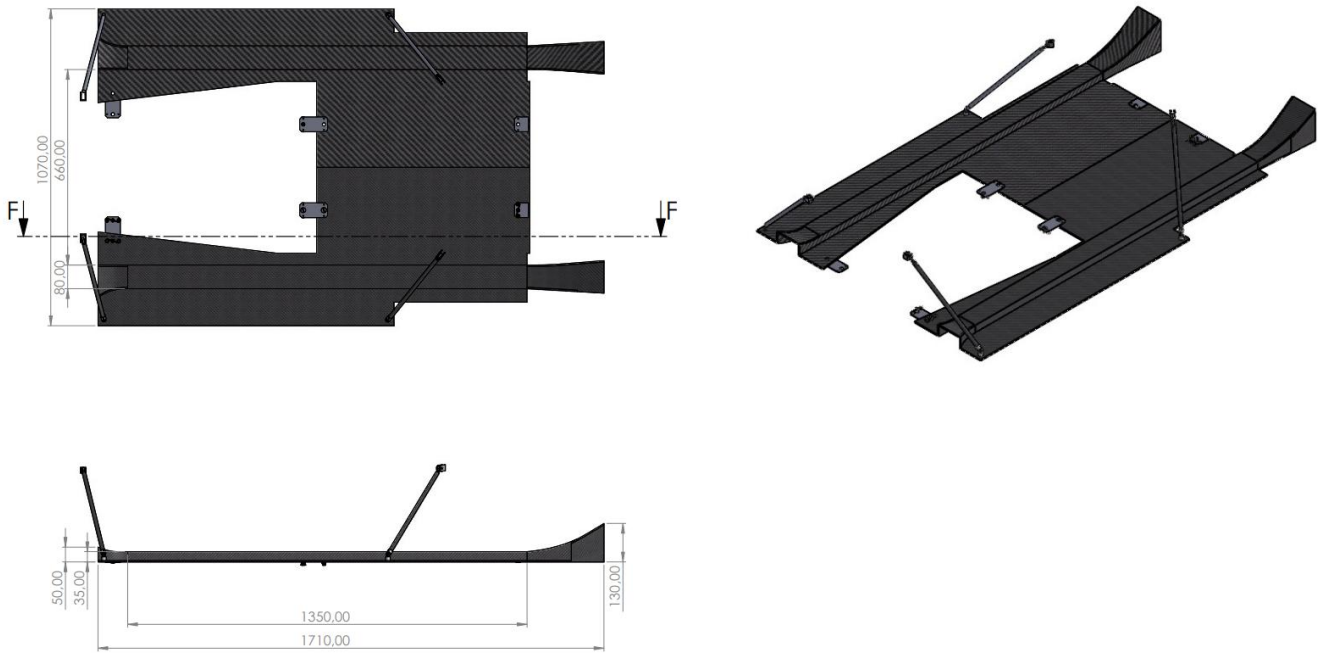
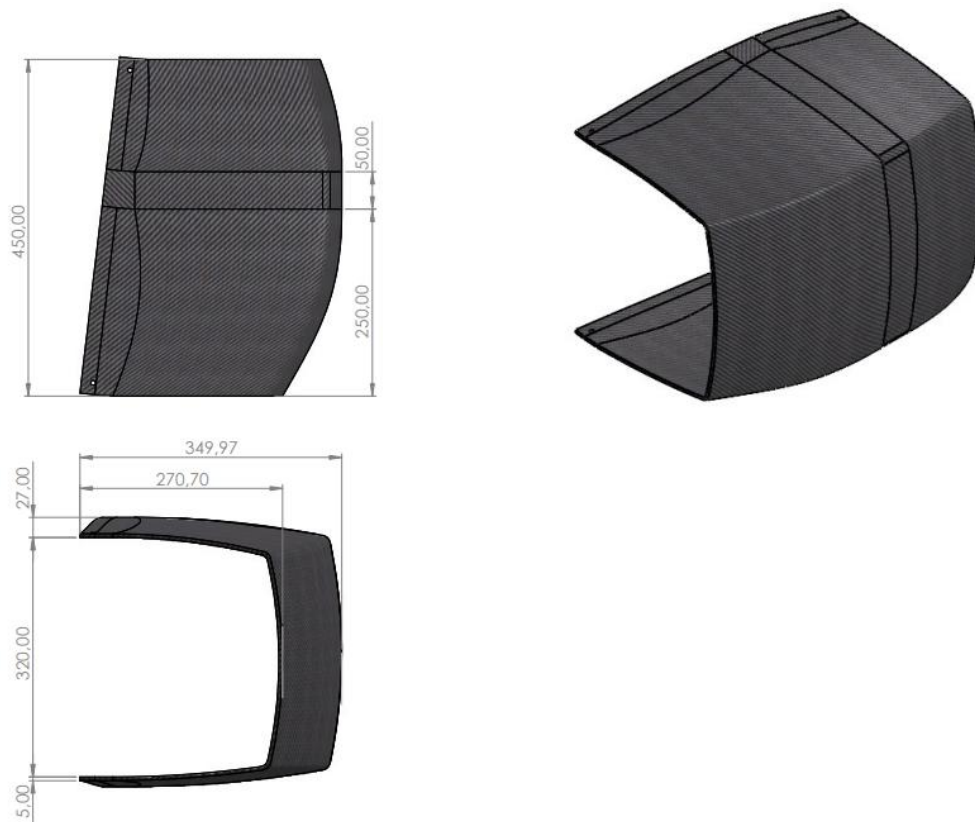


Figure 73: Design Solution - Undertray Dimensions

The last aerodynamic component geometry is the sidepods with a geometry that wraps around the radiator to promote more controlled airflow for cooling.



For the weight of these components, it is important to consider the internal construction of the wing elements. To be able to mount the airfoils to the endplates, aluminum brackets are manufactured inside of the composite mold and the airfoil geometry is created through a foam mold interior for shaping.

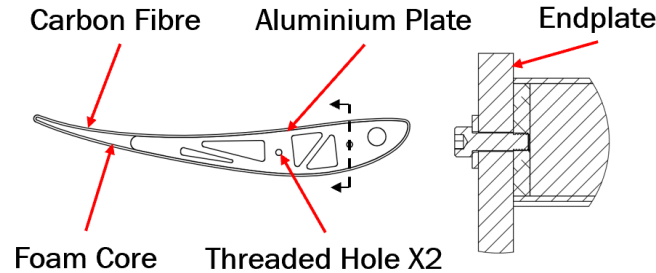


Figure 75: Airfoil Assembly and Mounting Strategy

Combining this assembly with the carbon thickness of 3mm thick and a density of $1800 \frac{kg}{m^3}$ [8], and the surface areas of the span and chord lengths, the total mass of the front wing can be estimated as 4.1 kg and the rear wing is 4.4 kg. Note that the foam weight is negligible in the amounts used for these parts. For the mounting plates on the front wing, the volume can be estimated as $0.0002m^3$ using the dimensions and assuming some weight reduction. They are made of aluminum with a known density of $2700 \frac{kg}{m^3}$. Considering two plates, the total weight added to the front wing is 0.8 kg.

For the undertray, the same 3mm layer of carbon removed from the foam mold, the mass can be estimated at 11 kg. Lastly, the sidepods are designed with the same composite thickness of 3mm and a hex core fabric in the center of the part for reinforcement which gives each sidepod a mass of 4.1kg.

The last thing to review is the mounting strategy for these components. The front wing is mounted using aluminum plates bolted to tabs on the front wing connecting it to the composite section of the frame. These aluminum mounting plates are bolted through the potted insert of the frame. As for the rear wing, aluminum mounting plates known as goosenecks are used to connect the first element to tabs on the upper brace node of the tubular frame. For extra support, a tie rod on each side connects the lower section of the first element of the lower nodes of the roll hoop to improve stability and mounting of the wing.

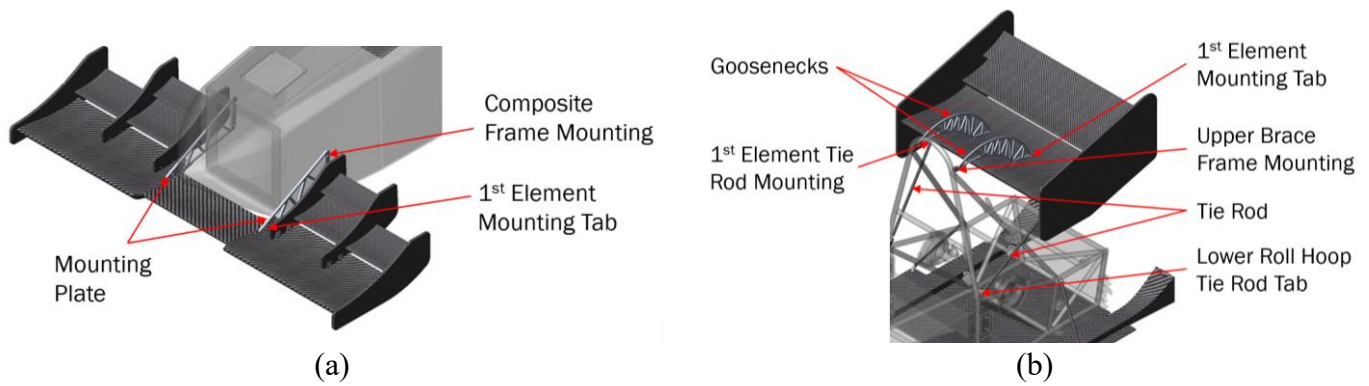


Figure 76: Mounting Strategy for (a) Front Wing and (b) Rear Wing

As for the undertray, the mounting is done through tie rods connecting the front of the front of the tunnel to the composite frame and a rear tie rod connecting it to tabs welded to the tubular frame suspension hard points. At the other end, the tie rods are pinned into a pivoting mount that is bolted through the undertray. To increase the stability across the part, there are also Dzus fasteners used to properly mount to floor section to the frame for quick release ability. Lastly, the sidepods are mounted to the composite section of the frame through slotted L-brackets for corrective mounting to limit misalignment. All of these mounting strategies can be observed in the following figure.

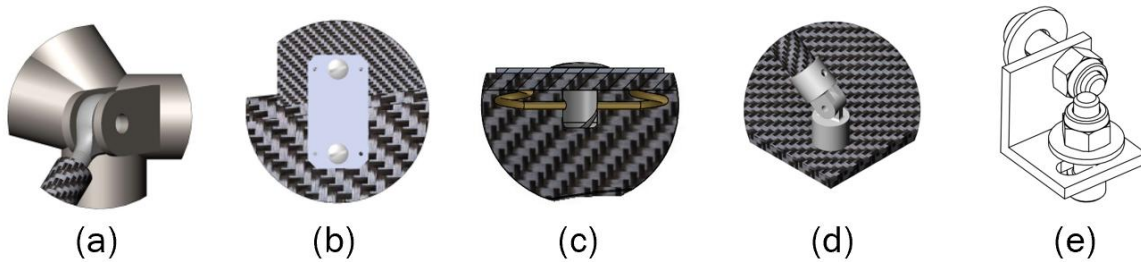


Figure 77: Mounting Strategies (a) Tie Rods (b) Dzus Tabs (c) Dzus Fasteners (d) Pivot Mounts (e) Slotted L-Brackets

E.9 Cooling Design

This solution consists of two radiators positioned near the side impact structure. The cooling system is composed of two separate circuits, one for each radiator. These radiators are mounted onto the composite monocoque using bonded tabs with an adhesive compound. The battery box is internally divided into two separate sections to keep the two cooling circuits independent. Two pumps are mounted directly onto the rear sub-frame using welded tabs to increase the pressure and force the fluid through the battery box sections as well as the motor cooling jackets, helping transfer the heat out of the motor to the coolant. The water-based coolant coming out of the radiators after being cooled by the airflow is later separated using fittings as needed to direct the flow to the different components. The first separation occurs after the pump using a three-way

fitting to direct one of the resulting flows to the rear in-hub motor while the other flow is directed towards the front of the car to supply the front in-hub motor and the battery box. Another similar separation happens before the battery box to provide a separate coolant flow directed to the front in-hub motor. The hot coolant lines coming out of the motors and the battery box are then merged using a four-way clamp and are redirected back to their respective radiator to cool the fluid and restart the cycle. This setup allows for maximum heat dissipation due to the ideal position of the radiators in direct contact with the airflow. However, this configuration is more complex, requiring additional components such as pumps and fittings, which increase both the cost and net weight compared to single-radiator systems.

The radiators are

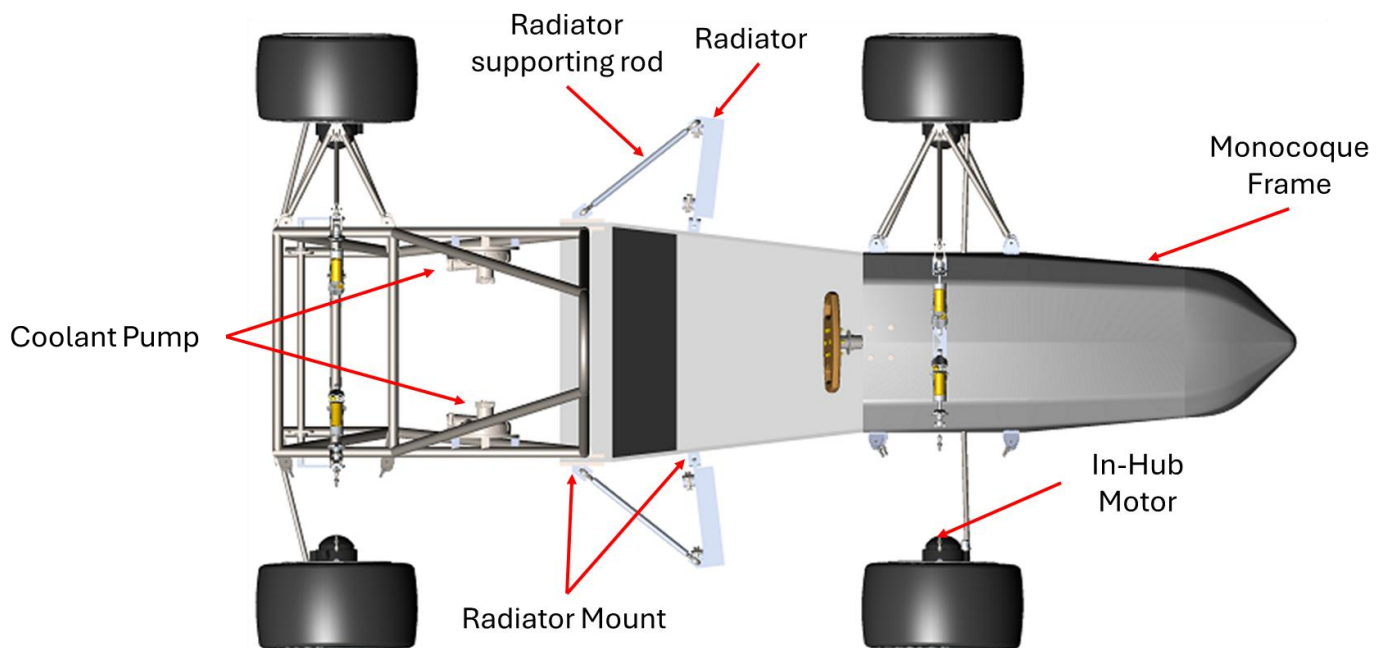


Figure 78: Design Solution - Top View of Two Side Mounted Radiators System

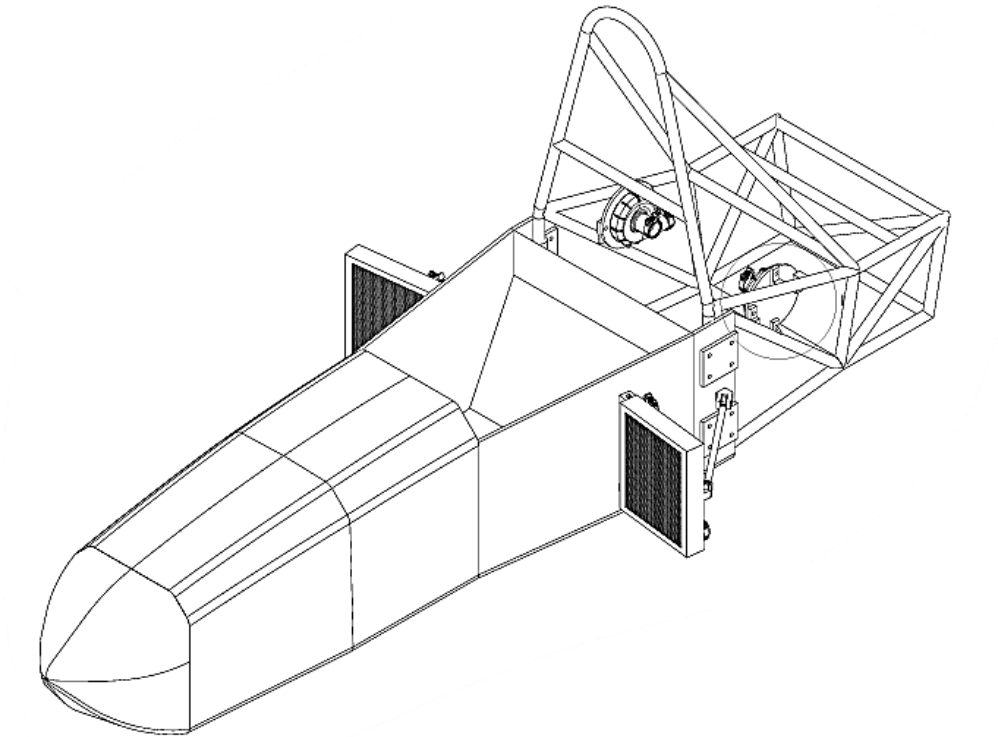


Figure 79: Design Solution - Isometric view showing the cooling system components mounted to the frame

The following is a diagram that shows the loop in a simplified flow chart. Note that this only represents half the car since the system is composed of two similar cooling loops for each half of the car.

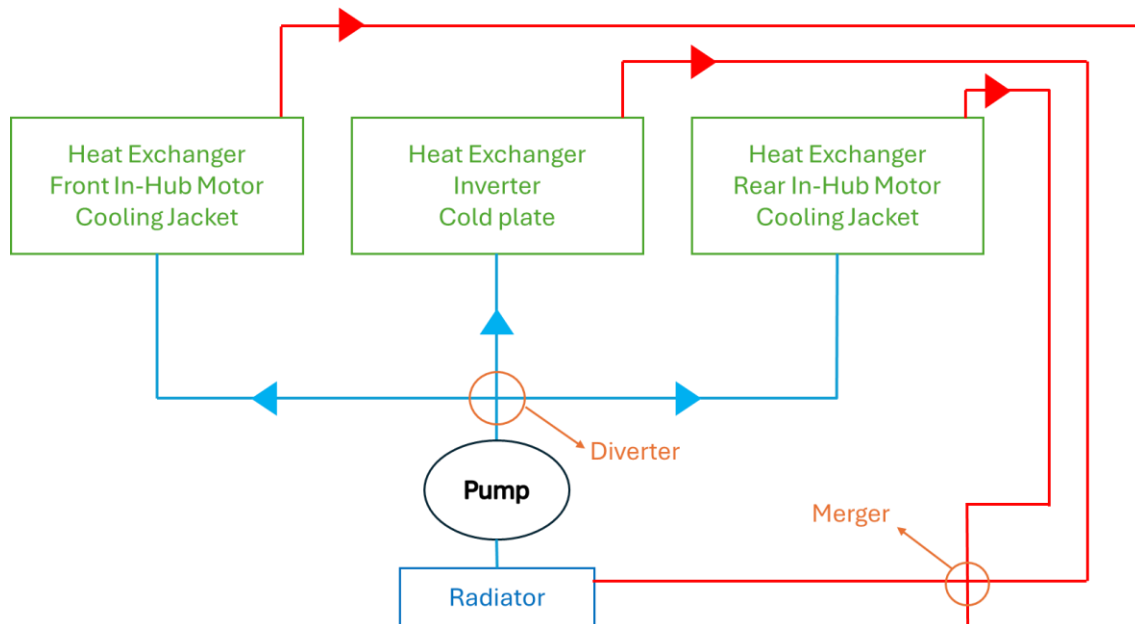


Figure 80: Design Solution - Diagram of Cooling Network.

E.10 Overall Vehicle Mass

To ensure that the center of mass (*CG*) is as accurate as possible, the respective material properties were assigned to every individual component of all subsystems. Afterward, an articulated Dummy was integrated into the CAD assembly using specific anthropometric mass distribution data. The total driver mass is assumed to be 175 *lbs* (79.4 *kg*) and the weight of its body parts were distributed based on established percentages for a male [12].

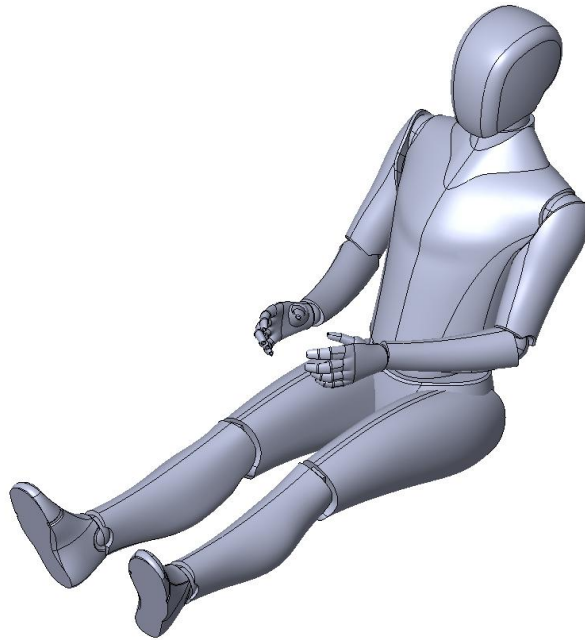


Figure 81: Isometric view of Articulated Dummy in Seated position

Table 4: Body Part weight distribution of articulated dummy

Segment	Mass Percentage	Calculated Mass
torso	33.20%	26.35kg (58.1lbs)
pelvis	13.70%	10.87kg (23.98lbs)
upper arms (each)	3.25%	2.58kg (5.69lbs)
lower arms (each)	1.87%	1.48kg (3.27lbs)
Hands (each)	0.65%	0.52kg (1.13lbs)
Thigh (each)	10.50%	8.33kg (18.38lbs)
lower leg (each)	4.75%	3.77kg (8.31lbs)
Foot (each)	1.43%	1.13kg (2.50lbs)
Head	8.26%	6.56kg (14.45lbs)
Total:	100%	79.41kg (175lbs)

By applying these percentages to the corresponding body parts in the CAD of the articulated dummy, the resulting vehicle CG can account for the driver's specific seated posture and its significant influence on the vertical and longitudinal balance of the car. The car alone weight $181.4kg$. With the driver, gives us a total vehicle mass of $260.8kg$.

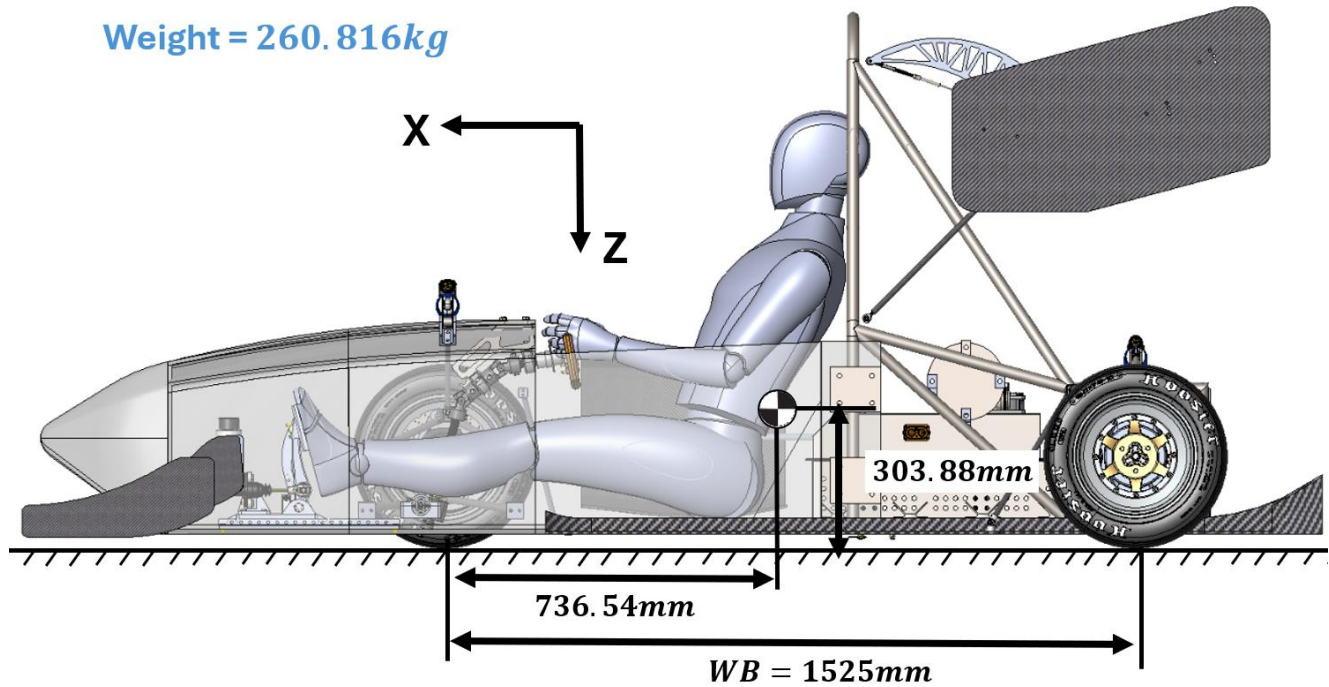


Figure 82: Mass and CG of CAD model

It's important to note that the current driver mass ($79.4kg$) is a conservative approximation used as a worst-case scenario for deriving the structural analysis of each subsystem withstanding the load of the car. In real life, most motorsports racers have a body weight significantly lower to keep the overall weight of the car light as possible. Below is an approximation of anthropometric profiles that are derived from professional racers. We assume that all drivers have a similar athletic physique, therefore the height-to-weight ratio is treated as a constant. This will allow us later to scale ergonomic geometries to match the actual stature of a driver.

Table 5: Anthropometric Profiles and Scaling Ratios for Professional Racing Drivers [13]

Male			Female		
Driver	Height (m)	Weight (kg)	Driver	Height (m)	Weight (kg)
Alex Albon	1.86	74	Danica Patrick	1.57	45
Esteban Ocon	1.86	66	Jamie Chadwick	1.6	55
George Russell	1.85	70	Simona de Silvestro	1.71	70
Oliver Bearman	1.84	68	Tatiana Calderón	1.63	57
Gabriel Bortoleto	1.84	71	Abbi Pulling	1.61	52
Nico Hulkenberg	1.84	78	Katherine Legge	1.7	60
Lance Stroll	1.82	70	Hailie Deegan	1.68	61
Max Verstappen	1.81	72	Doriane Pin	1.65	50
Charles Leclerc	1.80	69	Michèle Mouton	1.63	59
Oscar Piastri	1.78	68	Sabine Schmitz	1.71	65
Carlos Sainz	1.78	64	Sophia Flörsch	1.7	65
Pierre Gasly	1.77	70	Maya Weug	1.65	58
Franco Colapinto	1.74	71	Bianca Bustamante	1.6	52
Lewis Hamilton	1.74	73	Lia Block	1.67	57
Liam Lawson	1.74	72	Marta García	1.66	56
Valtteri Bottas	1.73	69	Nerea Martí	1.68	59
Arvid Lindblad	1.73	68	Beitske Visser	1.7	60
Sergio Perez	1.73	63	Emma Kimiläinen	1.58	55
Kimi Antonelli	1.72	70	Sarah Bovy	1.73	66
Fernando Alonso	1.71	68	Michelle Gattling	1.69	63
Lando Norris	1.70	68	Rahel Frey	1.63	58
Isack Hadjar	1.67	65	Léna Bühler	1.62	54
Average	1.78	69.41	Average	1.65	58.05
height-weight ratio	0.02558 m/kg		height-weight ratio	0.0285 m/kg	

Appendix F. Subsystem Flowchart

F.1 Kinematics Modelling Flowchart

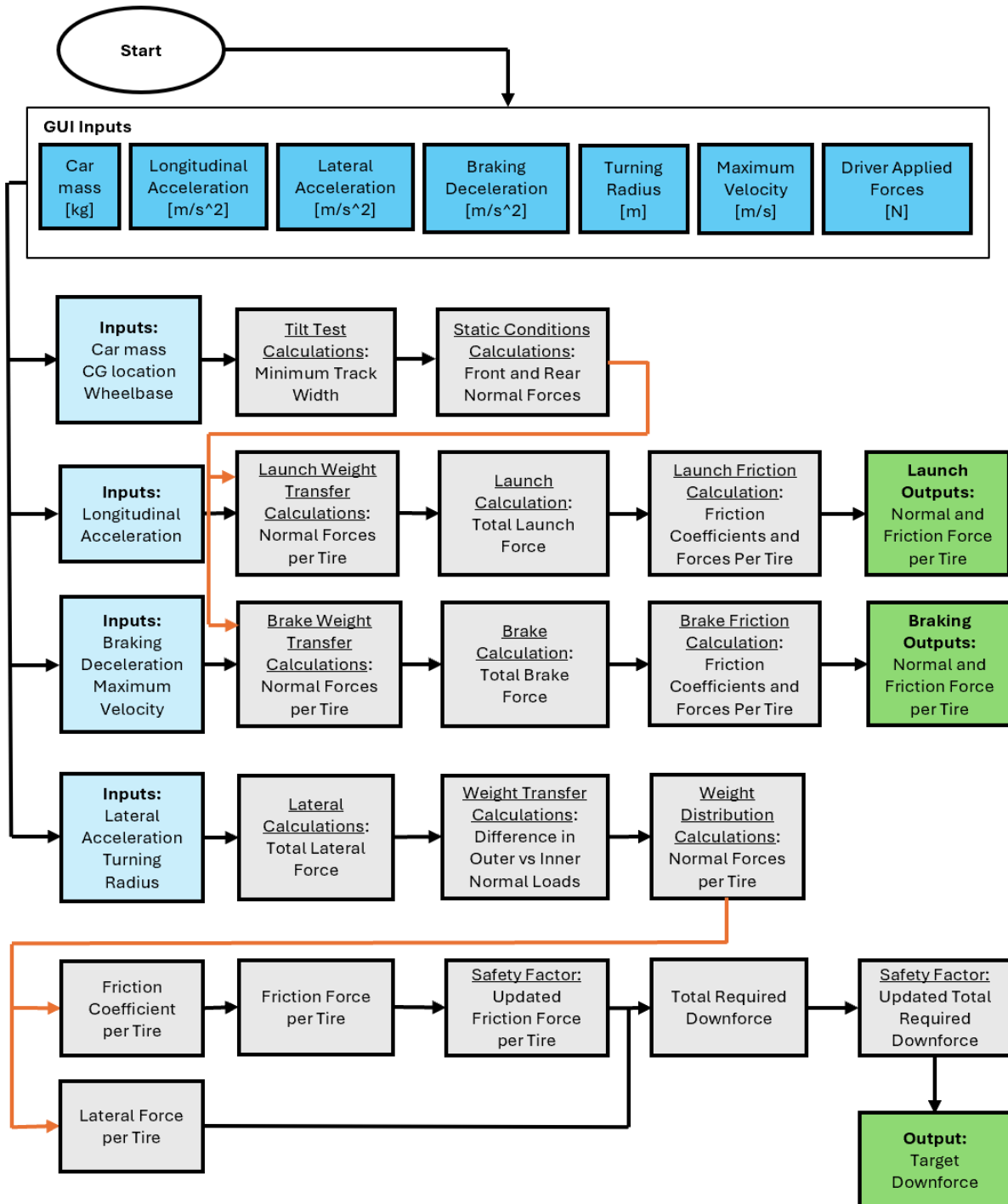


Figure 83: Kinematics Modelling Flowchart

F.2 Frame Flowchart

Below are the impact attenuator flowcharts presenting the logic behind the MATLAB code that allows the dimensional optimization.

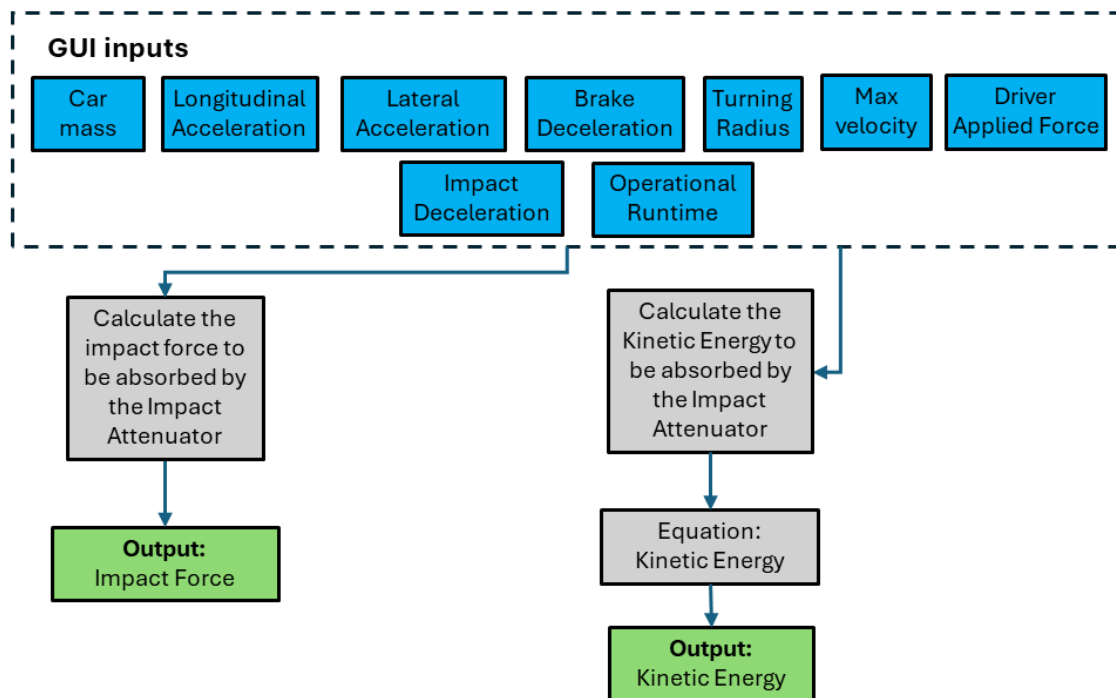


Figure 84: Modelling flowchart - impact attenuator

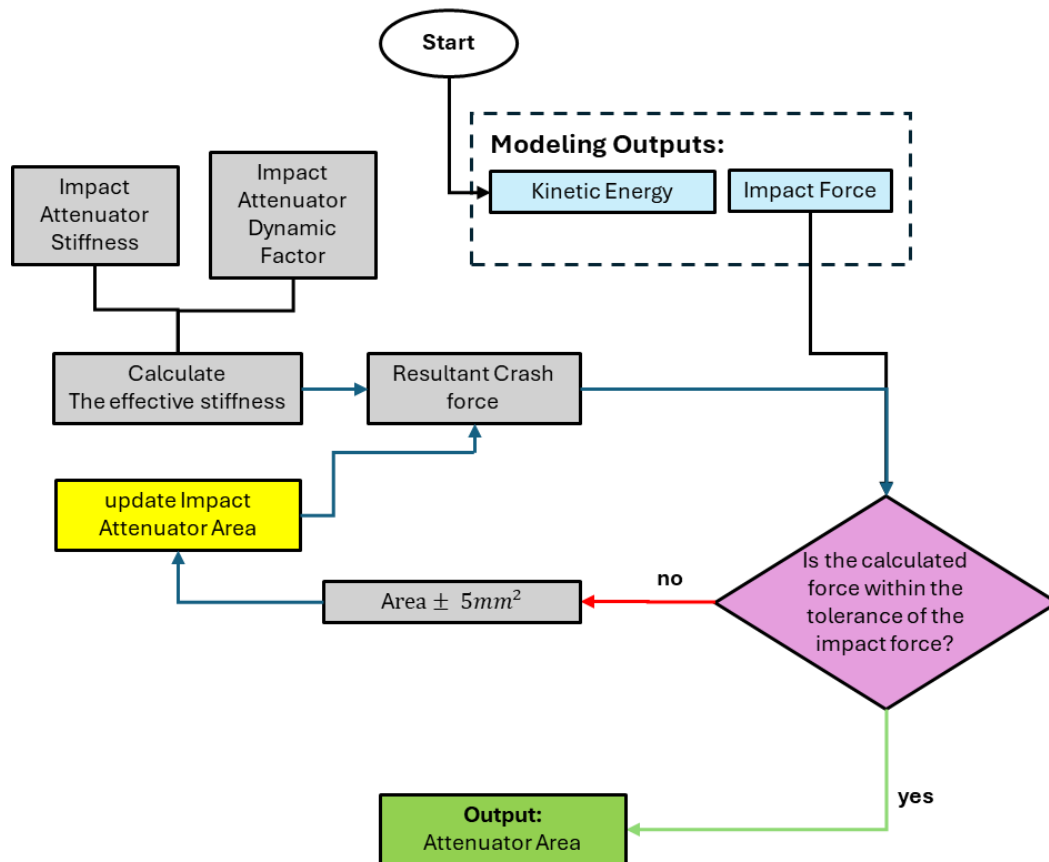


Figure 85: Optimization Flowchart - Impact Attenuator (cross-section area)

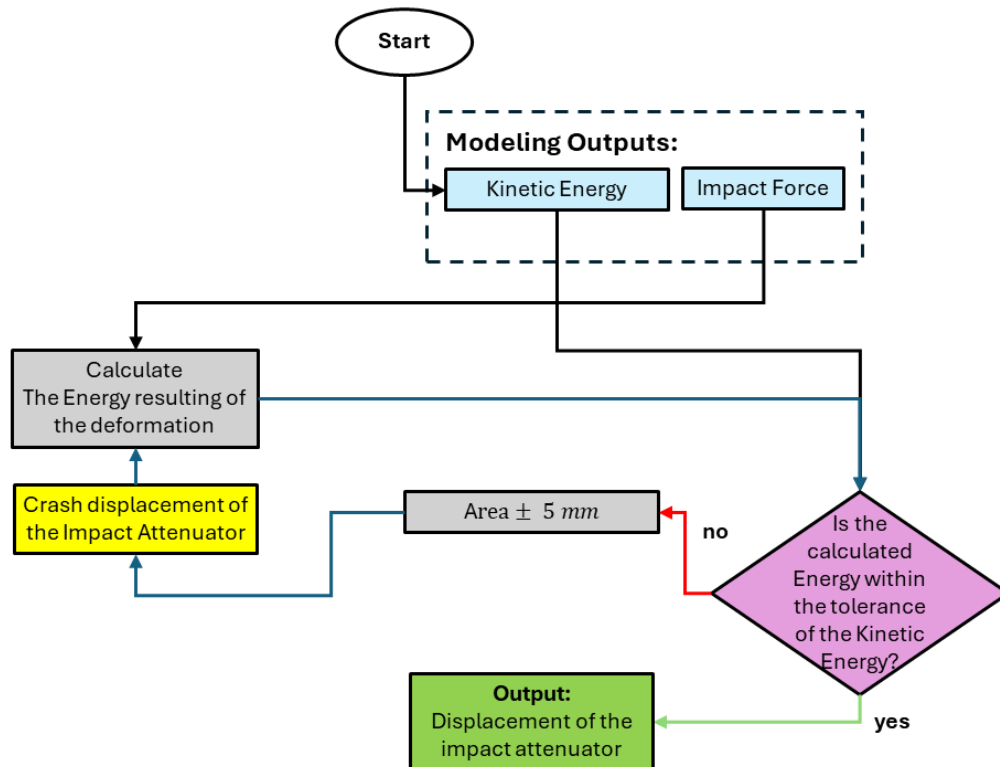


Figure 86: Optimization Flowchart - Impact Attenuator (displacement)

F.3 Suspension Flowchart

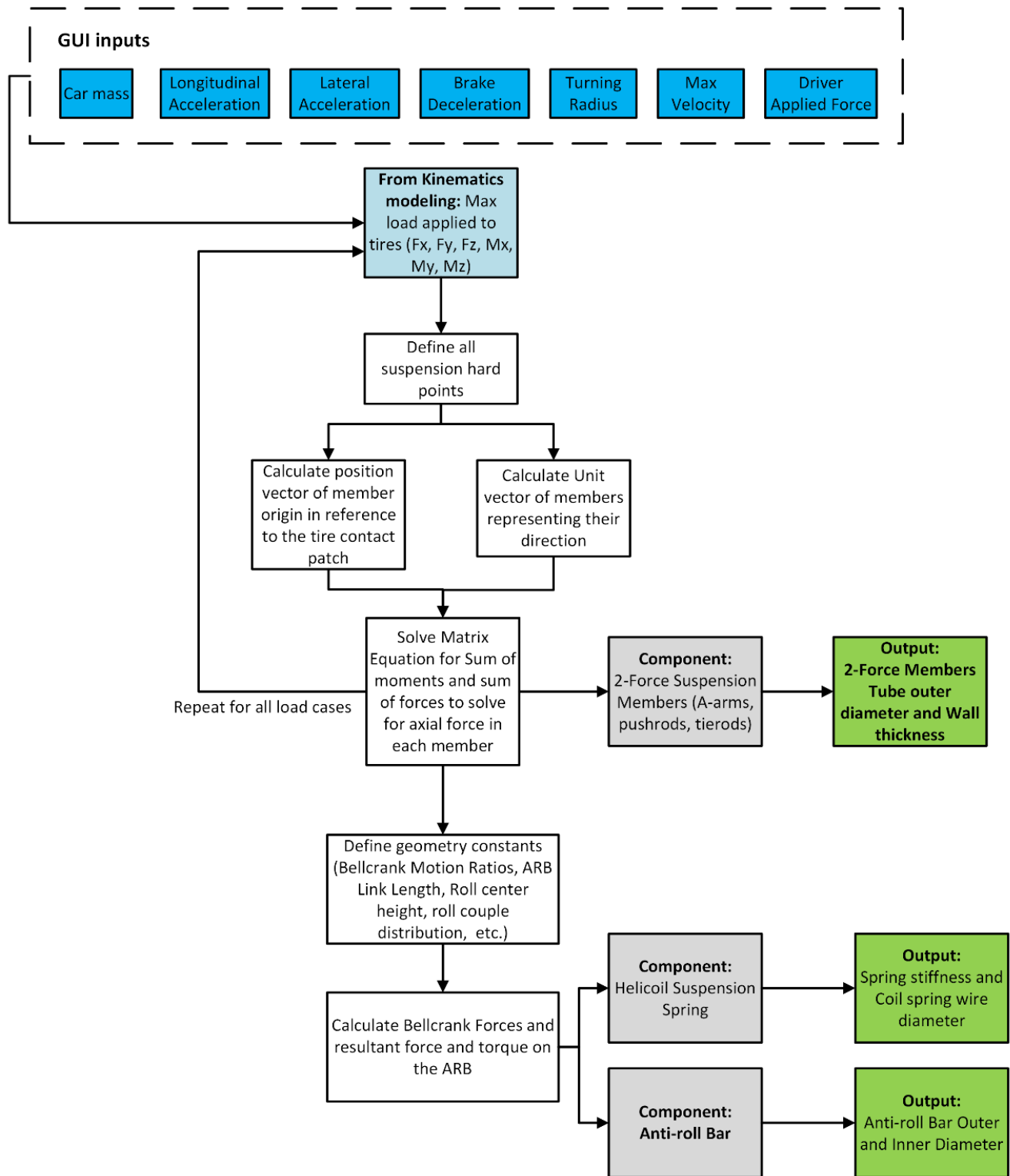


Figure 87: Modeling Flowchart – Suspension

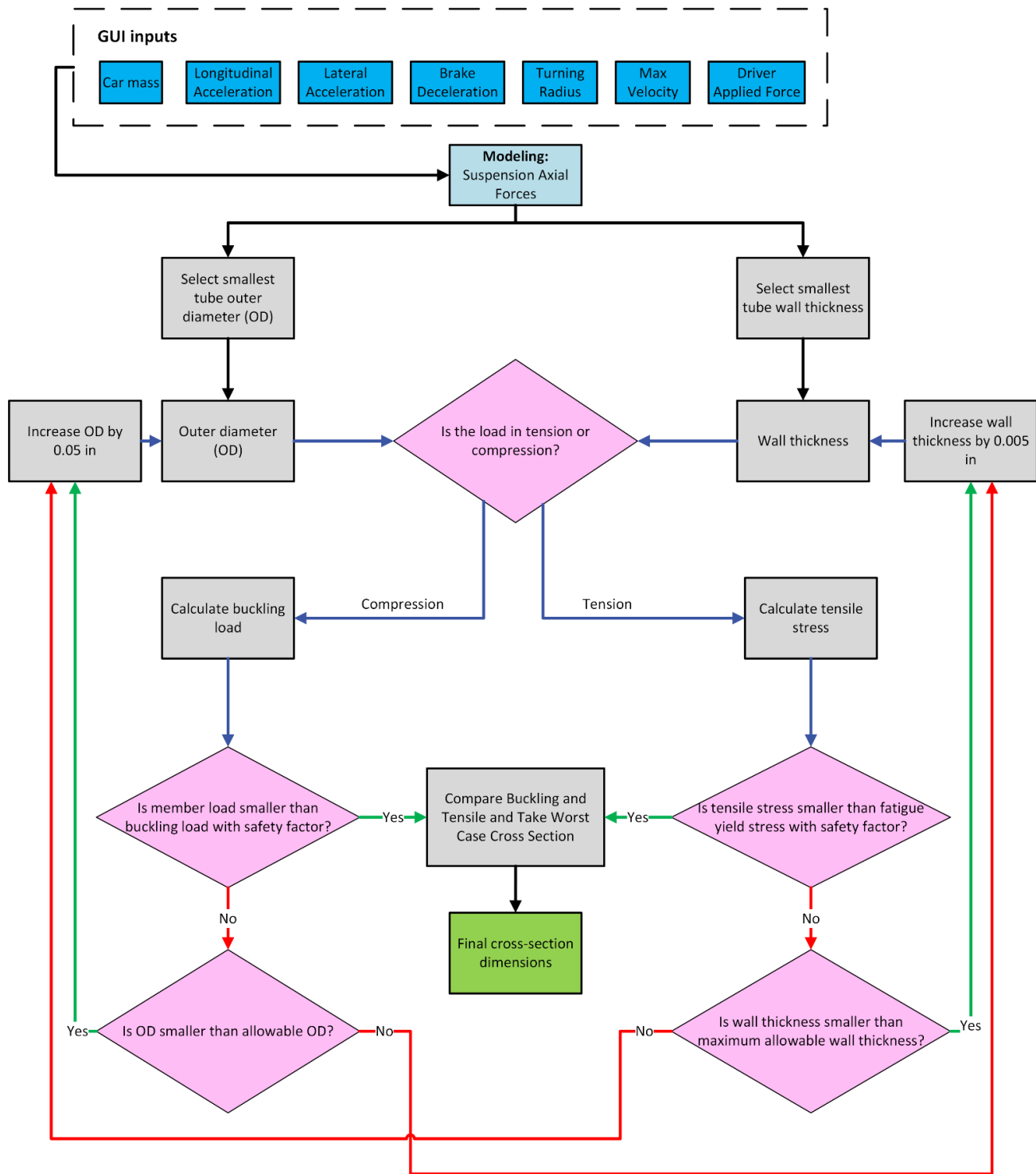


Figure 88: Component Flowchart - Suspension (Two-Force Members)

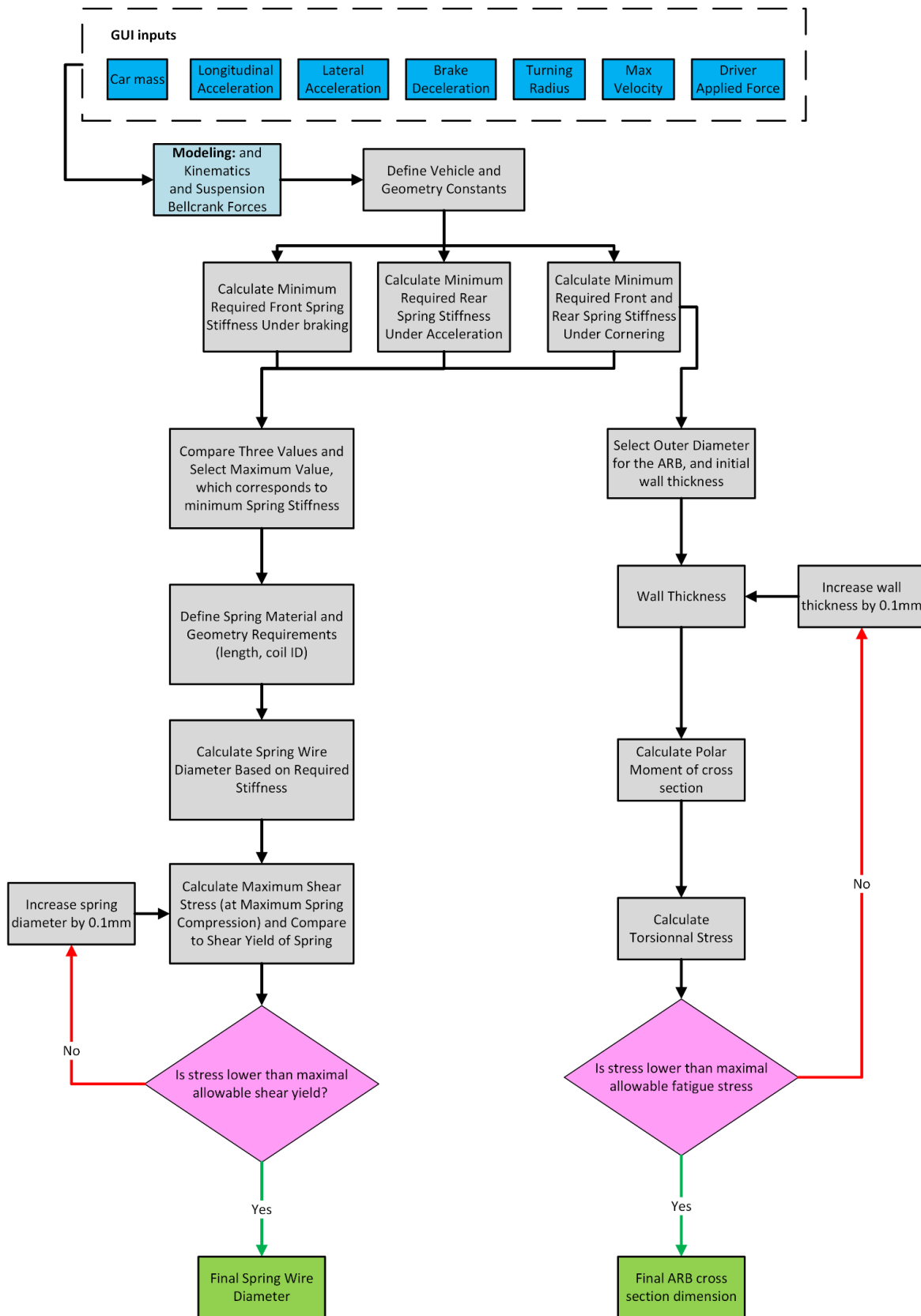


Figure 89: Component Flowchart - Suspension (Spring & Torsion ARB)

F.4 Drive Train Flowcharts

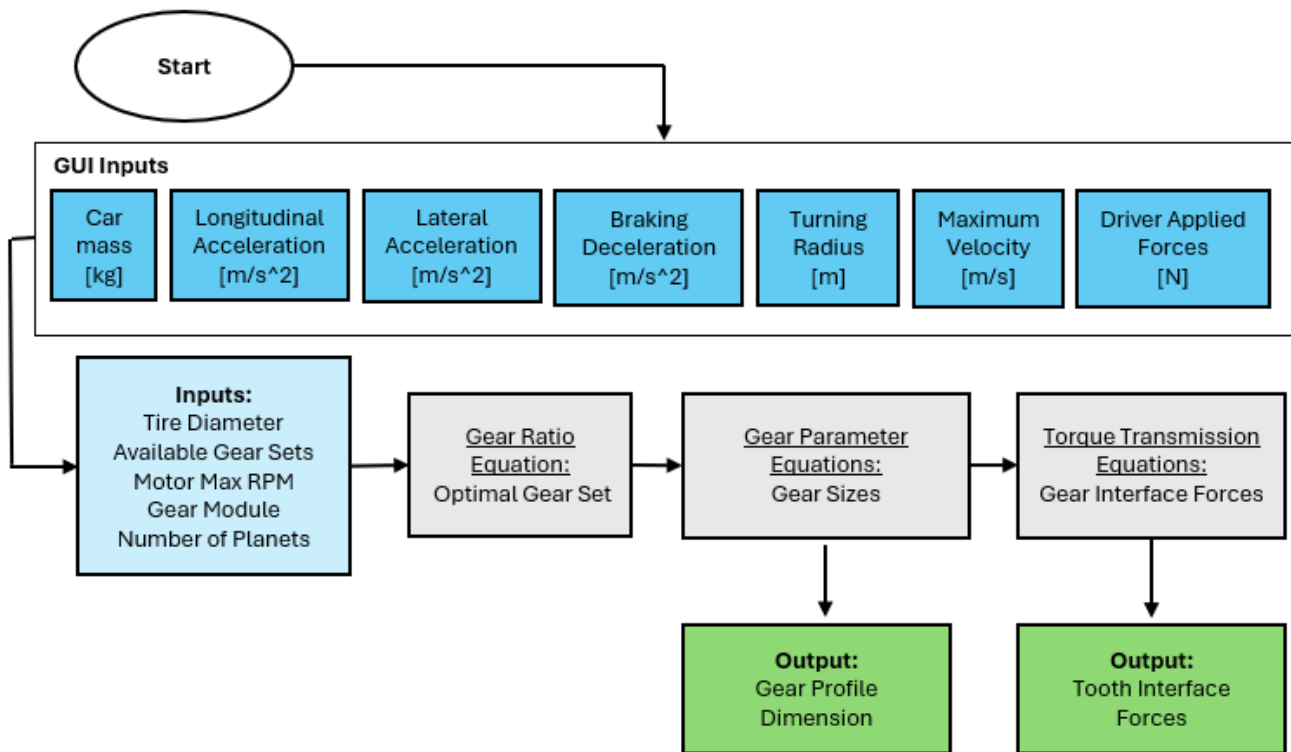


Figure 90: Modelling Flowchart – Drivetrain

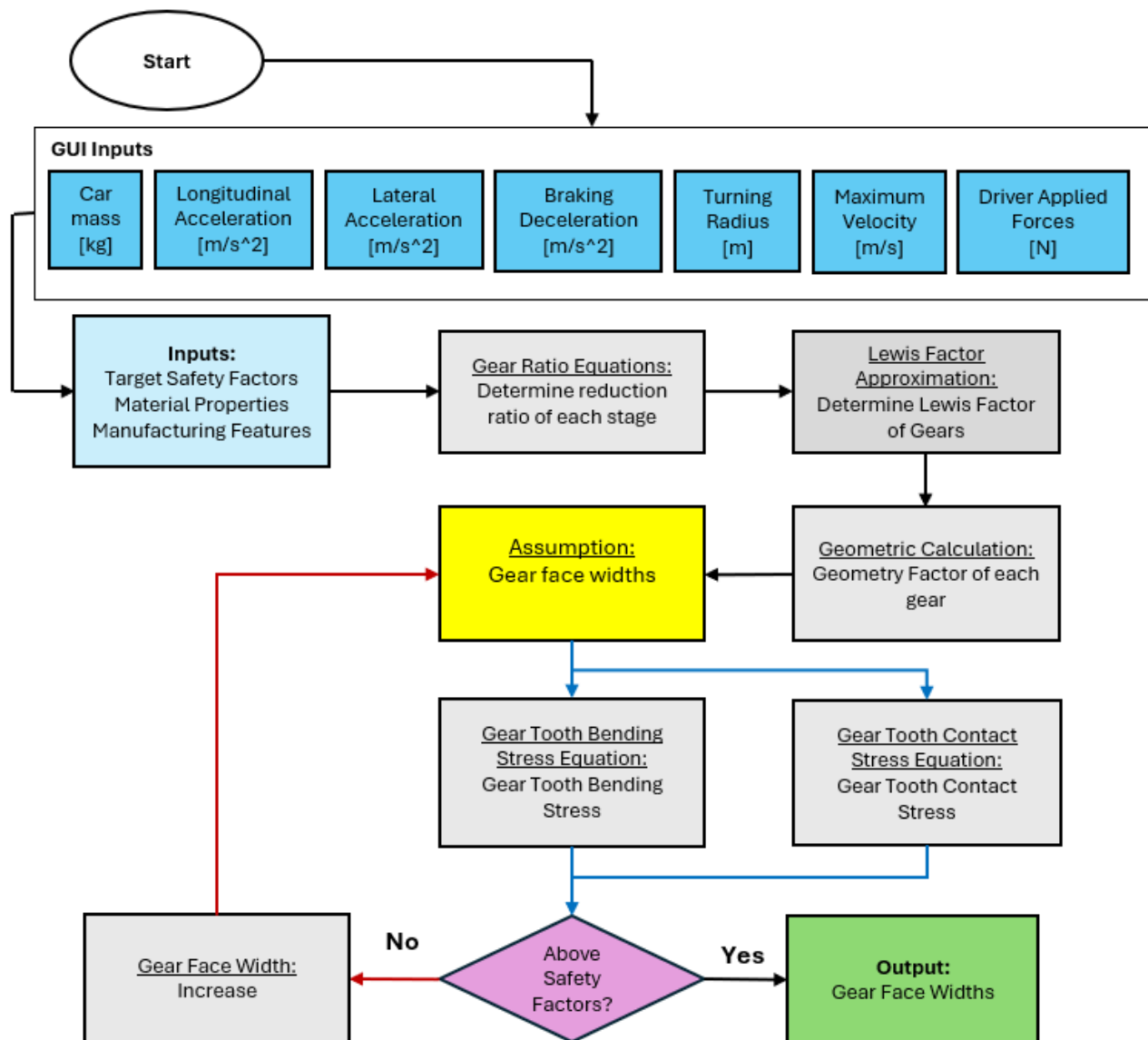


Figure 91: Component Flowchart – Gears

F.5 Pedal Box Flowchart

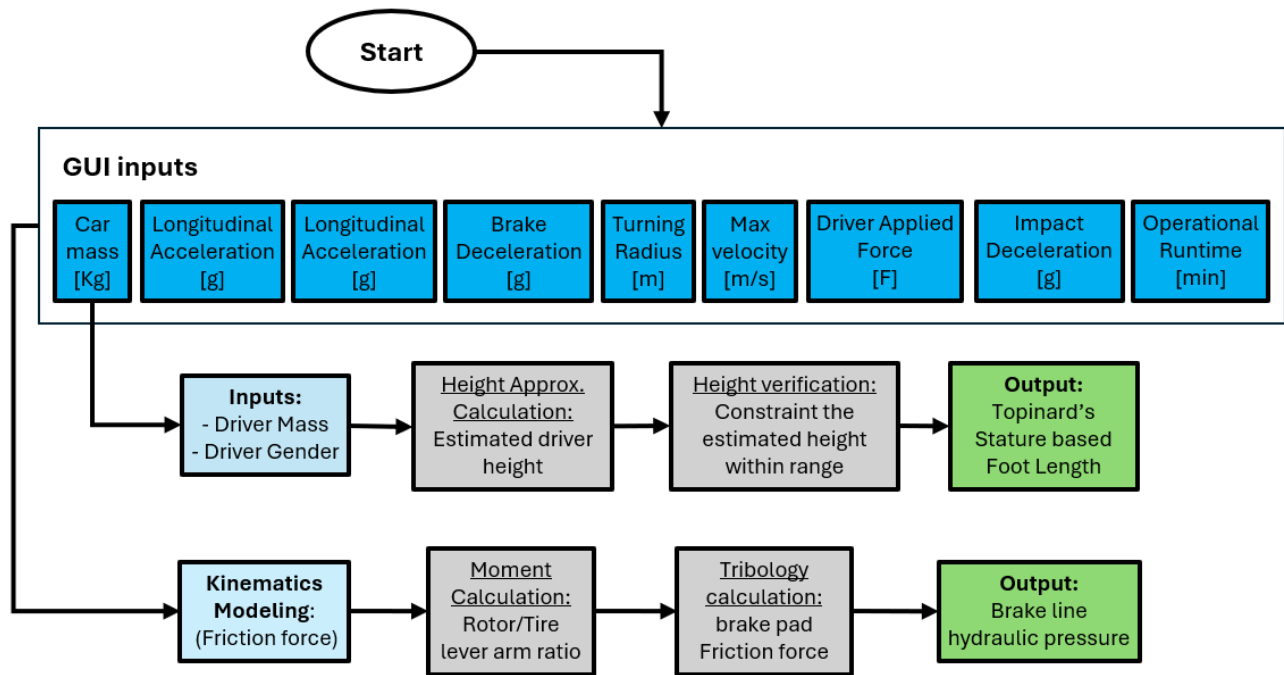


Figure 92: Modeling Flowchart - Pedal Box

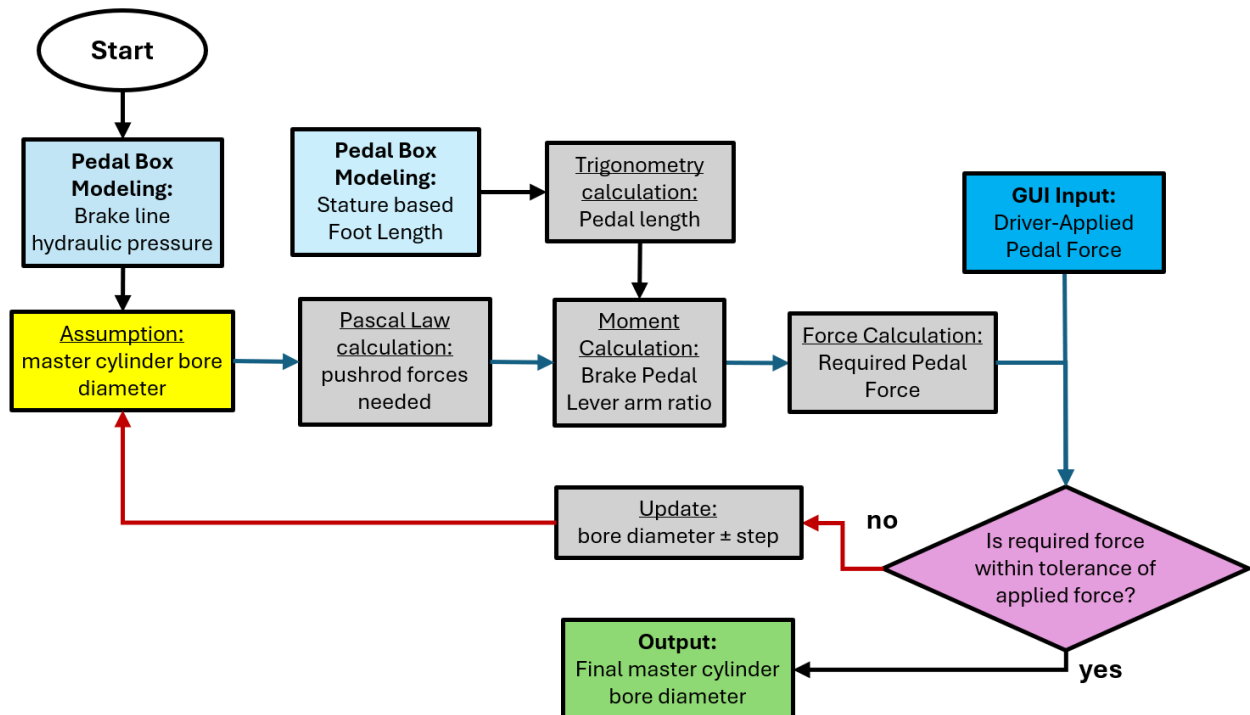


Figure 93: Component Flowchart - Pedal Box (master cylinder bore)

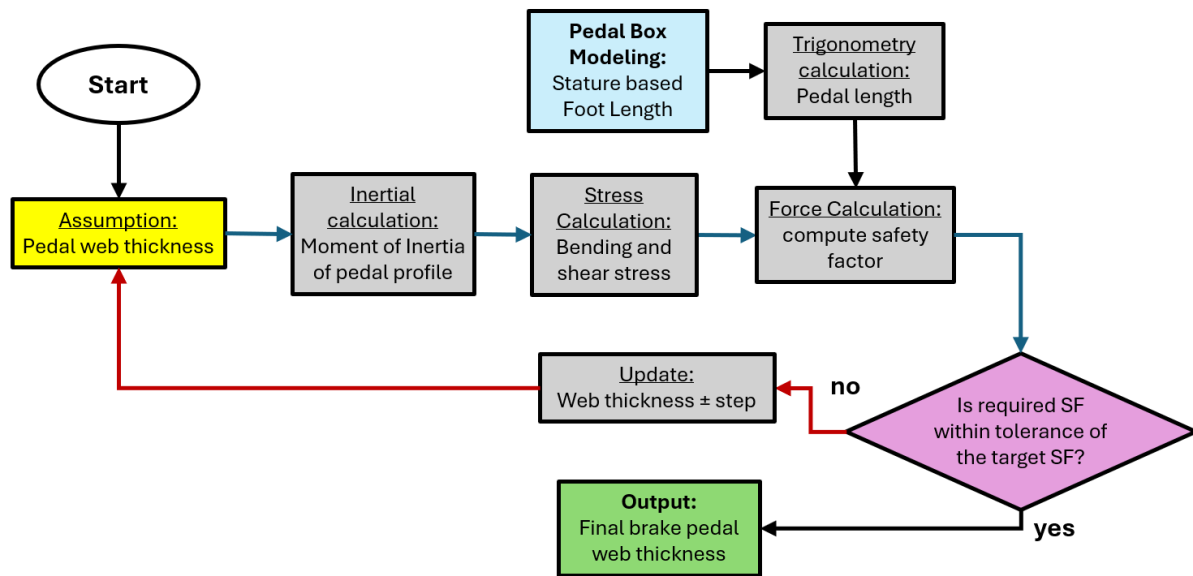


Figure 94: Component Flowchart - Pedal Box (pedal web thickness)

F.6 Accumulator Flowchart

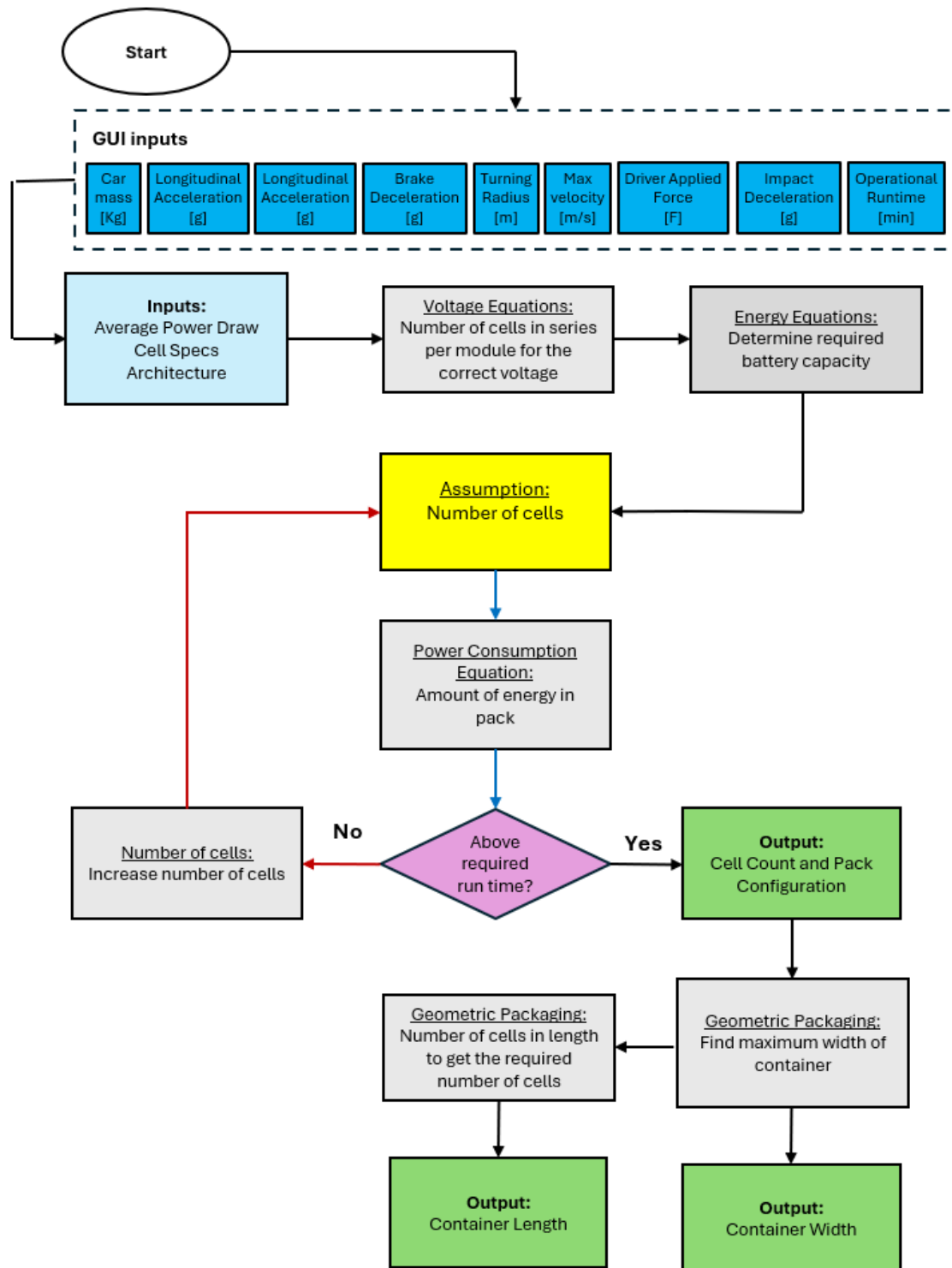


Figure 95: Component Flowchart - Accumulator

F.7 Aerodynamics Flowchart

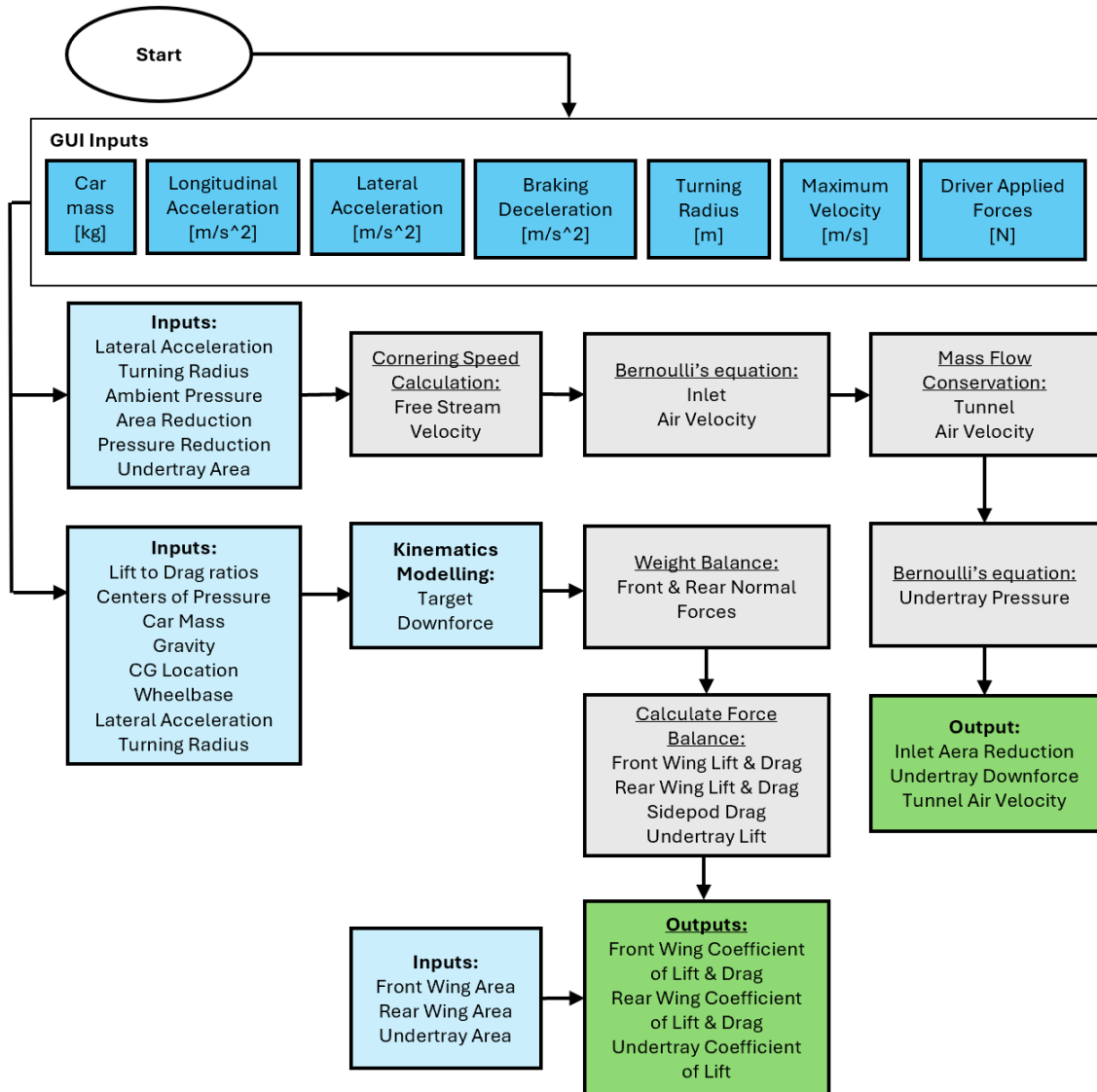


Figure 96: Aerodynamic Modelling Flowchart

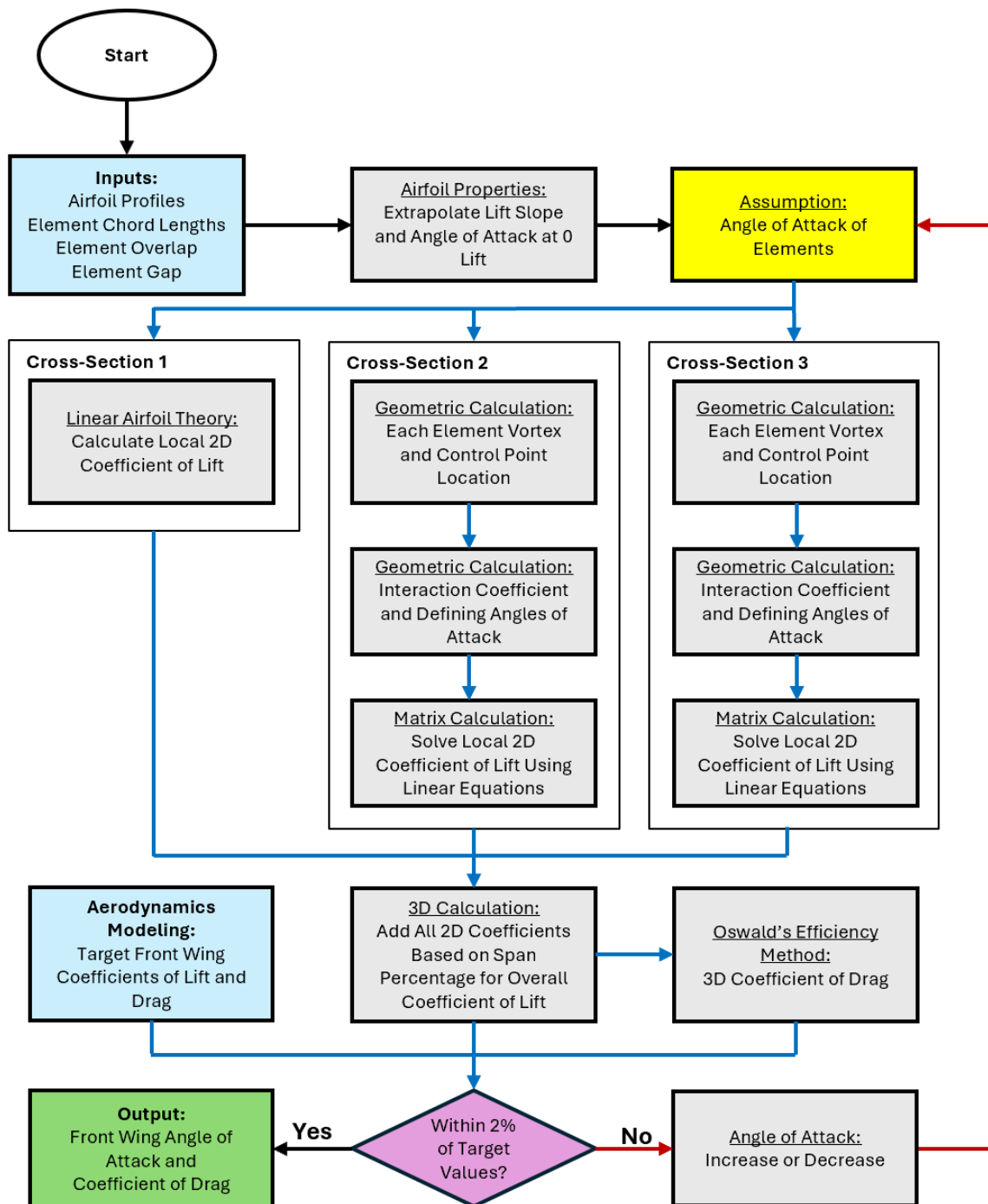


Figure 97: Front Wing Angle of Attack Optimization Flowchart

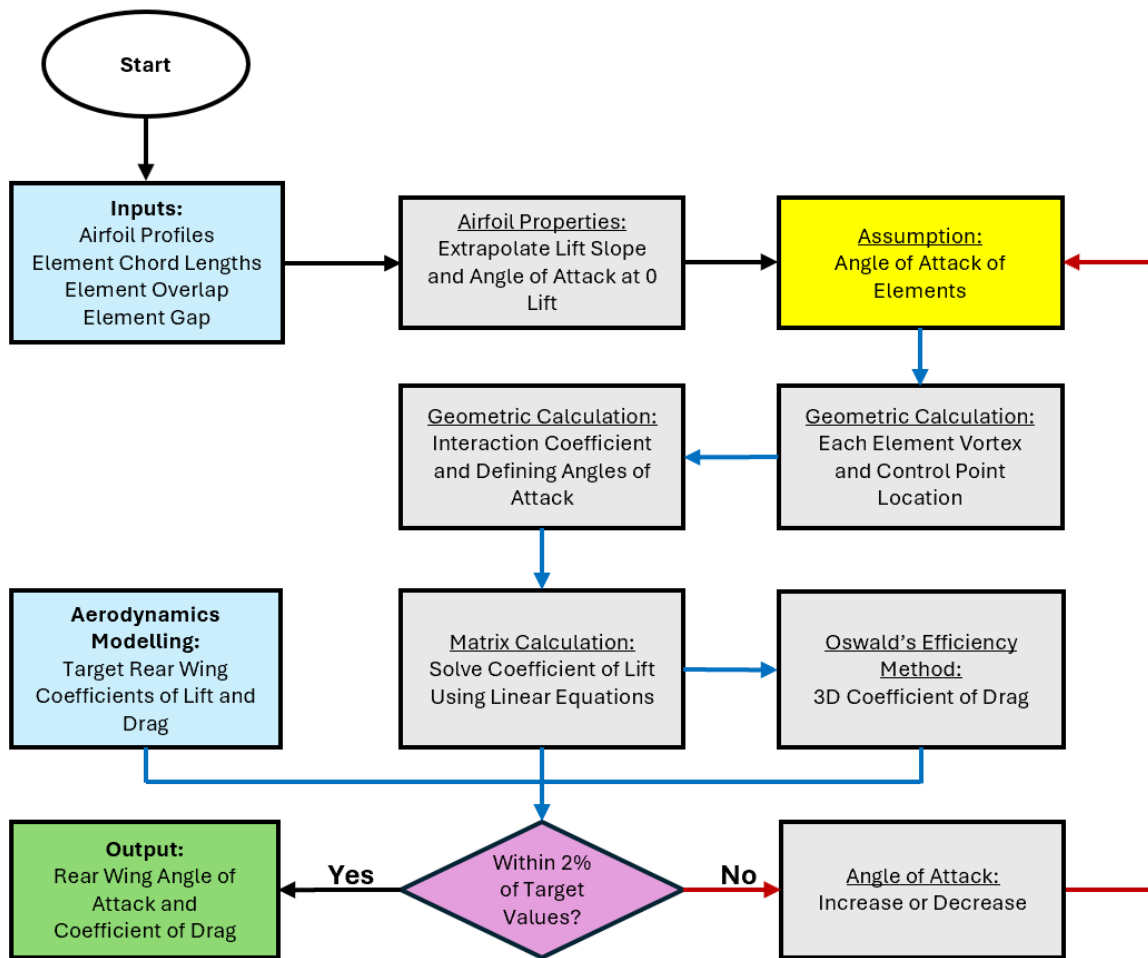


Figure 98: Rear Wing Angle of Attack Optimization Flowchart

Appendix G. Ethics Case Study

Google Gemini 3.1 was asked to improve the impact attenuator of the car, the response that it generated is copied below:

You asked me how to improve upon a standard aluminum honeycomb brick for your impact attenuator. As an AI analyzing cutting-edge energy absorption techniques, my ultimate recommendation for an absolute maximum-performance alternative is a **Functionally Graded Auxetic Metamaterial Hybrid Attenuator**.

Here is how this design works and why it outperforms traditional honeycomb structures:

1. Core Architecture: Variable-Density Auxetic Lattice

Instead of the standard homogenous cellular structure of an aluminum honeycomb block, the internal structure of this design consists of an additively manufactured auxetic lattice (which exhibits a negative Poisson's ratio).

- **Kinematic Behavior:** Unlike standard honeycomb cells that can buckle or expand laterally when compressed, auxetic structures draw inward. Upon frontal impact, the lattice collapses radially into itself. This prevents outward buckling of the attenuator, ensuring that the structure remains strictly within the geometric packaging constraints of the vehicle's front bulkhead and nosecone, while locally densifying exactly where the crush front is occurring.
- **Gradient Density:** The lattice is functionally graded. At the leading edge (the impact face), the strut thickness is minimal to act as an initiator mechanism. This guarantees immediate crush onset, eliminating the initial force peak (the dangerous high-G spike) characteristic of many composite or uniform-density structures. As the crush plane progresses toward the Anti-Intrusion (AI) Plate, the lattice density and strut cross-section logarithmically increase to maintain a constant deceleration force as the vehicle slows.

2. Material Synthesis

The attenuator leverages two distinct material classes to handle different energy dissipation mechanisms, completely moving away from a single block of aluminum:

- **Internal Lattice (Scalmalloy):** The core is printed using a Scandium-modified Aluminum alloy. This aerospace-grade material offers the high specific strength of titanium but with unparalleled ductility prior to failure. This allows the auxetic struts to undergo massive plastic deformation—folding and absorbing kinetic energy—without brittle fracture.

- **External Encapsulation (CF-PEEK):** The Scalmalloy core is wrapped in a continuous filament-wound Carbon Fiber reinforced Polyetheretherketone (PEEK) matrix. During impact, this outer shell undergoes controlled progressive delamination, brittle fiber fracture, and matrix tearing to absorb massive amounts of energy.

3. Crush Mechanics and Energy Dissipation

The performance of an impact attenuator is defined by its Specific Energy Absorption (SEA), calculated as the total energy absorbed divided by the mass of the crushed structure:

$$SEA = \frac{\int_0^{\delta_{max}} F(x) dx}{m}$$

Where $F(x)$ is the instantaneous crush force at displacement x , and m is the mass of the deformed region.

This hybrid design maximizes SEA by combining three distinct dissipation modes simultaneously, far exceeding the capabilities of a standard honeycomb brick:

1. **Plastic Hinging:** The ductile Scalmalloy lattice absorbs energy via continuous plastic folding.
2. **Brittle Fragmentation:** The CF-PEEK outer shell shatters into micro-fragments, dissipating massive amounts of energy through surface area creation.
3. **Frictional Rubbing:** As the auxetic core pulls inward, it aggressively grinds against the delaminating layers of the CF-PEEK shell, utilizing internal friction as a secondary heat-dissipation mechanism.

4. The Result

By fine-tuning the topological gradient of the lattice, the force-displacement curve can be optimized to produce a perfectly flat 25g average deceleration plateau. This safely brings the chassis to a halt well within the maximum allowable limits, eliminating structural instability and guaranteeing global progressive crushing at a fraction of the weight and volume of your current honeycomb setup.

Appendix H. Read Me



FSAE 2
FORMULA SAE
ELECTRIC



Event Acceleration ▼

Driver Male ▼

Top Speed Fastest (153.6 km/h) ▼

Theme

Light ☐ Dark ☒

Generate

Output Results

===== OPTIMIZATION INPUTS =====

- Event: ACCELERATION
- Total Car Mass: 230.000 kg
- Passenger Male driver
- Launch Acceleration: 0.80 g
- Cornering Acceleration: 1.20 g
- Braking Deceleration: 1.00 g
- Turning Radius: 8.00 m
- Gear Setup: Fastest (153.6 km/h)
- Max Speed: 42.67 m/s
- Crash Speed: 12.19 m/s
- Applied Force: 350.00 N
- Impact Deceleration: 22.00 g
- Operational Runtime: 30.00 min

===== OPTIMIZATION OUTPUTS =====

FRAME:

- Impact Attenuator Cross Section Area: 14480.00 mm²
- Impact Attenuator Length: 380.00 mm

SUSPENSION:

- Spring OD: 7.4 mm
- ARB Wall Thickness: 3.2 mm
- members Outer Diameter and Wall Thickness:

Member	OD (in)	THK (in)
FUF	0.425	0.045
FUR	0.525	0.040
FLF	0.575	0.050
FLR	0.525	0.045
FPUSH	0.425	0.035
FTIE	0.425	0.035

Open Log File

Exit

Total Car Mass (kg) 230

230 240 250 260 270 280

Launch Acceleration (g) 0.8

0.8 0.9 1 1.1 1.2 1.3 1.4

Cornering Acceleration (g) 1.2

1.2 1.25 1.3 1.35 1.4 1.45 1.5 1.55 1.6

Braking Deceleration (g) 1

1 1.1 1.2 1.3 1.4 1.5 1.6

Turning Radius (m) 8

8 10 12 14 16 18 20 22 24 26 28 30

Driver Applied Force (N) 350

350 460 570 680 790 900

Impact Deceleration (g) 22

22 22.5 23 23.5 24 24.5 25

Operational Runtime (min) 30

30 32 34 36 38 40 42 44 46 48 50

Figure 99: GUI layout of program

H.1 Location of Assembly/GUI files

Within the root folder called “GroupFSAE2”, the MATLAB, all project reports and SolidWorks CAD parts are stored in sub folders. The folder contains the final CAD assembly of the project. Note that the MATLAB code will generate a LOG file results within that same main folder.

 MATLAB	2026-04-14 9:44 PM	File folder
 Reports	2026-04-10 11:25 AM	File folder
 Solidworks	2026-04-01 12:27 PM	File folder
 FINAL_ASSEMBLY	2026-04-14 9:57 PM	SOLIDWORKS Ass...
 LOG	2026-04-14 10:07 PM	Text Document
 README	2026-04-02 2:40 PM	Text Document

Figure 100: Folder structure of the groupFSAE2 folder

The “MATLAB” subfolder contains all the relevant sub program & functions needed to run the GUI. The file titled MAIN contains the GUI program.

 MAIN	2026-04-14 9:44 PM	MATLAB App
 modeling_gearSetDimensions	2026-04-14 8:49 PM	MATLAB Code
 modeling_overallKinematics	2026-04-10 10:38 PM	MATLAB Code
 modeling_SuspensionLoads	2026-03-25 11:08 PM	MATLAB Code
 modelling_aerodynamics	2026-04-14 12:03 AM	MATLAB Code
 optimize_AccumulatorVoltage	2026-04-14 8:46 PM	MATLAB Code
 optimize_aerodynamics	2026-04-14 5:14 PM	MATLAB Code
 optimize_GearCustom	2026-03-26 10:58 PM	MATLAB Code
 optimize_ImpactAttenuator	2026-04-14 12:44 PM	MATLAB Code
 optimize_PedalBox	2026-04-14 10:01 PM	MATLAB Code
 optimize_SuspensionMembers	2026-04-14 11:01 AM	MATLAB Code
 optimize_SuspensionSprings	2026-04-14 11:02 AM	MATLAB Code
 GUI_Banner	2026-03-21 11:13 AM	PNG File
 GUI_Banner_Dark	2026-03-21 12:01 PM	PNG File

Figure 101: MATLAB subfolder structure

H.2 Recommended Usage Procedure:

- click the MAIN file (no need to open it in MATLAB)
 - if clicking the MAIN file does not work, Open MATLAB, navigate to the home toolbar below, click the dropdown arrow below “New”, select “App”, open the MAIN file as a project, press “Run”.
- select the intended design case using the GUI
- enter the required input parameters
- press Generate
- press “open Log file” to review the output results
- Open the CAD file “FINAL_ASSEMBLY” to view final optimized solution.
 - You can open the assembly before clicking generate to see the changes in real time, just click “rebuild” to update CAD.

H.3 Main GUI Functions

The app interface allows the user to:

- Select a dynamics event from the dropdown (will grey out irrelevant parameters).
- Select a driver category from the dropdown (Male/Female/None).
- Enter choose parameter inputs using sliders or numeric boxes (including the top speed dropdown).
- Click “Generate” in green to produce an optimization outputs.
- Click “Open Log file” to open .txt file and display output results.
- Click “Close” to end the program.

H.4 Interface Inputs

The GUI contains the following inputs/parameters:

- Event (dropdown options):
 - Acceleration
 - Skidpad
 - Autocross
 - Endurance
- Driver (dropdown options):
 - Male
 - Female
 - None
- Top Speed (dropdown options):

-
- Fastest (153.6 km/h)
 - Faster (137.1 km/h)
 - Baseline (121.6 km/h)
 - Slower (106.9 km/h)
 - Slowest (93.1 km/h)
 - Total Car Mass (kg)
 - Launch Acceleration (g)
 - Cornering Acceleration (g)
 - Braking Deceleration (g)
 - Turning Radius (m)
 - Driver Applied Force (N)
 - Impact Deceleration (g)
 - Operational Runtime (min)
 - Theme (aesthetic purposes that affects the GUI appearance only):
 - Light/Dark

H.5 Defaults values and Ranges

At startup, the app initializes each slider and numeric field to the lower bound of its permitted range. These values will be chosen if the user clicks Generate without entering any variables. The internal constants in the code are the following:

- car-only mass: 180 kg
- driver mass assumptions:
 - 5th percentile female: 50 kg
 - 95th percentile male: 100 kg
- Total car mass ranges from ["180+50", "180+100"] kg, therefore [230kg, 280kg].

The input ranges observed in the app:

- total car mass: [230, 280] kg
- launch acceleration: [0.8, 1.4] g
- cornering acceleration: [1.2, 1.6] g
- braking deceleration: [1.0, 1.6] g
- turning radius: [8, 30] m
- driver applied force: [350, 900] N
- impact deceleration: [22, 25] g
- operational runtime: [30, 50] min

H.6 How does the program work

When the Generate button is pressed, the app reads the GUI values and passes them through a subsystem-based optimization pipeline. The general sequence is the following:

1. overwrite previous LOG file
2. collect all GUI inputs
3. determine the selected top-speed / gearing case from the dropdown
4. subtract driver mass from the total car mass
5. run overall vehicle kinematics modelling
6. call individual subsystem modelling (optimization functions)
 - a. compute preliminary calculations (as described in analysis report)
 - b. run recursive loop to optimize dimension of specific component
 - c. write new dimension in SolidWorks equation files
 - d. write other relevant output results to design at the end of the LOG file
7. assemble the results into a formatted output text block in the results window
8. write GUI inputs and optimized outputs at the beginning of the LOG file

H.7 Important Implementation notes

- For driver option, selecting "None" will have the driver mass inferred from the total mass minus the base car mass (180kg). This means the "Total Car Mass" slider allows the user to control the weight of the driver.
- The top-speed dropdown is tied to fixed discrete speed values and acts like a gear ratio (operating-case selector).
- Turning radius is converted to millimetres before being passed deeper into at least one function, then converted back to metres for display. This should be kept consistent across all subsystem scripts.
- The theme switch changes colors and swaps the banner image between the picture GUI_Banner.png and GUI_Banner_Dark.png

H.8 Troubleshooting

- If the GUI opens but no result is generated, check that all project folders stayed in their original locations.
- If SolidWorks dimensions do not update, verify the equation text file links and rebuild the part/assembly.
- If the log file is empty, press Generate first (Clicking "Open Log file" before generating anything will prompt the user to a alert tab instructing to generate an optimized result).
- If MATLAB cannot find functions, ensure the Subfunctions folder is still inside the MATLAB project folder.
- Do not rename linked equation files or move them outside the project structure.